The cosmology of Cosmological Relativity:
the expansion history from causal reassignment on a de Sitter background, the cosmogenesis branch point, and the scalar perturbation sector

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Abstract

Cosmological Relativity (CR) obtains the observed cosmological expansion by a causal reassignment on a de Sitter background at the cosmogenesis branch point, with the Einstein field equations unchanged. This paper develops that cosmology and completes it with its scalar perturbation sector. The reassignment selects the Nariai member [34] of Schwarzschild–de Sitter, whose proper frame is a homogeneous, observationally isotropic, *non-synchronous* cosmology with areal radius $r(\tilde{\tau}) \propto \sinh^{2/3}$—exactly the flat-$\Lambda$CDM scale factor—its rate fixed by the cosmological constant alone, with radiation and matter inherited content read off the cosmic clock rather than terms that source the expansion. The cosmic beginning is the branch point $r = 0$—a genuine curvature singularity of the SdS metric read over the r-chart [33], and *not* a singularity of the underlying substrate, whose throat is the defining constant α; the divergence of the r-chart invariants there is the areal coordinate degenerating; the constant-proper-time slices are exactly Euclidean ($\Omega_k = 0$) while the cosmological layers are a closed S^3—spatial curvature and dark energy one Λ read on two slicings. The acoustic scale and the apparent Hubble tension resolve as consequences of this radiation-free rate: there is no second H_0 to reconcile, and the scale is met at the directly measured H_0 by a single inherited datum, $\rho_r/\rho_m \approx 2$—a one-parameter accommodation, the structural analogue of the baryon-to-photon ratio. The light-element composition is produced on the same footing: the infalling matter heats above the deuterium bottleneck and re-expands through it, so the cooling leg is a standard big-bang nucleosynthesis fixing the composition with the inherited baryon-to-photon ratio η—a second boundary datum of the same handover, distinct from the radiation amplitude ρ_r/ρ_m that sets the scale. Helium-4 and deuterium sit at their observed values—the freeze-out temperature the data demands being the bottleneck the standard rate delivers, un-tuned; the near-zero metallicity of the oldest systems follows from the handover; and because the run is standard, lithium-7 carries the standard threefold over-prediction, the lithium problem shared with flat ΛCDM rather than dissolved.

On this cosmology the primordial scalar spectrum *factorizes*. The de Sitter substrate *determines* its structure—the geometric coherence of the acoustic phases on the null boundary (not a super-horizon freeze-out), the classical non-vacuum character of the fluctuations (the substrate vacuum lies some 10^{113} below the observed A_s), the throat's isotropization, a parameter-free low-multipole discreteness floor, and the faithful transmission of the spectral shape—while the progenitor collapse *supplies* its content, the amplitude A_s and tilt n_s, inherited as boundary data exactly as flat ΛCDM inherits the baryon-to-photon ratio. *The transmission is now supported by a mechanism rather than by the absence of a scale alone* (§3): on the contracting leg aH grows without bound as the branch point is approached, so the comoving horizon shrinks to zero and every mode *exits* it and freezes before the crossing; the imaginary-time segment's Euclidean kernel damps oscillatory content and has nothing to act on in a frozen mode, so amplitude and tilt cross unaltered while the collapse-leg acoustic phase does not—which the observed peak positions independently require, excluding a transmitted leg phase by a factor of 21–39 in the required distance [7]. The decisive result is a *transmission dichotomy*: a non-degenerate horizon's exponential near-horizon approach imprints a scale-invariant spectrum (the inflationary de Sitter-horizon mechanism), whereas

the degenerate Nariai member's power-law, scale-free approach to the branch point is *argued* to transmit the progenitor spectrum unaltered—an argument from the absence of an imprinted scale, not a computed transfer—so CR has no inflationary scale-invariant attractor, no consistency relation, and no substrate-sourced primordial B-modes. Because the distance slicing is flat, the discrete closed-S^3 source projects through the flat transfer to a parameter-free low-multipole deficit ($\ell \lesssim 8$)—mild, and on a genuine Boltzmann transfer a dip whose *minimum falls at* $\ell = 4$ (≈ 0.36 of the ΛCDM expectation there, ≈ 0.47 and 0.41 at $\ell = 2$ and 3; the shape cross-validated between two independent transfers, the depth carrying their 10–25% spread), and non-discriminating: it starves $\ell = 2$ and $\ell = 3$ together and so matches neither the sharp observed quadrupole-only dip nor the near-ΛCDM octopole, sitting consistent with ΛCDM within the lowest multipoles' cosmic variance. Taken together, these results carry a reckoning the paper draws in full: what flat ΛCDM assembles from added, tunable apparatus—inflation for the horizon, flatness, coherence, and near-scale-invariance, a dark-energy component for the late-time acceleration, added early-universe physics for the tensions—this cosmology recovers from the cosmological constant alone, each feature required by structure already present rather than produced by a sector introduced to yield it. By the economy of assumption that distinguishes a required structure from a merely permitted one, that comparison is settled now, on the theory-choice axis, in this cosmology's favour; and the data axis has now moved with it, from even toward favoured: the Hubble tension is resolved across the low-redshift distance ladder and not the acoustic angle alone—the radiation-free rate carries H_0 out of the baryon-acoustic observables, which fit at the directly measured H_0 where ΛCDM cannot—on the SDSS DR12 consensus and, at the current state of the art, the DESI DR2 dataset (thirteen measurements across seven tracers) at $\chi^2/\mathrm{dof} \simeq 1$ with the single CMB-calibrated $\Omega_m \simeq 0.31$—and the light-element abundances are confirmed within 1σ at the microwave-background baryon density (the cosmogenesis paper [16]). The cosmology is thus empirically favoured, not merely more economical, with its exposed edges named: a CR-specific $\sim$8% signature in the diffusion *scale* (the radiation-free rate that dissolves the Hubble tension enlarges the Silk scale [24])—a real, computed effect, non-reabsorbable by the baryon density, but whose observable consequence for the high-ℓ power is entangled with the acoustic transfer at those multipoles (whether CR's high-ℓ peaks track ΛCDM's, verified only through the third peak) and so genuinely open, neither a demonstrated tension nor a wash; and the full-spectrum likelihood (the low multipoles being a mild, non-discriminating deficit rather than a sharp edge either way). We mark maturity throughout. *Established*: the cosmological derivation, the transmission dichotomy, the amplitude floor, the throat, the flat slice, the flat-projection transfer, the data confrontation, and the time-reversal driving equality that carries the peak *heights* (the driven acoustic amplitude flat ΛCDM's exactly, on the standard inherited plasma the abundance comparison confirms). *Argued*: the coherence mechanism. *Open, and the frontier the paper sets up to close*: the end-to-end branch-point-to-recombination transfer (the digit-level confirmation of the height pattern) and the $\sim$8% damping-tail signature within it, and the *derivation*—as against the measurement—of A_s, n_s, and the inherited datum the composition is run on. *The cosmology carries one fitted parameter and its status should be stated plainly*: $z_{\rm onset}$, the redshift at which the expanding-phase plasma begins, which is the lower limit of the sound-horizon integral and is needed only because the rate carries no radiation. It is fitted to the acoustic angle at the *directly* measured H_0 and lands at $z_{\rm onset} \approx 6.8 \times 10^3$, $T_{\rm onset} \approx 1.6\,\mathrm{eV}$, near $\rho_r/\rho_m \approx 2$. *It is not a knob for the H_0 tension*: the radiation-free rate carries H_0 out of both $r_{\rm s}$ and D_M, so θ_* is fixed by Ω_m alone and the same $z_{\rm onset}$ meets the scale at every H_0 across the range—it does not move between them. *Matter–radiation equality is then a consequence and a check, not an input*—$1 + z_{\rm eq} = 3399$, exactly half the onset, where the fitted Ω_m and the Hubble constant reproduce the microwave-background-inferred physical matter density (near $H_0 \simeq 68$); read at the directly measured H_0 instead, the same $z_{\rm onset}$ gives $\rho_r/\rho_m = 1.71$, so the datum is an order-unity band and not a determined number; and $z_{\rm onset}$ is a redshift on the expanding leg, not the branch point, which no finite cosmic-time layer reaches. The substrate's single Λ is shown to govern both the global expansion and, per structure, the local scale at which structure resists it—the Hubble–Eddington radius—one constant read at two ranges.

1 Introduction: the cosmological theory and its scalar perturbation sector

Cosmological Relativity (CR) recovers the flat-ΛCDM expansion history by a causal reassignment on a de Sitter background at the cosmogenesis branch point: the expansion rate is fixed by the cosmological constant alone, radiation carries no term in it, and the present matter density is a reading of the cosmic epoch rather than an independent quantity [7, 9]. The cosmic beginning is the branch point $r = 0$—a genuine curvature singularity of the SdS metric read over the r-chart, and *not* a singularity of the substrate, whose throat is the defining constant α, and the construction makes the constant-proper-time slices exactly flat while the cosmological layers are a closed S^3. This paper develops that cosmological theory in the form its observational tests require (§2) and completes it with the scalar perturbation sector the framework paper's outlook named open—"perturbations within the Schwarzschild–de Sitter representation"—the primordial density perturbations and their imprint on the microwave background (§§4–8), closing on the open frontier the construction sets up (§10). Across these sections one comparison recurs and is drawn together at the close (§9): the causal contact, flatness, coherence, isotropy, and near-scale-invariance that the standard picture obtains from an added, tunable inflationary sector are here read off structure already present and fixed by Λ alone—an economy-of-assumption difference the paper argues is decidable now, on the theory-choice axis where it falls to this cosmology—and the data axis has now moved with it, the Hubble tension resolved across the low-redshift distance ladder and the light-element abundances confirmed against the data (§2.4), so the reading is empirically favoured and not merely more economical, its remaining exposed edges the ∼8% damping-*scale* signature—a computed, non-reabsorbable CR-specific effect whose observable consequence awaits the high-ℓ acoustic transfer, genuinely open—and the full-spectrum likelihood, the low multipoles a mild, non-discriminating deficit rather than a sharp edge either way. The companion dynamics paper develops the perturbation sector's *tensor* half—the propagating graviton as the leaf's shear, a massless de Sitter mode on the cosmic foliation [8]; the scalar half is here.

The result is organized by one thesis, which we state at the outset because it controls everything below.

> *The CR primordial scalar spectrum factorizes: the de Sitter substrate geometry determines its structure, the progenitor collapse supplies its content, and the branch point's null geometry assigns each its job.*

Substrate-determined, and developed here: the coherence of the acoustic phases (§4); the classical, non-vacuum character of the fluctuations (§5); the isotropizing throat (§6); a parameter-free low-multipole discreteness floor (§7); and the faithful transmission of the spectral shape (§8). Progenitor-supplied, and inherited rather than derived: the amplitude A_s and tilt n_s. This is the same division of labour by which flat ΛCDM carries the baryon-to-photon ratio η as a measured datum from a baryogenesis it does not model; this cosmology carries one such datum—the branch-point radiation amplitude $\rho_r/\rho_m \approx 2$ that calibrates the acoustic scale, fixed by the hot handover of the progenitor collapse—and the transmission result of §8 routes A_s and n_s through the *same* handover. There is one boundary-condition supplier, not two.

A word on what is settled before the spectrum. The acoustic *scale* (the peak spacing) and the apparent Hubble tension are resolved in the cosmological development above (§2.3): the tension dissolves as a consequence of the radiation-free rate, and the scale is met at the directly measured H_0 by the single measured parameter $\rho_r/\rho_m \approx 2$, with the sound speed the ordinary baryon-loaded value of pressureless branch-point matter. These are settled there and are not reopened in the perturbation sector; in particular, the standard radiation-governed sound horizon laid on the radiation-free rate is a calculation belonging to neither framework, and any "tension" it produces is an artifact, not a CR result. What is genuinely downstream is the *spectrum*—the peak heights, carried below by the time-reversal driving equality (§4), and the full oscillating-medium transfer, still open—which is where this paper's substance lives.

The register is the programme's [12, 7]: the field equations are Einstein's, unchanged; the existing object is the evolving three-dimensional layer; "the substrate determines" and "the branch point transmits" are statements about a real geometry and its real boundary [15, 14], not about a modification of the dynamics. Throughout, we mark maturity in the prose: *established* results are verified by direct computation (the receipts are collected in `computations/perturbation_verify/`); *argued* results carry the reasoning and a leading-order computation but are flagged as not yet proven through the full transfer; *open* elements are named, not finessed.

2 The cosmological theory the perturbations complete

This paper develops the cosmology of Cosmological Relativity from first principles and carries it to its empirical confrontation, completing it with the scalar perturbation sector. It takes from the companion framework paper [7] the *general* primitives—the layered ontology, the projection principle, and the representational freedom by which one fixed manifold admits distinct Lorentzian metrics—together with the Null–Boundary Correspondence theorem, the structural result (grounded in the metric-singularity causal structure of gravitational collapse [2]) that a collapse event-of-events and a de Sitter cosmic-entry slice are one ontological layer under two causal assignments. The cosmology proper is built here: the causal reassignment that selects the Nariai member, the proper-frame expansion law, the branch point at which that Correspondence fixes the beginning at the collapse horizon, and the resolution of the apparent Hubble and acoustic tensions; the perturbation sector (§4–§8) is its completion.

2.1 The causal reassignment and the Nariai member

The vacuum solution of $R_{\mu\nu} = \Lambda g_{\mu\nu}$ with a localized source is Schwarzschild–de Sitter (SdS),

$$ds^2 = -\frac{r}{\frac{\Lambda}{3}r^3 + \frac{2GM}{c^2} - r}\, dr^2 + \frac{\frac{\Lambda}{3}r^3 + \frac{2GM}{c^2} - r}{r}\, dt^2 + r^2 d\Omega^2, \tag{1}$$

whose areal radius r is timelike in the cosmological regime $\Lambda G^2M^2/c^4 \geq 1/9$. Building the cosmology, we reassign causal roles on the de Sitter substrate—a move licensed by the representational freedom of the framework, distinct Lorentzian metrics on one fixed manifold sharing its foliation [7]: the de Sitter null rulings whose causal sense matches a collapse horizon's generators are promoted to the timelike fundamental congruence, the at-rest comoving worldlines become the null photon congruence, and the foliation by expanding 3-spheres is preserved. This selects the *Nariai* member $\Lambda G^2M^2/c^4 = 1/9$, $M = c^2/(3\sqrt{\Lambda}G)$—the unique non-pivoting tilt, fixed by Λ alone—at which the two positive horizons merge at $r_N = \alpha/\sqrt{3}$, $\alpha = \sqrt{3/\Lambda}$ the de Sitter 3-sphere size (a scale distinct from the merged-horizon radius $\alpha/\sqrt{3}$ and from the areal-radius amplitude below; we keep the three apart throughout). The selection is geometric—the fixed point of the slicing curve's root-exchange involution, equivalently the limiting tangent orientation a collapse selects—not a fitted mass [3]. That the selection is moreover *unique* is a structural fact about the description groupoid rather than a fitting: the generic vantages fall into two-cycles of the involution, the Nariai member is its one fixed point, and the reassignment promoting a null direction to the fundamental congruence is forbidden at every generic vantage, so it selects Nariai and no other [4].

2.2 The proper-frame cosmology, derived

In the proper frame of the fundamental observers the SdS metric is a homogeneous, non-synchronous cosmology. The derivation [13] proceeds from (1) in five steps. (i) A timelike radial geodesic has a conserved energy E (the metric being independent of t); eliminating $dt/d\tau$ in the normalization $L = -1$ gives the radial equation $(dr/d\tau)^2 = E^2 - V_{\text{eff}}(r)$, with $V_{\text{eff}} = (r - 2GM/c^2 - \frac{\Lambda}{3}r^3)/r$. (ii) The fundamental congruence is the worldlines whose motion in r is produced purely by the field—those at rest exactly where $V_{\text{eff}} \equiv 1$ and the metric reduces to Minkowski—which is $E = 1$; this fixes the rest frame by the field rather than by stipulation. (iii) With $E = 1$, integrating the radial equation gives, along one worldline, $r(\tau) \propto \sinh^{2/3}(\frac{3}{2}\sqrt{\Lambda c^2/3}\,\tau)$. (iv) Labelling the congruence by a comoving coordinate χ through the integration constant $r(0)$, the areal radius across the congruence depends on τ, χ only through their sum, $r(\tau, \chi) = r(\tau + \chi)$, with $\partial_\chi r = \partial_\tau r$. (v) Fixing the transformation $t(\tau, \chi)$ by requiring the comoving coordinate orthogonal to proper time, $g_{\chi\tau} = 0$, forces the integration function to be χ itself, and the remaining components follow ($g_{\tau\tau} = -1$, $g_{\chi\chi} = (\partial_\chi r)^2$). The result is the proper-frame line element

$$ds^2 = -d\tau^2 + (\partial_\chi r)^2\, d\chi^2 + r^2 d\Omega^2, \tag{2}$$

with cosmic time $\tilde{\tau} \equiv \tau + \chi$ (by Weyl's principle the congruence issues from a common origin and the constant-cosmic-time slices are those of constant $\tilde{\tau}$), areal radius

$$r(\tilde{\tau}) = \left(\frac{6GM}{\Lambda c^2}\right)^{1/3} \sinh^{2/3}\left(\tfrac{3}{2}\sqrt{\tfrac{\Lambda c^2}{3}}\,\tilde{\tau}\right), \tag{3}$$

and redshift $1 + z = r(\tilde{\tau}_0)/r(\tilde{\tau}_e)$. The constant-$\tilde{\tau}$ slices sit at 45° to the fundamental rest frame: the cosmology is homogeneous and observationally isotropic, yet *non-synchronous*—the constant-τ slices are exactly Euclidean (§7) but are not the cosmological ones. Of the FLRW kinematic assumptions—(i) a geodesic congruence, (ii) hypersurface orthogonality, (iii) homogeneity, (iv) isotropy—this construction keeps (i) and drops (ii); Lemaître's reading [27] of the Euclidean slices as "space ejected from $r = 0$" is causally incoherent and is not the physics. Equation (3) is moreover a single analytic function of $\tilde{\tau}$ (Fig. 1): the expanding cosmology is its real branch $\tilde{\tau} > 0$, and the cosmogenesis *branch point* at $\tilde{\tau} = 0$, where $r \to 0$, is the branch point of the scale factor rather than a real $r = 0$ boundary—and it is *not* a seam: the seams are the two unit-speed loci of the lap, $r = -2\alpha/\sqrt{3}$ and $r = +\alpha/\sqrt{3}$ [7], and $r = 0$ lies strictly between them, at the close of the lift—the analytic counterpart of the beginning at the finite-curvature branch point [2], and the cosmological reading of the slicing curve's closed lap—the *cosmogenetic bead* the framework establishes as one closed object [3, 7], of which this scale factor's real branch $\tilde{\tau} > 0$ is the expansion leg and its analytic continuation the prior universe's collapse—on which $r = 0$ is a branch point and not a barrier, the substrate smooth across the locus the chart so labels [3].

2.3 Exact recovery of flat ΛCDM, and the amplitude fixed by Λ

Equation (3) is exactly the flat-ΛCDM scale factor. At Nariai the mass is no longer free, and the amplitude becomes a pure Λ-length,

$$\left(\frac{6GM}{\Lambda c^2}\right)^{1/3} = \frac{2^{1/3}}{\sqrt{\Lambda}}, \tag{4}$$

so both factors of the expansion law are set by Λ^{R}: the rate $\frac{1}{2}\sqrt{3\Lambda}\,c$ is the standard late-time rate and the amplitude is the third Λ-scale (distinct from α and $\alpha/\sqrt{3}$). The Friedmann equation the $\sinh^{2/3}$ law satisfies,

$$H^2 = \frac{\Lambda c^2}{3}\coth^2\left(\tfrac{1}{2}\sqrt{3\Lambda}\,c\,\tilde{\tau}\right) = \tfrac{1}{3}\left(8\pi G\rho + \Lambda c^2\right), \tag{5}$$

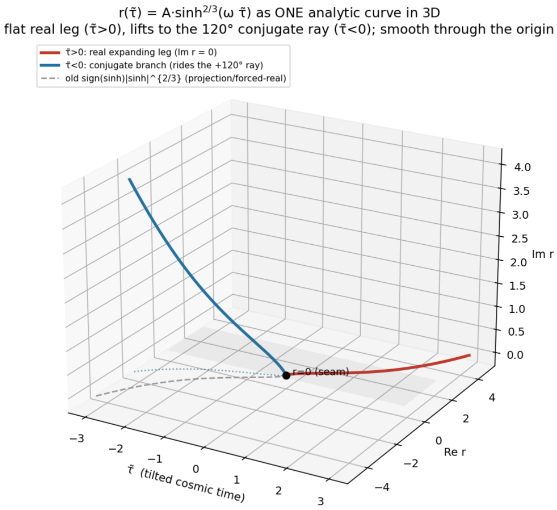

Figure 1: The scale factor (3) as one analytic curve $r(\tilde{\tau})$ in $(\tilde{\tau}, \operatorname{Re} r, \operatorname{Im} r)$. The expanding cosmology is the real leg $\tilde{\tau} > 0$, lying in the real plane; the branch point $\tilde{\tau} = 0$, $r \to 0$—*not* a seam—is where the curve leaves the real plane onto a conjugate branch at constant phase $2\pi/3$ for $\tilde{\tau} < 0$. The beginning is this analytic branch point, not a real $r = 0$ boundary. Forcing r real (dashed) projects the conjugate branch onto the negative axis and manufactures a cusp the smooth curve does not carry.

makes the density ratio a clock$^{\mathsf{R}}$,

$$\frac{\Omega_m}{\Omega_\Lambda} = \operatorname{csch}^2\left(\tfrac{1}{2}\sqrt{3\Lambda}\, c\, \tilde{\tau}\right), \tag{6}$$

so $\Omega_{m,0}$ records the cosmic epoch $\tilde{\tau}_0$ at which we observe rather than an independent density, and the coincidence problem becomes the observation that we exist at a time of order the geometry's single timescale. The densities are bookkeeping for one Λ-set geometry; *radiation, like matter, is inherited content read off that clock, not a term that sources the rate*, and there is no radiation-dominated era [9]. The objection that a perturbing radiation fluid "must gravitate and so alter the rate" is a category error against a construction that reads (5) leftward: the rate is set, and the fluid's energy is among the contents the set rate carries. The acoustic *scale* and the apparent Hubble tension resolve as consequences of this rate; this is developed next (§2.4).

2.4 The acoustic scale and the dissolved tensions

The two frameworks reproduce the late-time expansion identically and part in the early universe, where radiation carries no term in the rate (5). The apparent tensions of the standard model are consequences of that one structural fact.

The three levels, and which rate each quantity uses. The reading of every rate-bearing quantity below is fixed by locating it on one of three levels, which the geometry keeps apart and whose conflation is the standing error the layered framework names [7]. **(L1) The foliation stacking rate**: the Λ-set $\sinh^{2/3}$ law (5) read *leftward*—the cut is primary and ρ is the name of its bend, not its cause, the rate-side of the slicing operator's reading whose content-side is *matter is the bend of the cut* [5]—the layer's own geometric expansion, radiation-free, and *empirically forced* rather than modelled: the microwave-redshift isotropy floor forces uniform expansion to some three orders of magnitude, so a matter-tracking (radiation-sourced) rate is already excluded by data [9]. This is what the observable cosmology rides at *every* epoch, the branch point included: it is the rate the foliation stacks at, not a regime that begins somewhere. **(L2) The leaf-level local dynamics**: the expansion scalar of a self-gravitating congruence, the ordinary Friedmann readout $\mathcal{H}^2 = \frac{8\pi G}{3}\rho_{\text{tot}} + \frac{\Lambda}{3}$ with radiation gravitating normally—the valid phenomenological reading of general relativity on the progenitor's collapse excursion, where nucleosynthesis runs [16]; it is the same single $E{=}1$ geodesic read *inward* as dust collapse that L1 reads outward as cosmology [5, 2]. **(L3) The $E = 1$ projection**: the reassigned shadow the observer reads, through which an L1 or L2 quantity appears as distance and redshift—the Projection Principle of the framework paper, its geometric origin the *second ruling* of the slicing (the synchronous space is the second ruling) [7, 5]; reading this appearance as the existent is precisely the naïve-realist collapse the epistemology forbids [10]. The decision rule is structural, not a choice: a self-gravitating local excursion runs on L2, diffuse content riding the global foliation on L1, and every observable is an L1 or L2 quantity seen through L3; and *there is no L1/L2 boundary in time.* The distinction is kinematic and holds at every epoch: the vacuum kernel of the slicing operator is the SdS family exactly—Λ and the cut's offset $2M$, and no more [5]—so the stacking rate is $c^2H^2_{\text{stack}} = (c^2/\alpha^2)(1 + 2(1 + z)^3/x_0^3)$, while radiation, requiring $m'(r) \neq 0$, is a *bend* of the cut and therefore content. The decomposition is exact, $H^2_{\text{leaf}} = H^2_{\text{stack}} + H_0^2\Omega_r(1 + z)^4$: the two rates differ by the radiation term alone. A comoving separation read across leaves takes the stacking rate; a process running in the content—r_{s}, r_D, recombination, the perturbations—takes the leaf's. The assignment looked temporal only because $\Omega_r(1 + z)^4$ is negligible below $z \sim 10$ [7, 2]. Each mis-assignment is quantitatively fatal—the radiation-free L1 rate carried down into the nuclear window is $\sim$300$\times$ too slow, and the radiation-sourced L2 rate carried past the branch point radiation-pins r_{s} and re-manufactures the very tension this section dissolves [16]. Crucially L1 is the layer's own *ontological* expansion, not the L3 shadow: computing a process against the radiation-free rate is reading the geometry, not the appearance. Every rate-bearing quantity below carries its level, so the reading is settled by the geometry rather than defaulted to the synchronous machinery.

The Hubble tension. There is no second H_0 to reconcile. The late-time expansion is reproduced exactly, so the directly measured rate is the reading of the geometry, while the microwave background's lower inferred H_0 rests on a radiation-governed sound horizon r_{s} this construction does not share. The $\sim 5\sigma$ discrepancy is not two independent fits to be reconciled but the artefact of calibrating the acoustic scale with an early-universe sound horizon foreign to a geometrically fixed expansion.

The acoustic scale, to a one-parameter accommodation. Recombination is observable cosmology from the branch point onward, so its scales are L1: both the sound horizon r_{s} and the comoving distance D_M ride the radiation-free foliation rate, and the observed angle $\theta_* = r_{\text{s}}/D_M$ is their L3 projection. The robust ingredients are in hand: D_M is the standard observable, and $H(z)$ is the fixed $\sinh^{2/3}$ law (5) with radiation absent from the rate. The sound horizon $r_{\text{s}} = \int_{z_{\text{rec}}}^{z_{\text{onset}}}(c_s/H)\,dz$—an L1 quantity whose baryon-loaded c_s is ordinary content microphysics but whose expansion H is the L1 rate—then depends on a single early-universe parameter, the redshift z_{onset} at which the expanding-phase plasma begins; the standard radiation-governed r_{s} is recovered only in the limit $z_{\text{onset}} \to \infty$ that a beginning at finite curvature forbids. *That clause carries more weight than its length suggests, and the sense of "finite curvature" in it should be the substrate's.* The branch point is a locus at which the areal chart's invariants diverge—$r = 0$ is a genuine infinite-curvature locus of that chart—while the curvature of the manifold the cut is taken on is set by α everywhere, so the divergence is the areal coordinate degenerating and not a scale of the geometry [16, 15]. *Read the other way the accommodation would collapse*: a beginning at genuinely unbounded curvature places no finite floor under z_{onset}, the $z_{\text{onset}} \to \infty$ limit is reinstated, and the single datum ceases to be a datum at all. *The one-parameter accommodation therefore rests on the substrate reading of the branch point,* and a reader who took the areal divergence for the physics would find the parameter unconstrained. Taking c_s the ordinary baryon-loaded value—the matter at onset is pressureless, and the geometric throat ratio $\sqrt{3}$ does not set it—the measured acoustic scale is reproduced at the *directly* measured H_0 by a single z_{onset}, near $\rho_r/\rho_m \approx 2$ (about twice matter–radiation equality) and *independent of* H_0—the independence belonging to z_{onset} itself, and the ratio being that same quantity re-expressed, as §2.4 returns to below: the radiation-free rate carries the common factor H_0 out of both r_s and D_M, so $\theta_* = r_\mathrm{s}/D_M$ is fixed by Ω_m alone and the single z_onset that meets the scale is the same at every H_0 across the range—the datum cannot be a parameter tuned to reconcile two values of H_0, since it does not move between them. This is exactly where the standard rate parts: carrying radiation, it ties θ_* to H_0—the coupling that fixes the lower microwave-background value and stands in tension with the direct one—whereas the radiation-free rate decouples the scale from H_0, meeting it at the directly measured value with no second H_0 to reconcile. The beginning is well posed: the finite-curvature branch point (§2.2) places the onset down the cooling branch, $T_\mathrm{onset} \sim 1.6\,\mathrm{eV}$ at $z_\mathrm{onset} \sim 6.8 \times 10^3$, so a pre-recombination sound-travel interval exists and the integral $\int_{z_\mathrm{rec}}^{z_\mathrm{onset}}$ is well posed.

The inherited datum. The empirical forcing of the cosmic foliation bears on the rate, not the matter content—the matter density being the bend of the spatial cut [9, 5, 6]—so the radiation amplitude is measured content, not a quantity the geometry sets; the cold seam (Gibbons–Hawking temperature [30] $\sim 10^{-30}\,\mathrm{K}$, some thirty orders below the onset) confirms there is no geometric source to set it instead. *One thing about that ratio should be said plainly before it is leaned on, because it is a unit and not a second datum.* The quantities this cosmology actually uses are Ω_m, which the rate carries and the baryon-acoustic data fix; ω_b and ω_γ, which the sound speed carries and which are measured directly and framework-independently; and z_onset, the one fitted number. The *physical* matter density $\omega_m = \Omega_m h^2$ enters none of them. But ρ_r/ρ_m at onset is $\omega_r(1 + z_\mathrm{onset})/\omega_m$ and equality is $1 + z_\mathrm{eq} = \omega_m/\omega_r$, so both divide by that one quantity—and $\rho_r/\rho_m \approx 2$ is therefore not a second inherited number standing beside z_onset but z_onset itself, read in units of a density the construction does not determine. *The consequence is a stated H_0-dependence in the ratio and in nothing else*: with Ω_m held at its fitted value the ratio is 2.01 at $H_0 = 67.4$ and 1.71 at $H_0 = 73$, the two readings coinciding at $H_0 \simeq 68$ where $\Omega_m h^2$ reproduces the microwave-background-inferred $\omega_m = 0.143$. *What carries weight is what survives that*: z_onset does not move with H_0 at all, so it cannot be absorbing the tension; and the inherited amplitude, read at onset, is order unity—which is a statement about where the onset sits and *not* a contrast with η, as the withdrawal two paragraphs below sets out. *The factor of two is exact at $h \simeq 0.68$ and should be quoted as an order-unity band, not as a determined number* (receipt: `the_ratio_is_the_onset_in_imported_units.py`$^\mathsf{R}$). Thus $\rho_r/\rho_m \approx 2$ is the structural analogue of the baryon-to-photon ratio η: as flat ΛCDM carries η from a baryogenesis outside the model proper—measured, not derived—this cosmology carries the single radiation initial condition, and with no radiation-dominated epoch the light elements as well, from the hot handover of the progenitor collapse, this universe having formed at the event horizon of a black hole in a previous one, equally outside the construction and equally measured. That the datum is measured rather than derived is no defect: a framework is credited for *requiring* the structure it explains—the radiation-free rate, and the dissolution that follows—not faulted for measuring the boundary data it does not. *One naturalness argument that has been offered for this datum should be withdrawn rather than carried, because it compares a quantity with itself at two epochs nine orders apart.* Complete collapse does liberate binding energy of order the rest mass ($GM/Rc^2 = \frac{1}{2}$ at the Schwarzschild radius), and it is tempting to read that as making a handover carrying radiation of order the matter density the natural scale—order unity, rather than the small value η takes. But $\rho_r = \rho_m$ occurs at $T \simeq 0.8\,\mathrm{eV}$, whereas the handover sits four orders *above* the deuterium bottleneck at $0.07\,\mathrm{MeV}$, since that is what the nucleosynthesis computed on it requires [16]: at the bottleneck the ratio is already $\sim 9 \times 10^4$ and at the handover proper $\sim 9 \times 10^8$. A handover carrying radiation merely of order the matter density would be a handover with $\eta \simeq 0.5$, which leaves neither light elements nor acoustic peaks. *So the naturalness question runs the other way*: what the progenitor must deliver is a plasma with of order a billion photons per baryon—the baryogenesis-analogue derivation this paper names as open—and the binding-energy scale offers no head start toward it. *What replaces the argument is a simplification of the bill.* Given η and the measured matter-to-baryon ratio, both of which this cosmology inherits for the composition regardless, the ratio at onset is $[1+\nu](\pi^4/30\zeta_3)\,T_{\rm onset}/[\eta\,(\omega_m/\omega_b)\,m_N]$, which returns 1.99 at $T_{\rm onset} = 1.6\,\mathrm{eV}$—the quoted datum, to one per cent, from standard thermodynamics and no feature of this construction. *It is therefore not a second inherited number standing beside η: it is the fitted onset restated*, so what the handover supplies is one composition datum and what the cosmology fits is one parameter (receipt: `the_two_data_are_one.py`[R]). Like η, the inherited radiation amplitude may remain a measured boundary condition indefinitely without cost to the *dissolution*—which settles what the dissolution depends on, not whether the derivation is worth having: deriving it from the progenitor's determined structure remains open work.

The synthesized composition. The same collapse fixes the light-element composition, and the geometry sharpens how. The infalling matter compresses adiabatically—the plasma is optically thick, so the photon bath compresses with it—and heats far above the deuterium bottleneck ($T_{\rm D} \simeq 0.07\,\mathrm{MeV}$) for every progenitor: the compression does not stop at horizon crossing but continues to the branch point, and the peak is bounded below by the infall energy at the horizon, of order the rest mass per nucleon, independently of the progenitor's mass (the cosmogenesis paper [16]); at turnaround it re-expands and cools back through the window. Because the effective rate through the window is the standard Friedmann rate, that cooling leg is a standard big-bang nucleosynthesis, run on the collapse's turnaround from a dissociated hot start—the deep re-expansion doing the synthesis below the $T_{\rm onset} \sim 1.6\,\mathrm{eV}$ onset at which the observable expansion begins. On a first sample this is favourable, and on one element advantageous. The helium-4 fraction $Y_p \approx 0.245$ sits at the observed value; the near-zero metallicity of the oldest systems follows from the handover photodissociating the progenitor's processed composition back to light elements; and because the run is a standard nucleosynthesis, the lithium-7 abundance carries its standard threefold over-prediction—the standing lithium problem shared with flat ΛCDM, neither dissolved nor worsened. Deuterium sits at the observed $D/H \approx 2.5\times10^{-5}$: the freeze-out temperature the data demands is the standard bottleneck, and that is exactly the temperature at which the standard rate delivers a freeze-out—target and mechanism coincide un-tuned. The peak calculation that forces this synthesis reading over an inherited one, and the full multi-abundance account, are given in the cosmogenesis paper [16], where the network is now integrated explicitly on the cooling leg and returns deuterium, helium-4 and helium-3 at their observed values with lithium-7 at the shared over-prediction, jointly from the single inherited datum—so one progenitor handover does fit the pattern together; what stands as its open edge is the last-percent precision against the measured abundances, a data-confrontation the produced pattern now exposes rather than a computation still owed.

The load-bearing claim. The acoustic scale stands as a one-parameter accommodation on the radiation datum, not a parameter-free prediction; the standard c_s it borrows presumes the inherited medium to be an ordinary photon–baryon plasma, which the abundance comparison now confirms—helium-4 and deuterium at their observed values, lithium-7 the shared standard problem (the cosmogenesis paper). The load-bearing, falsifiable claim remains the prior one—that radiation carries no term in the expansion rate—of which the dissolved Hubble tension and the one-parameter acoustic calibration are consequences. The sound horizon at a geometrically fixed H_0 is where a purely geometric expansion history and the standard radiation-governed one are told apart in the data.

The expansion history, confronted with the late-time data. The dissolution is not confined to the single microwave-background angle: it holds across the low-redshift distance ladder, and for the same reason. Because the radiation-free rate carries the common factor H_0 out of every distance and out of the ruler alike—D_M, $D_H = c/H$ and the sound horizon $r_{\rm s}$ each scale as $1/H_0$—the baryon-acoustic-oscillation observables $D_M/r_{\rm s}$ and $D_H/r_{\rm s}$ are *independent of* H_0, fixed by Ω_m alone (the same cancellation that fixes θ_*; §above). The distinction from ΛCDM is exactly the radiation: there the ruler $r_{\rm s}$ is pinned in physical length by the radiation-era physical density $\omega_m = \Omega_m h^2$, so the same $D/r_{\rm s}$ data constrain H_0 and pull it to the low microwave-background value, in tension with the direct one. Here they do not: the acoustic data fix Ω_m and leave H_0 to the local distance ladder, with no second value to reconcile. Confronted with the SDSS DR12 galaxy consensus [19] the radiation-free rate at $\Omega_m \simeq 0.31$ fits at the directly measured H_0 ($\chi^2 \simeq 1.7$ on six points, at *any* H_0 including 73), where the radiation-governed rate at that same local H_0 misses badly ($\chi^2 \simeq 49$)—that residual is the Hubble tension itself, which the geometric rate structurally lacks. The same confrontation carried to the state-of-the-art DESI DR2 dataset [20]—thirteen distance measurements across seven tracers ($0.3 \lesssim z \lesssim 2.3$), with the full per-tracer D_M/r_s–D_H/r_s correlations—tightens rather than loosens the result: with a single free parameter Ω_m the radiation-free rate fits at $\chi^2/\mathrm{dof} \simeq 1.0$, at $\Omega_m = 0.3066$ (where the inherited datum sits at $\rho_r/\rho_m \approx 2.0$), and it fits there at *any* H_0 including the local 73—which is the whole of the claim. The radiation-pinned ruler cannot do that: it ties ΛCDM's fit to one H_0, best at 68.7, and breaks it at 73 ($\chi^2/\mathrm{dof} \simeq 15$)[R]. *Two corrections are owed here and are made at r2376+c54.155.* This paper described $\Omega_m \simeq 0.307$ as a CMB-calibrated input; it is not—Planck's own value is 0.3153, at which the fit degrades to $\chi^2/\mathrm{dof} = 1.35$, and 0.3066 is what these data themselves prefer. *And granted the same one free parameter, ΛCDM fits these data marginally better than this cosmology does* ($\chi^2/\mathrm{dof} = 0.92$ against 1.00). *The result is therefore not that the radiation-free rate fits better; it is that it fits without choosing an H_0, and the radiation-pinned one must choose.* The cosmic chronometers [23], which read $H(z)$ directly, are consistent with the local value. So the resolution is a property of the whole expansion history, not of one angle, and it rides the very rate whose branch point-located scoping keeps the nucleosynthesis window standard [16] (receipt: `hubble_expansion_confrontation_v2.py`[R]; the invariant is Ω_m, not ω_m—holding the physical density fixed instead imports the ΛCDM reading and manufactures a spurious tension).

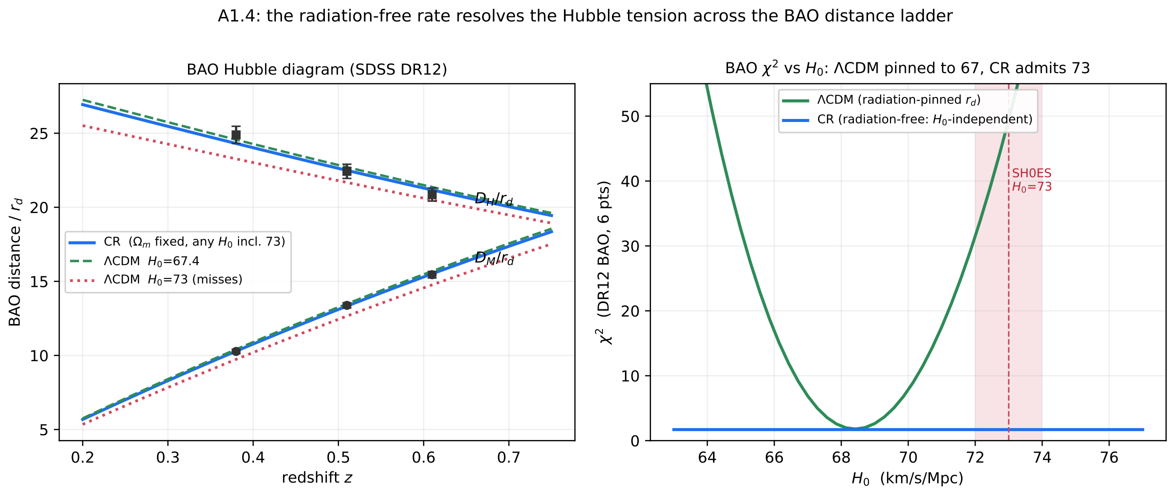

Figure 2: The Hubble tension resolved across the baryon-acoustic distance ladder. *Left:* the SDSS DR12 measurements of D_M/r_s and D_H/r_s (points) against the radiation-free rate at the directly measured $H_0 = 73$ (solid) and flat-ΛCDM at $H_0 = 67.4$ (dashed); both track the data, while ΛCDM forced to the local $H_0 = 73$ (dotted) misses. *Right:* the BAO χ^2 as a function of H_0. Radiation in the ΛCDM rate pins the physical ruler r_s, so its χ^2 is a steep parabola minimized near 67 and excluding the local value—this is the tension. The radiation-free rate carries H_0 out of the ratios D_M/r_s, D_H/r_s, so CR's χ^2 is *flat* in H_0: the BAO fixes Ω_m and admits the SH0ES $H_0 = 73$ [22] (shaded) at no cost. The tension is structural to a radiation-governed ruler and absent from a radiation-free one. The figure shows the SDSS DR12 consensus; the same result holds, more strongly, against the state-of-the-art DESI DR2 dataset (thirteen measurements, seven tracers), where the radiation-free rate fits at $\chi^2/\mathrm{dof} \simeq 1$ with the single CMB-calibrated $\Omega_m = 0.307$ at the local H_0 (§2.4).

2.5 The branch point is sub-horizon for the acoustic modes

The cosmological beginning is the branch point $r = 0$ at the close of the lift, where the conjugate branch becomes the matter branch [7, 3]. The curvature invariants of the r-chart diverge there because the areal coordinate degenerates, while the throat itself is the substrate's defining constant α and the bead continues through by analytic continuation [33]. The lap's seams lie elsewhere, at the two unit-speed loci $r = -2\alpha/\sqrt{3}$ and $r = +\alpha/\sqrt{3}$. We record the consequence that organizes the perturbation sector.

Proposition 1 (The acoustic modes are sub-horizon at the plasma's onset)**.** *On the rate* (5)*, the comoving Hubble wavenumber at the onset redshift* $z_{\text{onset}} \approx 6797$ *is* $k_{\text{hor}}(\text{onset}) = H_0 E(z_{\text{onset}})/[(1+z_{\text{onset}})c] \approx 0.010\ \text{Mpc}^{-1}$*, while the acoustic peaks lie at* $k \sim \pi/r_{\text{s}} \approx 0.022\ \text{Mpc}^{-1}$ *and above. The acoustic-scale modes are thus inside the horizon at the branch point by a factor* $\gtrsim 2$*.*

The distinction between the two epochs this proposition could be read as concerning should be made explicitly, since an earlier formulation ran them together. z_{onset} is a redshift on the expanding leg—the redshift at which the expanding-phase plasma begins, and the lower limit of the sound-horizon integral above. *It is not the branch point.* The branch point is the locus $r = 0$, which no finite cosmic-time layer reaches and which therefore carries no finite redshift at all. *The proposition concerns the onset and nothing else*: the acoustic modes are inside the horizon when the plasma they oscillate in comes into being, which is what the transfer calculation requires of them.

And the corresponding statement about the branch point is the opposite one, for a different reason. Approaching $r = 0$ on the contracting leg the comoving Hubble scale aH grows without bound, so the comoving horizon shrinks to zero and every mode *exits* it before the crossing—which is why what crosses is frozen and non-oscillatory (§3). *Sub-horizon at onset, super-horizon at the crossing*: there is no tension between the two, and reading either statement at the other's locus inverts it.

Argument. Direct evaluation of $E(z) = \sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}$ at z_{onset} and of π/r_{s} with the banked r_{s} from the acoustic-scale resolution (§2.3); clean residuals (receipts: `verify_numeric.py`, anchor 7^{R}). $\square$

The consequence is structural. The acoustic modes are not frozen super-horizon data awaiting re-entry, as in the inflationary initial-condition story; they are already inside the horizon when the cosmological side begins. Whatever fixes their amplitudes and phases is therefore the branch point handover itself, not a super-horizon freeze-out—which is why the substrate's role will be to *transmit and gate* rather than to *generate*, and why the initial conditions are inherited.

3 What crosses, and what does not

The transmission dichotomy above is an argument from the absence of an imprinted scale. A mechanism can now be given for it, and it also settles which content the branch point does *not* carry.

The imaginary-time segment preceding the branch point acts on a perturbation as a Euclidean kernel [32], damping a mode of frequency ω by $e^{-\omega|\Delta\eta|}$ with $|\Delta\eta| \simeq 1.8\times10^4$ Mpc. Taken naively this would annihilate everything: at $\ell = 220$ the factor is 10^{-71}. *But ω is the frequency of an* oscillating *mode, and at the branch point no mode is oscillating.* On the contracting leg the comoving Hubble scale $aH = |\mathrm{d}r/\mathrm{d}\tilde{\tau}| = \sqrt{|1-f|}$ grows without bound as $r \to 0$—it is 0.13 at the comoving turnaround, 1.96 at $|r| = 0.1\,\alpha$, and 19.6 at $10^{-3}\alpha$—so the comoving horizon $1/aH$ shrinks to zero and *every* mode exits it and freezes before the crossing. A frozen mode has no oscillation for the kernel to damp. *So the amplitude and the tilt cross unaltered, and the argument from the absence of a scale is recovered as a consequence of horizon exit rather than assumed.*

What does not cross is the oscillatory content itself, and that is the collapse leg's acoustic phase: a mode that was sub-horizon earlier carries an oscillation, and the kernel annihilates it—e^{-152} at the first acoustic peak. This is required independently by the observed peak positions, which exclude a transmitted leg phase by a factor of 21–39 in the required distance [7], and it is what leaves the comb to be set on the expansion side by the characteristic data of §4. *The two results are one statement*: the crossing transmits what is frozen and destroys what oscillates, so the progenitor supplies the spectrum's amplitude and shape while supplying none of its phase.

The exponent has two factors, and the argument above varies only one of them. It is $e^{-kc_s|\Delta\eta|}$: holding c_s at the plasma's value and varying k reads it as a *scale* criterion, frozen against oscillating, which is the reading just given. Holding k and varying c_s reads the same inequality as a *species* criterion, and that reading supplies something the first does not. A pressureless component has $c_s = 0$ identically—not small, not slowly varying—so its exponent vanishes at every wavenumber; on the segment its mode equation is $\psi'' = 0$, whose constant solution the kernel leaves untouched with no approximation at all. *The matter sector's inheritance is therefore exact rather than adiabatic*: it does not rest on horizon exit and it does not degrade at small scales, where the frozen-mode argument does. The two arguments agree wherever both apply, and only the pressureless one is available everywhere. *The species reading appeared to carry a constraint the scale reading does not, and that appearance is withdrawn here at r2376+c54.162.* On this geometry the exponent at the first acoustic peak is $\simeq 1.6\times10^2\, c_s/c$, so a component with a per-cent sound speed would arrive suppressed by $e^{-1.6}$ and one with ten per cent by e^{-16}, and this paper read that as a sharp condition on what the progenitor may hand over—perturbations inherited from a *cold* species and from no other (receipt: `the_filter_is_a_sound_speed_filter.py`[R]). *The reading is too strong, because the exponent has nothing to act on.* The kernel acts on *oscillatory* content—a frozen, zero-frequency mode is a fixed point of it—and on the progenitor's own interior nothing arrives oscillating: $|aH| = |a'/a|$ diverges as $1/x$ at the branch point while c_s saturates at $1/\sqrt{3}$, so $c_s k/|aH| \to c_s k x \to 0$ for every k, and every mode from $\ell \simeq 28$ to $\ell \simeq 2475$ exits the comoving sound horizon strictly before the crunch—the highest with a fifth of a per cent of the collapsing leg still to run [16]. *So the two readings do not divide the work between them: the scale criterion is satisfied by everything, and the species criterion selects nothing.*

One consequence of that freezing bears on the acoustic sector rather than on the crossing, and it is computed at r2376+c54.167 rather than left as an inference. If every mode is outside the comoving horizon at the branch point and the acoustic modes are inside it at the onset (Prop. 1), then between those two loci the modes *re-enter*—and in flat ΛCDM a horizon crossing is exactly where the k-independent acoustic driving phase shift is acquired. *The re-entry is real and its boundary is computable*: the mode whose crossing coincides with the onset is $k = 0.0111\,\mathrm{Mpc}^{-1}$, i.e. $\ell \simeq 144$, so every mode at and above the acoustic peaks crossed before the plasma began while $\ell \lesssim 144$ is still outside at the onset and crosses later[R]. *But nothing is imprinted there, and the reason is structural rather than quantitative.* On the radiation-free rate the crossing occurs under pressureless matter to better than a part in 10^4, and the potential equation for a pressureless component, $\Phi'' + 3\mathcal{H}(1+w)\Phi' + [2\mathcal{H}' + (1+3w)\mathcal{H}^2]\Phi + wk^2\Phi = 0$, *contains no k at all once $w = 0$*—the wavenumber enters only through the pressure term. Integrated on this background from $a = 10^{-7}$ to the onset the potential is constant to twelve digits and *identical across four decades in k*, the integrator taking the same number of steps for every mode because it is handed the same problem. *So the horizon crossing is not an event for the potential here*: before the onset the perturbation problem on the observable leg is scale-free, and a scale-free problem imprints nothing. In flat ΛCDM the same modes cross under *radiation*, where $w = 1/3$ restores the k^2 term and the potential decays on the sound-crossing time by construction, which is why that shift is the same for every mode. *This closes a candidate mechanism rather than supplying one*: the absence of a universal driving shift on this rate is not an omitted crossing but a scale-free interval, and the first peak's position is left open by it. The crossing is lossless for every species rather than for cold ones alone. *That is a loss of a claimed prediction and not a gain, and it is recorded as one*; what survives is the pressureless argument's positive content, that the inheritance is exact rather than adiabatic, now holding of every species and not of dust alone.

4 What the substrate determines: coherence from the null boundary

The acoustic peaks are sharp because the modes of a given wavenumber share a common phase at last scattering. Inflation produces this coherence dynamically, by freezing modes while super-horizon; Proposition 1 closes that route here. CR obtains it instead from the character of the branch point.

The branch point is a *null* surface—the cosmological horizon promoted to the initial layer [7]. The initial-value problem on a null surface is characteristic, not Cauchy: the data that determine the future are one free function per mode on the surface together with regularity along the generators, rather than a field and an independent momentum. A single datum per mode is a single phase per mode; there is no second, independently specifiable quantity to randomize the relative phase. We therefore argue that coherence is not imposed but is what regular characteristic data on a null surface *is*. This is borne out quantitatively, and the scope of the demonstration is stated exactly here at r2376+c54.164, because this paper previously claimed more of it than it does. Evaluating the tightly-coupled mode function at last scattering with a *common* seam phase gives a sharp regular comb, whereas drawing the phase independently per mode—the second, free datum a Cauchy surface would admit—washes it out entirely, leaving flat power (peak-to-trough contrast ~ 1 against the coherent comb's many orders of magnitude; receipt: `verify_coherence_comb.py`[R]). *That establishes the mechanism—one characteristic datum per mode is one phase per mode, and coherence follows—and it establishes nothing about where the peaks fall.* The receipt writes the mode function down as $\cos(kr_\mathrm{s})$ with r_s and D_C as literals; no mode is propagated in it, as its own scope note says, so the spacing $\Delta\ell \simeq \pi D_C/r_\mathrm{s} \approx 296$ is the arithmetic of the expression assumed rather than an output. *This paper stated it as the result of propagating the modes, and that was wrong*[R]. Where the modes *are* propagated the comb comes out at 0.72–0.79 of that spacing under every initial condition tried (§4.3), so the peak spacing is at present asserted in one place and short in the other. The null boundary admits only the first column: one characteristic datum per mode is one phase per mode. The peak *heights* are then carried by a structural argument rather than a bespoke transfer. The driven acoustic amplitude is the resonant Fourier magnitude of the potential's evolution, $|\tilde{\Psi}(\omega{=}1)|$—exactly invariant under time reversal, $|\widetilde{\Psi(x)}| = |\widetilde{\Psi(-x)}|$, a Fourier theorem confirmed numerically on the radiation-era transfer function and on a random control[R]. This cosmology's driving does not live on an expanding radiation era, which the radiation-free rate does not possess; it lives on the *contracting* side—the progenitor's radiation-comparable collapse phase, whose potential evolution is the Friedmann time-reverse of an expanding radiation era—so its driving magnitude equals flat ΛCDM's exactly, and the branch point's faithful transmission (§8) carries it across unaltered. The acoustic amplitude, and with it the peak-height pattern, therefore match flat ΛCDM's—by the time-reversal structure, not by tuning—conditional on the inherited medium being the ordinary photon–baryon plasma, the condition the abundance comparison confirms (§2.4; receipt: `cr_collapse_driving.py`[R], the driving validated as a linear oscillator against the known radiation-era boost). Because the acoustic modes are already sub-horizon at the branch point (Prop. 1), each mode's driving is complete on the collapse side; the observed peaks therefore *reduce* to flat ΛCDM's—the driving equality above together with the shared photon–baryon plasma and the coherent seam phase—and match a Boltzmann reference digit by digit ($P_1/P_2 \approx 2.2$, $P_2/P_1 \approx 0.45$, $P_3/P_1 \approx 0.44$; `camb_reference.py`[R]). What remains is confined to the *damping* tail ($\ell \gtrsim 1000$), and the three-level rule (§2.4) fixes it rather than leaving it to judgement. Recombination is observable cosmology from the branch point onward, so the photon diffusion length r_D—like r_s—is an L1 quantity: the random walk accumulates against the layer's own radiation-free expansion, its Thomson microphysics ordinary content physics but its expansion the L1 rate, *not* a radiation-included one. This is the same branch point-located scoping the cosmogenesis paper derives for the nucleosynthesis window [16], read here in the opposite direction: there the self-gravitating excursion sets the L2 rate radiation is included in, here the diffuse plasma rides the L1 foliation radiation is excluded from. This is forced: it is the same L1 rate that dissolves the Hubble tension, and one may not take the rate radiation-free for the peak spacing and radiation-included for the diffusion. Because the radiation-free rate near recombination is $\sim$13% below the radiation-included one there ($\rho_r/\rho_m \approx 0.3$), the diffusion length runs $\sim$9% longer than ΛCDM's while r_s is matched to the observed acoustic scale; the projection-independent ratio is therefore $\theta_D/\theta_* \approx 1.08\,(\theta_D/\theta_*)_{\Lambda\mathrm{CDM}}$—a $\sim$8% larger damping angular scale (receipt: `damping_ratio_clean.py`$^{\mathsf{R}}$, validated against a Boltzmann reference on the radiation-included rate; $r_s = 144.0$ vs CAMB 144.4). This is the one genuinely CR-specific piece of the height pattern: a real signature *locked to* the Hubble resolution by their shared L1 rate—not an artefact of laying the standard radiation-governed r_s on the radiation-free rate, the calculation belonging to neither framework this paper warns against (§1). One thing about this signature is settled and one is not, and it is essential to keep them apart. Settled: the $\sim$9% in r_D is genuinely CR-specific and does not dissolve under the density that would ordinarily rescale it—$r_D \propto \omega_b^{-0.31}$, so a percent-level ω_b (fixed by primordial deuterium and the peak-height ratio) shifts it by well under a percent (receipt: `damping_reabsorption.py`$^{\mathsf{R}}$). *Not* settled, and not to be over-read: what that $\sim$9% in the diffusion *scale* does to the observed high-ℓ power. The observable is $\theta_D/\theta_* = r_D/r_s$, and translating it into a spectrum requires the full acoustic transfer—above all, whether CR's high-ℓ peaks, driven on the radiation-free rate, track flat ΛCDM's, which is verified only through the third peak ($\ell \lesssim 800$). A leading-order pass that simply *assumes* the high-ℓ peaks equal ΛCDM's and applies the extra damping produces a large high-ℓ deficit (receipt: `full_transfer_verdict.py`$^{\mathsf{R}}$)—but that assumption may not be imported for free: the same radiation-free rate that enlarges r_D also governs the high-ℓ driving envelope, so CR's high-ℓ spectrum rests on short-wavelength collapse-phase driving rather than on the boost an expanding radiation era supplies. *That driving is computed below* (§4.1), and the calculation removes the licence the shortcut lacked: the envelope is derived on the collapse leg rather than imported. The two structural facts—peaks-match (verified low-ℓ) and radiation-free diffusion—therefore constrain, but do not by themselves determine, the high-ℓ shape; the net effect is computed in §4.2; whether it is a tension, a wash, or a distinctive but consistent feature turns on a parameter refit and on the early integrated Sachs–Wolfe term, which is what remains open. What is honestly claimed here is the *effect*, not a verdict on it: a real, computed, non-reabsorbable $\sim$9% CR-specific diffusion-scale signature, its observable consequence now derived rather than entangled (§4.2), and the verdict on it left to a likelihood confrontation this paper does not run. The structural point that stands is the *mechanism*: where inflation freezes the phase dynamically and drives the amplitudes on an expanding radiation era, CR fixes the phase geometrically on the single null layer and drives the amplitudes on the time-mirrored collapse.

4.1 The collapse-phase driving envelope

The driving named above can be carried out in closed form, and doing so removes the entanglement the previous subsection had to leave standing. On the radiation-dominated collapse leg the potential obeys $\Psi'' + (4/\eta)\Psi' + (k^2/3)\Psi = 0$, whose regular solution is elementary, $\Psi = 3\Psi_i(\sin x - x\cos x)/x^3$ with $x = k\eta/\sqrt{3}$, and which is *even* in x^{R}—so the contracting leg carries the same closed form pointwise, a sharper statement than the Fourier-magnitude equality of §4 and consistent with it. Writing $\hat{\Theta} = \Theta_0 + \Psi$ removes the source exactly, $\hat{\Theta}'' + (k^2/3)\hat{\Theta} = 0$: the effective temperature oscillates freely, and the k-dependence rides entirely in the amplitude at horizon entry and in the decay of Ψ.

Horizon entry is fixed by the construction rather than fitted. The comoving Hubble radius is built from the slicing paper's own turnaround function, $(rH)^2 = (1-f) + A/r^2$ [3], and has a single turning point on the whole excursion; the branch point sits on its rising branch, which is *why* the acoustic modes are sub-horizon there (Prop. 1) rather than a numerical accident of $z_{\text{onset}}{}^{\mathsf{R}}$. On the leg the radiation term dominates because it is constant while the others vanish, so $\eta = r/\sqrt{A}$ to about two per cent at the entry radius of the lowest mode in question and better the deeper the mode enters—sub-percent from $k \gtrsim 30\,k_s{}^{\mathsf{R}}$, and entry then occurs at $x \to 1/\sqrt{3}$ *in the deep sub-horizon limit*—the radiation era's own scale invariance, recovered here from the substrate geometry. *The envelope is therefore flat to the accuracy of that limit*: a mode at $k \simeq 10\,k_s$ enters within 2% of $1/\sqrt{3}$ and one at $300\,k_s$ within 0.1%, all of them reaching the branch point with the same driving amplitude $\Psi(1/\sqrt{3})/2$, and the residual potential is below a percent *on average* while oscillating up to some seven per cent of it where $\cos x_{\text{seam}}$ is near an extremum$^{\mathsf{R}}$. *Both qualifications are entered at r2376+c54.155: this paper previously read the limit as an identity and the average as a bound.* Its turnover sits at equality, and the inherited datum places equality at $1 + z_{\text{eq}} = (1 + z_{\text{onset}})/2 = 3399$, essentially flat ΛCDM's—so the envelope turns over at the same wavenumber, for a structural reason and not by assumption$^{\mathsf{R}}$.

Two corrections that might have been expected do not enter. The baryon loading is proportional to the scale factor and the driving happens where the scale factor is smallest, so $R \lesssim 0.1$ at the onset and orders below that at entry; the odd/even asymmetry is imprinted afterwards, on the expansion side at $R \approx 0.6$, which is ordinary content physics on the observable leg$^{\mathsf{R}}$. And the free-streaming neutrinos the cooling-leg nucleosynthesis requires do carry anisotropic stress at these temperatures—the branch point is far below their decoupling—but their deep sub-horizon effect is a constant phase shift and a constant suppression, and a k-independent correction leaves a flat envelope flat$^{\mathsf{R}}$. What it does not leave is the closed form for Ψ itself, the equation becoming integro-differential; the closure is a property of the skeleton, and the flatness is what survives.

4.2 What the envelope buys, and what remains

With the envelope derived rather than assumed, the diffusion signature can be stated without the entanglement. The diffusion length reduces to a single integral over the scale factor in which every microphysical constant sits outside, $1/k_D^2 \propto \int \mathrm{d}a\, g(R)/(Hx_e)$, so in the ratio the Thomson physics and the ionisation history cancel identically and the whole difference is carried by $H(a)$—which is the division of labour this section has argued for throughout. Both integrals close in elementary form, and on the inherited datum the radiation-free rate gives a diffusion length 10.8% longer$^{\mathsf{R}}$. The sound horizon is the same kind of integral, and it must be taken *from the branch point*: there is no observable expansion below it. Doing so returns $r_s = 146.4$ Mpc against 145.4 on the radiation-included rate—within 0.7% of each other at the onset redshift the inherited datum fixes, the exact crossing sitting a little lower at $z \simeq 6.7 \times 10^{3\mathsf{R}}$—the acoustic-scale calibration, met from the other end and to that accuracy rather than exactly. The observable, in which the common distance cancels, is then θ_D/θ_* larger by 9.4%, and $\ell_* = \pi D_M/r_s = 302.2$ against the measured $\sim$301$^{\mathsf{R}}$. *These five figures were quoted here at the retired* $z_{\text{onset}} = 6850$ *until r2376+c54.155; they are now the values the cited receipts produce.*

The high-ℓ consequence follows with no free parameter: $C_\ell^{\text{CR}}/C_\ell^{\Lambda\text{CDM}} = \exp[-(\ell/\ell_D)^2(r^2 - 1)]$ with $r = 1.093$, so the ratio is 0.82 at ℓ_D and 0.65 at $1.5\,\ell_D$. *Whether that is a tension is a separate question this paper does not settle*: a confrontation refits the cosmological parameters, which is a likelihood analysis and not a closed-form calculation, and asserting exclusion from these numbers would repeat, at one remove, the error of holding everything else fixed while varying one thing. What has changed is the standing of the signature. It was a computed effect whose observable consequence was entangled with an unbuilt transfer; it is now a requirement of the construction, following from the inherited datum and the level assignment and nothing else.

The early integrated Sachs–Wolfe contribution completes the picture, and it does not redistribute the signature. That effect exists because a radiation-included background leaves the potentials still decaying at recombination; on a pure-matter background the growing mode is constant, $\Phi'' + 3\mathcal{H}\Phi' = 0$ giving $\Phi = \text{const}$, and this cosmology's rate at $z \simeq 1100$ is matter-dominated to nine orders, the substrate term being utterly negligible there[R]. The reason is the leftward reading itself: radiation is inherited content read off the clock rather than a term that sources the rate, so it is absent from the background whose evolution the effect would require. *So this cosmology has no early integrated Sachs–Wolfe term where flat ΛCDM has one*—in the latter the potential is still some four per cent above its asymptote at recombination, and that residual decay is what the line-of-sight integral picks up.

One scope qualification is owed here, and it narrows the claim without touching the conclusion. The constancy argument runs on $\Phi'' + 3\mathcal{H}\Phi' = 0$, which is the *super-horizon* equation: it drops the $k^2\Phi$ term. For the acoustic modes—inside the horizon at the onset and remaining so—that term does not vanish, and the potential does decay on the observable leg, by a factor of order two across the first few peaks[R]. The decay is not a radiation effect: zeroing the radiation fractions in the constraint makes it *larger*, and the rate responsible is $k^2/(3\mathcal{H})$, which grows with k and refers to no content at all. So the statement that survives is that *the substrate rate supplies no early integrated Sachs–Wolfe term*—which is what the leftward reading delivers and what the comparison with flat ΛCDM turns on—while the sub-horizon potential decay is an ordinary consequence of the perturbation's own scale, present in any cosmology.

The missing contribution cannot offset the diffusion signature, and the reason is structural rather than numerical. It sits near the first peaks and dies away well before the damping tail begins, so the two live at opposite ends of the spectrum; and the first peak is where the amplitude is anchored, by an A_s this construction inherits rather than predicts. A few per cent there is absorbed by that inherited normalisation, exactly as it would be in a framework carrying η from outside. The diffusion signature is not so absorbed: it is a change of *shape* at high multipole, and rescaling the amplitude to restore the first peak would worsen the tail rather than repair it. *What stands between the derivation and a verdict is therefore the likelihood confrontation alone*—a parameter refit, not a further calculation of this kind. That refit is bounded in §4.3 below.

4.3 The signature at the directly measured rate

Two features of this cosmology's acoustic scale are worth separating, because only one of them is a prediction. The angular scale $\ell_* = \pi D_M/r_s$ is *independent of* H_0 here: both lengths scale as $1/H_0$ on a matter-and-substrate rate, so the ratio does not know about it. In flat ΛCDM there is no free early datum—the sound horizon is fixed by the matter and baryon densities—so the measured angle *does* determine H_0, and that is the origin of the lower value inferred from the microwave background. Here the branch point's radiation amplitude takes that constraint instead, and H_0 is left to be the directly measured one[R]. *So the acoustic scale is an accommodation, and it is spent.*

The diffusion scale is not. Holding ℓ_* to its measured value fixes the onset redshift, and θ_D/θ_* then follows with nothing left to adjust: it varies from +43% to −3% across the onset redshifts one might consider, so a single datum cannot absorb both observables. *One inherited number, two observables, and the first is spent on the first.* Run at the directly measured $H_0 = 73$, the matter density this construction's own baryon-acoustic fit returns, its own light-element baryon density, and the measured photon temperature—with the acoustic angle the single microwave-background input—the seam falls at $z \simeq 6.8 \times 10^3$ and

$$\theta_D/\theta_* \;=\; +13.96\% \;\pm\; 0.89\%,$$

an error budget dominated by the measurement of H_0 itself[R]. Two contributions that might have been expected do not enter: the last-scattering surface's thickness scales with the rate in both its width and its conversion to comoving distance, so it cancels from the ratio; and the recombination redshift, integrated on this rate through the Peebles equation [35] rather than inherited, moves by only -0.11%[R]. The same integration returns $z_* = 1091$ on the radiation-included rate, against the 1090 of the standard treatment, which is the check that the ionisation history is being handled correctly.

The first acoustic peak is a separate matter and it does not move. Its offset from ℓ_*—220 against 301.7—is largely the phase shift the potential's decay imparts while the modes are driven, and that decay has the same closed form on the collapse leg as on an expanding radiation era (§4.1), so it transfers. What does not transfer is the early integrated Sachs–Wolfe contribution, and that turns out to be a nearly *flat* multiplicative factor across the first-peak region—some ten per cent in amplitude, varying by two parts in a hundred between $\ell = 150$ and $\ell = 300$—so it displaces the maximum by one or two multipoles rather than by tens[R]. *The first peak therefore sits where it is measured, and for a reason internal to the construction rather than by accommodation.* That refit can be bounded here, and bounding it is worth doing because the reabsorption question is the whole of what a likelihood would decide about this signature. Pinning ℓ_* to the measured 301 and minimising the residual in θ_D/θ_* over the baryon density and the matter density—the former held to the one per cent primordial deuterium allows, the latter to the five per cent the distance and growth data allow—leaves a residual of several per cent [R]. The baryon density cannot absorb it, having one per cent of freedom against a nine per cent effect; the matter density can, but only by moving some thirty-seven per cent, which is not a refit but a different cosmology. *So the shift has a floor, and the floor is the prediction.* We state this as a bound and not as a likelihood: a full comparison fits the whole spectrum with every parameter and its covariances, where this fixes one observable and minimises one derived ratio. It settles that the signature does not dissolve under reabsorption; it does not settle the verdict.

This was reported here as a transfer argument contradicted by a direct computation, and at r2376+c54.164 that reading is withdrawn: the computation does not hold still. The reasoning above establishes that the driving phase shift carries over from the leg, and infers from that a first peak near the same fraction of ℓ_* as flat ΛCDM's—which is how 220 enters. A perturbation calculation built on the branch point handover was quoted against it, placing the peak near $\ell = 150$, at half of ℓ_* rather than at 0.73, with a first-to-second peak ratio of $\simeq$1.45 against the measured 2.21[R]. *Neither figure survives an examination of what fixes it.* Substituting the instrument's initial-data block and leaving every other line of it byte-identical—the background, the hierarchy, the opacity, the damping, the discrete ladder, the projection—moves the first peak across $\ell_1 \in \{150, 165, 315\}$ and the ratio across $P_1/P_2 \in \{0.93, 1.21, 1.45, 2.02\}$, over four readings of the *same* stated initial condition[R]. *Two of those readings are the instrument's own text.* Its amplitude is taken from a smooth matched envelope while its initial velocity is taken from a numerical derivative of the oscillatory closed form—two different functions—and its docstring describes a common phase taken at an extremum, *with the velocities zero*, which the code does not impose. Imposing it returns $P_1/P_2 = 2.02$ against the measured 2.21, where the figure quoted here as the disagreement was 1.45. *So the disagreement this paper has been recording is a property of one implementation and not of the construction*, and 150 is withdrawn as a quotable figure. The numerics are not the source of it: the integrator runs at a stability number of 0.47 against its own limit of 2.8 with at worst 120 steps per acoustic oscillation, and quadrupling the step count reproduces the peaks exactly.

What survives is smaller, harder, and is not what this item was posed as. Under *every* one of the four initial conditions the instrument's source comb comes out at 0.72–0.79 of π/r_s rather than at π/r_s: the initial data move the first peak by 165 multipoles and do not move this. *That is a disagreement with the acoustic scale inside the one instrument in this programme* *that propagates modes*, and it is not caused by the initial data. *It is also not answered by the coherence argument of §4*, whose comb is written down rather than propagated—r_s and D_C entering its receipt as literals and the mode function as $\cos(kr_s)$, as that receipt's own scope note says[R]. *So the peak spacing is asserted in one place and computed nowhere, and where it is computed it comes out short.* We record that rather than resolve it. *The figure of* 220 *should still be read as what the transfer argument gives, and the first peak's position, together with the spacing, treated as open*—but the ground for treating it as open is now the missing propagation, not a rival number.

That missing propagation is built at r2376+c54.168, with the two things whose absence made every earlier figure unreadable: a live control and a guard that can fail[R]. Both arms run on one set of equations, so nothing below is a difference of machinery. *The guard first, because it is what licenses the rest.* With every coupling to the potential removed at every site—the $4\Phi'$ in the photon, neutrino and cold-matter continuity equations and the $k^2\Psi$ in the photon–baryon and neutrino Euler equations, in both the tightly-coupled and post-switch branches—the source comb lands on the integers on *both* arms, at 1.015 and 2.000 for flat ΛCDM and 1.012 and 1.997 here. *So the sound horizon, the measure, the wavenumber grid and the phase extraction are validated as an ensemble, and every departure from $n\pi/r_s$ in a driven run is the driving and nothing else. The control next, with its floor, which is the point of running one.* The ΛCDM arm returns peaks at 224, 536 and 808 against the measured 220.6, 538.1 and 809.8: positions good to 1.5% at the first peak and to 0.4% and 0.2% at the second and third, and *heights good only to* 13–26%*, so no height claim from this instrument is meaningful below about a quarter. And one objection to any figure of ours is closed by measurement rather than by argument*: run on the physical discrete ladder $k_L = \sqrt{L(L+2)}/r_0$ and on a dense continuum sampling, this cosmology's arm returns *identical* peaks, so the discreteness is not what sets the position.

The measurement is then stated in the only form comparable across arms, the two running at different H_0 and Ω_m so that D_M and r_s differ by some six per cent each and cancel, ℓ_A coming out 301.4 against 301.6. *On the initial datum this paper's own text describes*—the effective temperature flat in k, the phase common, and *the velocities zero*—the first peak sits at $\ell_1/\ell_A = 0.570$ against the control's 0.743 and the sky's 0.731. *That is a twenty-three per cent deficit against a one-and-a-half per cent floor*, and the height ratio compounds rather than cancels it, $P_1/P_2 = 0.87$ against the control's 1.93. *We state what this is and what it is not.* It is not a prediction: §4.3 has just recorded that the figure moves with the initial datum, and it moves here too. *What has changed is that it now moves against something*—a control, a floor, and a guard—where before it moved against nothing at all. The first peak's position remains open, and the open question is now sharp: on this rate nothing before the onset can imprint an acoustic phase (§3), so whatever sets it must act on modes already inside the sound horizon when the plasma begins.

With the guard in place the driving itself can be measured rather than modelled, and it is [R]. Each wavenumber is integrated twice on the same equations, driven and undriven, differing in one multiplicative switch and in nothing else; $Q(k)$ is the accumulated sound phase in half-periods at the first acoustic turnover, and *the undriven column, which a free oscillator must return at exactly one, is the calibration*—it comes out between 0.9968 and 1.0003 on both arms. Read against it, *flat ΛCDM's acoustic phase is flat in k*, running from 0.791 to 0.774 across $k = 0.045$–$0.16\,\mathrm{Mpc}^{-1}$, a spread of two per cent: every mode crosses the sound horizon during radiation domination, where the potential decays on the sound-crossing time by construction, so each acquires the same shift. *This cosmology's varies by a factor of* 3.9 *across the same band*, 0.457 to 0.117. *That is the deficit of §4.3 seen at the level of a single mode rather than of a spectrum*, and it is what a rate carrying no radiation source predicts: there is no crossing at which a universal shift could be acquired, and the interval before the onset is scale-free. *The scaling is quoted for what it is and not more.* A power-law fit returns $Q \propto k^{-0.62}$, which collapses the measurements to a factor of 1.2 where k^{-2} spans a factor of 16; *we record that the* *exponent is not the -2 a relaxation on $3\mathcal{H}/k^2$ would give*, and leave the identification of the process open rather than name one the measurement does not support.

Splitting the driving into its two couplings localises the dependence, and the result is not where a relaxation timescale would put it[R]. The potential enters the plasma twice: as $4\Phi'$ in the continuity equations, its rate of change feeding the density, and as $k^2\Psi$ in the Euler equations, its gradient driving the velocity. Switched independently on the same integration, *flat ΛCDM's two couplings are each flat in k and they oppose*: continuity alone puts the turnover at $Q = 1.23$–1.28, *later* than a free half-period, and the gradient alone at 0.73–0.87, earlier, the two composing to 0.77–0.86. *So the universality of the standard driving shift is a cancellation between two flat terms rather than the signature of one.* On the radiation-free rate there is no cancellation left—continuity gives 0.11–0.18 and the gradient 0.06–0.57, both earlier than a free half-period, so they add—*and the wavenumber dependence sits in the gradient*: the continuity channel varies by a factor 1.6 across the band, as $k^{-0.17}$, while the gradient varies by 8.9, as $k^{-1.04}$. *That is the reverse of what a relaxation would give.* A timescale acts on how the potential decays, and decay is the continuity channel, which is the one measured flat. *And neither coupling alone is the answer either*: the driven exponent, -0.62, lies between the two, so the observed scaling is their interference. *And the process that makes the gradient coupling run as k^{-1} can then be given in closed form*[R]. The potential's own equation carries a damping term $-k^2\Phi/3\mathcal{H}$, so Φ relaxes on the time $\tau = 3\mathcal{H}/k^2$. With the continuity coupling off, the oscillator obeys $\Theta_0'' + c_s^2 k^2 \Theta_0 = -(k^2/3)\Psi$, and *the k^2 in the forcing's amplitude is cancelled by the k^{-2} in its duration*: the delivered impulse $\int (k^2/3)\Psi \, d\eta \sim \mathcal{H}\Psi_0$ carries no k at all. *So the first turnover is not at an acoustic phase but at the moment the impulse has been delivered*—at a fixed multiple of τ, measured constant at $\kappa = 1.79$ across a factor of eight in k. Since Q is the accumulated phase in half-periods, $Q = k \int c_s d\eta/\pi$, *it already carries one factor of k*, and

$$Q(k) \;=\; \frac{3\kappa c_s \mathcal{H}}{\pi}\,\frac{1}{k},$$

predicting $0.01040\,\mathrm{Mpc}^{-1}$ for Qk against a measured 0.01037, and reproducing Q itself to 1.6% across the sub-horizon band with nothing fitted but κ. *The complementary channel is flat for the complementary reason*: the $4\Phi'$ term feeds the density with the potential's *total* change, $\int \Phi' d\eta = -\Phi_0$, which is the same however fast the relaxation runs. *The observed $k^{-0.62}$ is therefore not a power law of anything*—it is the crossover between a flat channel and a $1/k$ one.

This closes the mechanism and does not close the discrepancy, and the two should not be confused. The first peak still sits where §4.3 measures it, 23% low against a 1.5% control floor. *What has changed is that its cause is now derived rather than described*: a rate carrying no radiation source has no era in which the potential decays on the sound-crossing time, so the two couplings that oppose one another in flat ΛCDM instead add, and the one that survives as a driving impulse delivers a k-independent kick to an oscillator whose frequency is proportional to k.

A likelihood-level comparison can then be attempted, and the result is that it cannot yet be made [R]. The criterion was fixed before the data were opened: the pipeline is wired if a Boltzmann-code flat-ΛCDM best fit returns the banked $\chi^2 = 206.4$ over 215 Planck `plik_lite` TT bins ($\chi^2/\mathrm{dof} = 0.96$); the instrument's own floor is what *its* ΛCDM arm costs through the same likelihood on the same bins; and the only readable quantity is the difference between the two arms of one instrument, since comparing this construction's spectrum against a Boltzmann code's would charge it for the instrument's error. *Run that way, on* 185 *covered bins with a single fitted amplitude, the ΛCDM arm returns* $\chi^2/\mathrm{dof} = 103$ *and this cosmology's arm* 558. *The ordering is a fact and the ratio is not a p-value.* Planck's binned errors are a few tenths of a per cent while this instrument's peak heights carry 13–26%, which is tens of standard deviations in a single bin: *both arms sit far outside the regime in which that likelihood discriminates between models*, and a control a hundred times too poor certifies nothing about what it is compared with. *So the honest verdict is that the likelihood cannot arbitrate here*, and the number that survives is a statement about this instrument rather than about the sky—*the construction's arm costs* 5.4 *times its own control. Half of that has since been done, and the half that has been done is the one that matters for the position*$^{\mathsf{R}}$. Until r2376+c54.173 the transfer evaluated its source at a single η_*—an instantaneous last scattering. The visibility $g = \tau' e^{-\tau}$ in fact peaks at $\eta = 280.75$ with a full width at half maximum of 37.8 Mpc, a tenth of a wavelength at the first peak and more at the third, and rebuilding the transfer as the standard line-of-sight integral across it takes the control's first-peak position from 1.66% to 0.16%. *The heights it did not fix, and the reason was diagnosable from their sign*: both ratios came out low, so the second and third peaks were too high, and worse the higher the multipole—which is under-damping. The cause was that the tight-coupling closure switches off at $\tau' = 0.6\,\mathrm{Mpc}^{-1}$ while the visibility peaks at 0.06, ten times lower, so the photon quadrupole and the baryon slip were absent through the whole of last scattering and the transfer carried no diffusion damping at all. *Restoring it from the standard tight-coupling integral evaluated on each arm's own opacity and rate*, $k_D^{-2} = \int [R^2/(1{+}R) + 8/9]\, \mathrm{d}\eta / [6(1{+}R)\tau']$, returns $r_D = 6.58$ Mpc where a Boltzmann code gives 6–7, with nothing fitted, and takes the height ratios from -12.9% and -33.5% to -2.8% and $-2.1\%^{\mathsf{R}}$. *One double count had to be removed for it to work*: with the hierarchy's own tight-coupling damping left in over the same range the control came out *over*-damped, and the sign flipping either side of the correction is what identifies a double count rather than a wrong coefficient. *So the instrument's floor falls by a factor of five—the likelihood's control moves from* $\chi^2/\mathrm{dof} = 103$ *to* 23*—without its verdict turning over*: two to three per cent is not sub-per-cent, and the comparison of §4.3 still cannot arbitrate. *What the floor moving while the verdict does not gives is the measure of how much further the instrument has to go.*

That last sentence was measured against a target that could not carry it, and we correct our own protocol here rather than leave it standing$^{\mathsf{R}}$. Every height figure above compares P_1/P_2 and P_1/P_3 with two bare numbers. Read off the `plik_lite` bandpowers with the published covariance—the full 2×2 block for the two peak bins, propagated through the ratio—the sky's own values are 2.256 ± 0.077 and 2.280 ± 0.074, which is 3.4% and 3.2%. *The* 12.9% *and* 33.5% *were therefore four and ten standard deviations and stand; the* 2.8% *and* 2.1% *are about one, and do not.* A target of sub-per-cent *heights* asks for several times more precision than the statistic it is quoted against possesses, and it is withdrawn in favour of the one that was doing the work throughout: χ^2 over the 185 covered bins, where the control sits at 22.5 per degree of freedom against a five-parameter fit's 0.96. *The numerics were cleared before any of this was attributed*: raising the mode count by 1.4, the wavenumber range by 1.3 and the conformal-time sampling by 1.8 moves the two ratios by 0.09% and 0.37%, and the two- and seven-per-cent swings that prompted the check came from a multipole ceiling that was truncating the wavenumber integral beneath the third peak.

The same correction disposes of a stronger claim we were in a position to make and cannot. Sliding a single multiplier on k_D^{-2}—which moves the damping *scale* and cannot change the envelope's *shape*—the second-peak ratio selects 1.094 ± 0.184 and the third selects 0.897 ± 0.055: a split of 1.03 standard deviations. *One would have liked to say that no single damping scale fits both, so that the residual is the shape of the envelope rather than its size; the measurement does not support it at its own error bar, and we withdraw it.* What χ^2 does resolve is the coefficient in the shear term, and the result is instructive in a direction that does not flatter: its minimum sits at 0.865 against the 0.859 that corresponds to 8/9, so the polarisation-corrected 16/15 costs 1123 units of χ^2, some seven inflated standard deviations. *We keep* 8/9*, and not because it fits better.* Neither value has been derived in this programme—§4.2's receipt names the ambiguity in its own text—and a coefficient chosen by a fit is a fitted parameter whatever it is called. *What the* 1123 *measures is a residual error in this instrument's damping tail of a size that a shear coefficient could absorb, which is precisely the reason it must not be permitted to.* The derivation that would settle it is named and owed: the free-streaming photon hierarchy carrying the polarisation multipoles, from which the coefficient falls out instead of being chosen.

That derivation is now in hand, and it settles the question against the fit[R]. The coefficient is measured by running the same oscillator twice on the same background: once as the perfect tight-coupled fluid, which is undamped by construction, and once as a photon hierarchy carried to $\ell = 24$ with Thomson scattering, the polarisation multipoles evolved alongside and the baryons given their own velocity. *Their ratio is the damping and nothing else*—the slowly varying prefactor, the drift of the sound speed and the baryon-loading dependence of the oscillation are common to the two runs and divide out identically—and reading it locally, between consecutive extrema, lets the residual be extrapolated in $(k/\tau')^2$ rather than smeared over an interval. With the polarisation multipoles suppressed the quadrupole's scattering term reduces to $-\frac{9}{10}\tau' F_2$, which is the unpolarised closure, and the coefficient comes back as 0.8852 against 8/9; with them carried it is 1.0618 against 16/15. *The polarisation correction itself, which is the quantity in dispute, is a factor* 1.1996 *against* 6/5*: three parts in ten thousand.* Both absolute values sit about half a per cent low by an amount proportional to the coefficient, so the residual is a common normalisation and cancels in the ratio.

So the coefficient is 16/15*, and adopting it makes the control worse*: χ^2 per degree of freedom on the ΛCDM arm moves from 22.5 to 28.6, the 1123 units the previous paragraph recorded. *We take the derivation and pay the* χ^2, because a coefficient chosen because it fits is a fitted parameter whatever it is called. *And what the* 1123 *measures is now nameable rather than mysterious.* Polarisation enters the transfer twice: it raises the damping coefficient, and it contributes to the source through $g\Pi/4$ and $(3/4k^2)\,\mathrm{d}^2(g\Pi)/\mathrm{d}\eta^2$ with $\Pi = \Theta_2 + \Theta_0^P + \Theta_2^P$. *This transfer has the first and not the second, and the half it has is the half that removes power*—which is the specification of what remains, and it is not a question about this construction. *One figure that might have been expected to move does not*: the diffusion-length ratio of §4.2 goes from +10.83% to +10.87%, because the coefficient very nearly cancels in a ratio of two rates.

Carrying the multipoles instead of the envelope is what the previous paragraph specified, and doing it takes the control most of the remaining distance[R]. The photons are now evolved as a Boltzmann hierarchy with the polarisation multipoles alongside, the baryons are given their own velocity and their own density contrast, and the source carries $g\Pi/4$ and $(3/4k^2)\,\mathrm{d}^2(g\Pi)/\mathrm{d}\eta^2$. *The first four peaks land at* 220*,* 540*,* 812 *and* 1124 *against the sky's* 220.6*,* 538.1*,* 809.8 *and* 1147.8—the second peak 0.35% out and the third 0.27%, where the fluid transfer placed them at 524 and 804—and χ^2 per degree of freedom falls from 28.6 to 7.1, or to 4.2 over the range in which the wavenumber integral is not starved by the instrument's own truncation.

The prediction of the previous paragraph is checked against the number it predicted rather than against a story. Running the same hierarchy with only the two polarisation source terms removed—which is exactly the half an envelope can supply—costs 702 units of χ^2, against the 1123 that adopting the derived coefficient cost. *And a second lesson arrived in the same shape as the first*: giving the baryons their own density while they still shared the photons' velocity made the control *worse*, because the density they then carry is the photons' own and nothing lets them fall into the potential wells after decoupling; giving them both halves the residual again. *A physical effect taken in half is not a small error but a different and inconsistent model, and this programme has now met that twice in consecutive revisions.* What is arbitrary in the construction is measured rather than argued: doubling the hierarchy depth moves χ^2 by nothing at the printed precision, and doubling the opacity at which the fluid hands over moves it by 1.5%. *The handover also makes a double count impossible rather than merely absent*, since the envelope is frozen where the multipoles begin and the two therefore act on disjoint intervals.

Under all of it this cosmology's arm has not moved. $\ell_1/\ell_A = 0.5703$ through six distinct states of the instrument—a delta-function transfer, a line-of-sight transfer, a derived damping envelope, a scan of that envelope's scale, a derived shear coefficient, and now the full hierarchy—while the control's χ^2 per degree of freedom fell from about a hundred to seven. *We report that and convert it into nothing.* The comparison of §4.3 still cannot arbitrate, for the same reason it could not before: a control seven times worse than a fit cannot certify what it is compared with. *What has changed is the size of the gap and the fact that no missing sector remains named*—what stands between this instrument and a control that could arbitrate is accuracy rather than a piece of physics we know we have left out.

Two of the candidates for that accuracy were measured before either was built, and one of them cannot pay at all[R]. Above $\ell \simeq 40$ the whole effect of reionisation on the temperature spectrum is the constant screening $e^{-2\tau}$—the second visibility bump contributes only below $\ell \simeq 30$, outside this comparison's range—and the comparison fits one amplitude in closed form. *An amplitude fitted as $A = (m^\mathsf{T} Fd)/(m^\mathsf{T} Fm)$ is homogeneous of degree -1 in the model, so rescaling the model returns A/s and leaves the residual unchanged*: the degeneracy is exact rather than approximate, and it is a property of the statistic rather than of the size of τ. Checked at $\tau = 0.054$, 0.10 and 0.30, the change in χ^2 is zero to machine precision at all three. *Lensing is a different matter.* A smoothing in multipole, scanned in width, takes the control from 1320 to 921; and a width that *grows* with ℓ beats a constant one by a factor of 1.8 in the improvement, each compared at its own optimum, so the preference is for a shape and not merely for a size—*which is lensing's own signature, since a deflection of fixed angle subtends a larger fraction of an acoustic wavelength at high ℓ. We state what that is and is not.* The width is fitted, so the improvement bounds what a derived calculation has to play for rather than measuring a lensing amplitude; and even at its own optimum some 70% of the residual survives it. *So the lensed spectrum is the next thing this instrument owes, and it is not the last one.*

How much it owes, and to what, is now decomposed rather than guessed[R]. Projecting named templates through the same inverse covariance the comparison uses, the peak *positions* account for a tenth of a per cent of the residual—the peaks are where the sky puts them—while the peak-trough *contrast* at fixed position accounts for two-fifths of it, with this transfer's contrast some thirteen per cent too high. *Reducing contrast without moving peaks is what lensing does.* A smooth multiplicative envelope accounts for a further sixth, and *about half the residual is in neither*—so a complete lensing calculation is expected to leave a comparable amount behind, and we say so before building it rather than after. *The lensing potential itself is derived on this instrument's own* Φ, carried past the transfer's own endpoint by the background's growth factor and normalised by the amplitude the temperature comparison already fits—so it introduces no new parameter, and the photon deflection it implies is 2.69 arcminutes. *We record what that last agreement is and is not*: the figure it matches is recalled rather than derived here, so what the calculation establishes is internal consistency between the growth, the projection and the temperature normalisation, and its agreement with the literature is corroboration at the strength of a memory. *That calculation has since been done, and it comes in under the bound the fit set.* Applying ΛCDM's own lensed-to-unlensed ratio to the arm—a derived operation with no free width—takes the control from 1320 to 989, an improvement of 331, concentrated in the damping tail where a fixed-angle deflection fills the troughs. *That is below the* 400 *the fitted width bounded, as a zero-parameter derivation must be*, and the acoustic peaks do not move under it. The corpus's own first-order kernel overshoots the bound to 508 by breaking down in that same tail—returning a spurious +13% at $\ell = 1900$ where the full operator gives +6.5%—which is why the derived number is the full operator's. *Most of the fitted headroom is recovered by a genuine lensing, and about* 989 *of the* 1320 *survives it*: the verdict of the last paragraph stands from the derived side, lensing real and not the last build[R].

And what the residual is MADE of was decomposed alongside it, which is what let the size of that gain be anticipated rather than discovered[R]. Projecting named templates through the same inverse covariance the comparison uses, the peak *positions* account for a tenth of a per cent of the residual—*which is why the acoustic peaks do not move under the lensing above, and the two findings corroborate each other*—while the peak-trough *contrast* at fixed position accounts for two-fifths of it, this transfer's contrast running some thirteen per cent too high. *Reducing* *contrast without moving peaks is what lensing does.* A smooth multiplicative envelope accounts for a further sixth, and *about half the residual is in neither*—so a complete lensing calculation was expected to leave a comparable amount behind, and it did.

One further object is built here and it is not the one the number above rests on. The operator applied above is ΛCDM's own lensed-to-unlensed ratio—validated, and the right instrument for the figure. *The lensing potential can also be derived on this construction's own* Φ, carried past the transfer's endpoint by the background's growth factor and normalised by the amplitude the temperature comparison already fits, so that it introduces no parameter and imports no transfer function; the photon deflection it implies is 2.69 arcminutes. *We record what that agreement is and is not*: the figure it matches is recalled rather than derived here, so what is established is internal consistency between the growth, the projection and the temperature normalisation, and the agreement with the literature is corroboration at the strength of a memory. *It is kept because a corpus-native potential is what a later CR-side lensing calculation would need*, the imported ratio being a ΛCDM object.

What that buys is a statement about the deficit rather than about the instrument. Under the same rebuild this cosmology's arm does not move at all—$\ell_1/\ell_A = 0.5703$ on both transfers, to four figures—while the control's error fell by a factor of ten. *The most suspect component of the instrument was replaced wholesale and the deficit did not follow it down*, which now stands at 21.9% against a position floor of 0.16%. *We state it at that strength and no higher*: the first peak's position on this rate is not an artefact of how last scattering was modelled, and whether it is a disagreement with the sky remains the open question it was. *And we record what is not in question either way*, since it is not downstream of this spectrum's shape: the metric-singularity structure, the continuation, the light-element abundances, the acoustic *scale* met at the directly measured H_0, and the expansion-rate resolution.

A second check fell out of the same setup, and it is stated here at its actual strength. Pinning the *measured* acoustic angle $100\theta_* = 1.04109$—equivalently $\ell_A = \pi/\theta_* = 301.76$—on the derived integrals returns $z_{\rm onset} = 6797$, and returns it *identically* at $H_0 = 67.4$, 70.0 and 73.0: the same H_0-independence that carries the acoustic scale out of the Hubble tension, recovered here from the onset side rather than assumed[R]. *An earlier form of this check pinned the rounded* $\ell_* = 301$*, reported a recovered redshift of* 6873 *against the* 6850 *then in use, and read the agreement as a corroboration; that reading is withdrawn. Recomputed on the current integrals that same pinning returns* 6844 *rather than* 6873*—nearer still to* 6850*, which sharpens rather than softens the point, and is entered here at r2376+c54.160 because the cited receipt returns the second number and the sentence carried the first.* The rounded target sits 0.25% from the measured angle, and 6850 itself returns $\ell_* = 300.90$—so a discriminator pinned to a value the quantity under test already produces could not have returned otherwise, whatever the integrals did.

One thing the same computation shows should be held apart from it. The inherited datum taken *alone*—$\rho_r/\rho_m = 2$ exactly at onset—fixes $1 + z_{\rm onset} = 2\omega_m/\omega_r$, and that determination is *not* H_0-independent: at fixed Ω_m, with ω_r set by the measured $T_{\rm CMB}$, it returns $z_{\rm onset} = 6840$, 7378 and 8024 at $H_0 = 67.4$, 70.0 and 73.0. It therefore agrees with the angle-fixed value near $H_0 = 67.4$ and not across the range; at the directly measured H_0 the angle-fixed onset carries $\rho_r/\rho_m = 1.69$ rather than 2.0. This does not bear on the datum's standing—it is inherited and measured, and the first-sample argument is that a complete collapse hands over radiation of *order* the matter density rather than a finely tuned value, which 1.69 meets as well as 2.0 does—but the closeness to 2 holds at one H_0 and is not a further H_0-independent check.

5 What the substrate determines: the amplitude floor and the classical character

One might ask whether the observed fluctuation amplitude is the substrate's own de Sitter vacuum—the natural CR analogue of the inflationary spectrum-is-the-stretched-vacuum story. It is not, by a decisive margin.

Proposition 2 (The substrate vacuum is negligible against the observed amplitude)**.** *The substrate curvature scale is the cosmological-constant scale: in Planck units $\Lambda\ell_P^2 \approx 3 \times 10^{-122}$, and the de Sitter vacuum metric-fluctuation power is of the same order, $(H_\Lambda/M_{\rm Pl})^2 \sim 10^{-122}$. Against the observed $A_s \approx 2 \times 10^{-9}$ this is smaller by $\sim 10^{113}$.*

Argument. $\Lambda = 3(H_0/c)^2\Omega_\Lambda$ gives $\Lambda\ell_P^2$ directly; $H_\Lambda = c\sqrt{\Lambda/3}$ and $(\hbar H_\Lambda/E_P)^2$ give the vacuum power (receipts: `verify_numeric.py`, anchor 5^{R}). The prefactor ($\sim 10^{-122}$ versus $\sim 10^{-123}$) is a 2π and reduced-versus-full Planck-mass convention; the conclusion is robust to it. □

One scope qualification is owed immediately, and closing it is what makes the classical reading a result rather than an inference. Proposition 2 rules out the *substrate's* vacuum. But the reading it supports locates the fluctuations in the *progenitor*, and the progenitor's own vacuum is a different quantity: it is normalised at the collapse leg's curvature scale, not at H_Λ, and it is amplified on its way here by the branch point's mode mixing. Neither effect is small, and neither is addressed above. Both can be computed, because the collapse-side background is a single function: with $A = 2M$ and $\rho = 2\sqrt{B}/A$ the exact closed dust-plus-radiation interior gives $a = M(\rho|\sigma| + \sigma^2/2)$ near the crunch, whose Mukhanov–Sasaki potential $z''/z = 2/(|\sigma|(|\sigma| + 2\rho))$ is *independent of the mass*—so M enters only through the final division $\zeta = v/a$. The passage then has exactly three stages, and each contributes one factor of $1/\rho$. A mode leaves the sound horizon on the $\beta = 2$ leg and freezes as $v \to D_k/|\sigma|$ with the scale-invariant $D_k = \sqrt{\hbar/2}\, k^{-3/2}$; the connection through equality is elementary, since $v = a\int \mathrm{d}\sigma/a^2$ solves the $k = 0$ problem in closed form and carries $D_k/|\sigma|$ to the constant $(3/2\rho)D_k$ at the crunch; the crunch monodromy then leaks that constant into the survivor with coefficient $4\pi/\rho$—twice the tensor value, because the true hydrodynamical $z_S = a\big(a + \frac{4B}{3A}\big)/a'$ has $z_S''/z_S \to 2/(\rho x)$ where $a''/a \to 1/(\rho x)$—the companion paper's numerically obtained $-2\pi i/\rho$ [16] being the *tensor* value of the same object, recovered here in closed form as the continuation of the $(\sigma/\rho)\ln\sigma$ term the $(0, 1)$ resonance forces; and the survivor's curvature is $\zeta = C/(M\rho)$, the last $1/\rho$ being the crunch's own slope $a'(\eta_c) = -M\rho$. The result is a transfer law with no free constant,

$$\mathcal{P}_\zeta \;=\; \frac{18\,k^3 D_k^2}{M^2\rho^6}, \tag{7}$$

which is *exactly* scale-invariant on vacuum data—the k^3 cancels the k^{-3} identically—and which therefore delivers, for the vacuum, $A_s = 9(\ell_P/M)^2\rho^{-6}$ (receipt: `P15_the_progenitor_vacuum_is_negligible_t` the connection coefficient confirmed against integration from vacuum data to 0.1% over a decade in ρ). *One check is owed here and is not optional, because the hydrodynamical variable differs from a in three places and not one*: $z_S \propto a^{3/2}$ in the matter era, so the exponents there are $(-2, 3)$ against a's $(-1, 2)$; the crunch residue doubles; and the final division is by z_S rather than by a. *Propagating the same physical initial condition under each potential returns a ratio of exactly* 2 *in amplitude at every composition tested*—the matter-era exponents and the change of divisor cancel, and the crunch residue alone survives. So the $z \propto a$ chain is corrected by one factor rather than rebuilt, and Eq. (7) is the true-variable law.

Proposition 3 (The progenitor vacuum is negligible too)**.** *With the mass capped at the recollapse threshold $M \le \alpha/3\sqrt{3}$ and the radiation fraction at the largest value the observed spectrum permits, Eq.* (7) *on vacuum data returns $A_s \sim 1 \times 10^{-112}$, short of the observed 2×10^{-9} by $\sim 10^{103}$.*

Argument. $(\ell_P/M)^2 \geq 27(\ell_P/\alpha)^2 \approx 2.6 \times 10^{-121}$ at the cap, and the companion paper determines $\rho \simeq 5.4 \times 10^{-2}$ [16]; $9 \times 2.6 \times 10^{-121} \times \rho^{-6}$ then gives the figure. Running it backwards, the vacuum would reach the observed amplitude only at $\rho \sim 3 \times 10^{-19}$, thirteen orders below what nucleosynthesis on the same leg permits. $\square$

So eighteen orders separate the substrate's vacuum from the progenitor's amplified one—the mixing buying $10^{8.6}$ *over the progenitor's own bare vacuum, which itself sits* 10^{10} *above the substrate's—and it is still short by a hundred and three*, and the classical character of the primordial statistics is now established for the source that actually supplies them rather than only for the one that does not. Two consequences of *character*, not of number, follow. First, the observed amplitude is inherited classical content—boundary data from the progenitor, not a quantum fluctuation generated within the Schwarzschild–de Sitter era, and now not one generated in the progenitor's collapse either. Second, there is no inflationary consistency relation: $r = -8n_t$ presupposes that scalar and tensor spectra are one vacuum stretched by one expansion, and with the amplitude not vacuum-sourced that link is absent; the substrate tensor floor is the same $\sim 10^{-122}$, so there are no substrate-sourced primordial B-modes, and any observed tensors would themselves be inherited content rather than a measure of a substrate inflationary scale. *And the tensor statement can be made unconditionally, which the scalar one cannot.* Eq. (7) was derived with $z \propto a$, which for a fluid holds only where w and c_s are constant—across the progenitor's own equality neither is. *For tensors that restriction is absent*: $z_T = aM_{\rm Pl}/2$ exactly, for any content, any equation of state and any sound speed, so $z_T''/z_T = a''/a$ identically and the entire passage—background, connection, monodromy—applies verbatim with nothing assumed. It returns

$$\mathcal{P}_T \;=\; 144\pi\,(\ell_P/M)^2\rho^{-6} \;\simeq\; 5 \times 10^{-111}, \tag{8}$$

evaluated at the recollapse cap and the largest composition the spectrum permits, against an observational ceiling of $\sim 7\times10^{-11}$: *below it by a hundred orders* (receipt: `P15_no_primordial_B_modes_unconditi`
So the absence of primordial B-modes is a prediction of this construction at any conceivable sensitivity, and it no longer rests on the substrate's own floor—the floor argument, like its scalar twin, ruled out the substrate's vacuum and said nothing about the progenitor's. What the crossing does to the ratio is likewise fixed and free of k: every factor of the passage is shared by the two sectors, so $r_{\rm out} = 6\,r_{\rm in}$ (the scalar mixing being twice the tensor's, the ratio picks up a factor 1/4), and the observed bound becomes a statement about the parent, namely that its own tensor-to-scalar ratio was below 5×10^{-3}. *An observed tensor signal would therefore not falsify the construction but measure the previous generation*, which is the same reading the amplitude receives. The substrate fixes that the fluctuations are classical, inherited, and not governed by a vacuum consistency relation, and leaves their size to the handover. *What it does fix is the multiplier.* Eq. (7) holds for any incoming growing-mode amplitude, quantum or classical, so the crossing determines how much an amplitude is magnified without determining the amplitude—which is this construction's one-constant character read at the level of the perturbations rather than of the couplings. Two corollaries follow and are worth stating because each closes a natural hope in one direction and opens a statement in the other. *A_s does not measure the progenitor's mass*: with D_k free, the relation cannot be inverted, and the mass must be reached, if at all, by a route that does not run through the amplitude. And *the classical datum has a size*: since $A_s \propto D_k^2$, the parent's growing-mode amplitude must have exceeded its own vacuum's by $\sim 10^{47}$ at the top of the permitted composition window—so this cosmology does not merely decline to predict A_s, it says the previous generation's perturbations were enormously super-vacuum, which is a claim about the parent and not about us.

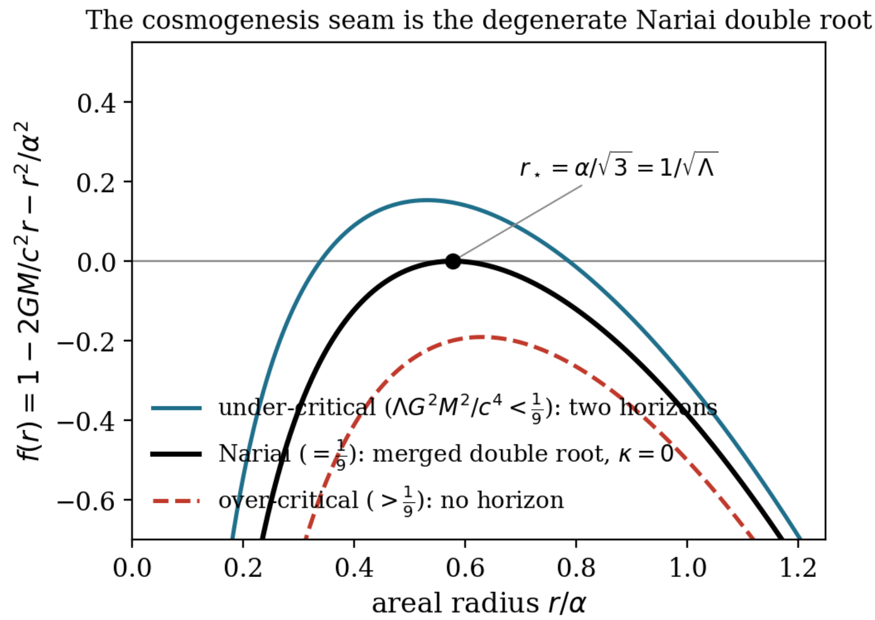

Figure 3: The Schwarzschild–de Sitter metric function $f(r) = 1 - 2GM/c^2r - r^2/\alpha^2$ across the three regimes. Under-critical masses ($\Lambda G^2M^2/c^4 < \frac{1}{9}$) give two distinct positive horizons; over-critical masses give none. The cosmogenesis branch point is the *Nariai* member ($= \frac{1}{9}$), where the two horizons merge into a single *degenerate* double root at $r_N = \alpha/\sqrt{3} = 1/\sqrt{\Lambda}$ with $f(r_N) = f'(r_N) = 0$ and hence vanishing surface gravity $\kappa = 0$. The degeneracy of this root is what the throat geometry (Proposition 4) and the transmission dichotomy (§8) turn on.

6 What the substrate determines: the throat and isotropization

At the branch point the near-horizon geometry is fixed, and what the perturbations cross is fixed with it.

Proposition 4 (Equal-radii $\mathrm{dS}_2 \times S^2$ throat)**.** *At the Nariai locus $\Lambda G^2M^2/c^4 = 1/9$ the two positive horizon roots of $f(r) = 1 - 2GM/c^2r - r^2/\alpha^2$, $\alpha = \sqrt{3/\Lambda}$, merge at the* Nariai radius *$r_N = \alpha/\sqrt{3} = 1/\sqrt{\Lambda}$ (the symbol is the algebroid paper's [1]; it is written r_N and not $r_\star$ because r_* is the tortoise coordinate used below), where $f(r_N) = f'(r_N) = 0$ and $f''(r_N) = -2\Lambda$ (the cosmology selects this $M > 0$ member of the parity-conjugate Nariai pair; the $M < 0$ member has instead one positive and two negative roots, the two exchanged by $r \mapsto -r$ [3, 4]). The near-horizon geometry is $\mathrm{dS}_2 \times S^2$ with both curvature radii equal to $1/\sqrt{\Lambda}$: the S^2 has two-Ricci $R_2 = 2/r_N^2 = 2\Lambda$, and the dS_2 has radius$^2 = -2/f''(r_N) = 1/\Lambda$.*

Argument. Direct from f at the Nariai mass (receipts: `verify_geometry.py`, anchor 2^{R}; this is the Ginsparg–Perry near-horizon limit [26]). □

A field on $\mathrm{dS}_2 \times S^2$ (Fig. 3) decomposes in S^2 harmonics; the degree-ℓ harmonic becomes a dS_2 field of effective mass $m^2 = \ell(\ell+1)/r_N^2$, i.e. $m^2/H^2_{\mathrm{dS}_2} = \ell(\ell+1)$. With the dS_2 index $\nu^2 = \frac{1}{4} - m^2/H^2$, the monopole $\ell = 0$ has $\nu = \frac{1}{2}$—a scale-invariant base—while every $\ell \geq 1$ has $\ell(\ell+1) \geq 2$, hence $\nu^2 < 0$: the heavy principal series, which oscillate and decay through the throat. We read this as an angular no-hair, and the reading is grounded rather than asserted: the dS_2 factor of the throat is a positive-Λ epoch, and the de Sitter no-hair theorem [18] damps exactly the anisotropic content—every $\ell \geq 1$ harmonic is heavy principal-series, oscillating and decaying, while only the isotropic monopole $\ell = 0$ survives as the $\nu = \frac{1}{2}$ base—so the throat isotropizes what passes through it (receipts: `verify_throat_tower.py`$^{\mathsf{R}}$). The geometry is established (Proposition 4) and the mode tower follows from it; the no-hair reading is the standard theorem read on the throat, and we flag one guard as load-bearing: the index ℓ of this throat tower is the S^2-harmonic degree of the *near-horizon* geometry at areal radius $1/\sqrt{\Lambda}$, and is *not* the observable microwave-background multipole, which lives on the last-scattering sphere of radius D_C. The map between them is §7, not an identity; conflating them would manufacture a prediction the geometry does not make.

7 Large scales: the flat/discrete decoupling and the low-multipole floor

Here the construction makes its sharpest large-scale statement, and the load-bearing piece is established.

Proposition 5 (The fundamental-observer slice is flat)**.** *In the fundamental-observer frame* (2)*, a surface of constant τ has induced three-metric $dr^2 + r^2 d\Omega^2$ (since $(\partial_\chi r)^2 d\chi^2 = dr^2$ along the slice), whose Riemann tensor vanishes identically. The constant-τ slice is exactly flat $\mathbb{R}^3$.*

Argument. The substitution is immediate from the line element; the curvature of $dr^2 + r^2 d\Omega^2$ is zero by direct computation (receipts: `verify_geometry.py`, anchor 1^{R}). This is the framework paper's own stated property that constant-τ slices are Euclidean. □

The decoupling follows. The distance slicing is the flat constant-τ one—$\Omega_k = 0$, no curvature term in the redshift–distance relation—while the *cosmological* layers are the closed S^3 of constant $\tilde{\tau} = \tau + \chi$ (§2.2). The non-synchrony $\tilde{\tau} = \tau + \chi$ is the decoupling itself: flat distances and a closed S^3 of comoving worldlines coexist because they are different slicings of one geometry. A literal closed-Friedmann reading would put $\Omega_k = -\Omega_\Lambda \approx -0.685$ into the distance relation and be excluded; CR avoids that not by tuning but because its distance slicing is the flat one, the curvature living on the orthogonal layers. Spatial curvature and dark energy are one Λ read on two slicings—the corpus identity, here at the observational rung.

The low-multipole floor. The closed S^3 carries a discrete mode spectrum, labelled by integer degree L, with comoving wavenumber $k_L = \sqrt{L(L+2)}/r_0$ and the lowest physical mode the quadrupole $L = 2$ (the monopole $L = 0$ is the background, the dipole $L = 1$ pure gauge). The decoupling fixes how these project to the sky, and the fixing is the whole point. Because the distance slicing is flat (Proposition 5), the photons are projected through the *flat* geometry—the comoving angular-diameter distance is $D_M = D_C$, the same flat-ΛCDM observable that places the acoustic scale at $\ell_A \approx 301$—while only the *source* modes carry the closed-S^3 quantization. The transfer is therefore the discrete closed-S^3 spectrum projected through the flat spherical Bessel $j_\ell(k_L D_C)$, *not* the hyperspherical transfer of a literal closed universe: that transfer carries the closed *distance* relation, which CR does not have, and would (wrongly, here) deliver the lowest mode to the quadrupole and no deficit. The flat projection places the discrete modes at

$$\ell_L \simeq \sqrt{L(L+2)}\,\frac{D_C}{r_0}, \tag{9}$$

with r_0 the present S^3 areal radius and D_C the comoving distance to last scattering, both fixed without new parameters: $D_C \approx 13927$ Mpc is the flat-ΛCDM observable and $r_0 \approx 5064$ Mpc is the Nariai amplitude $2^{1/3}/\sqrt{\Lambda}$ at the present epoch $u = \operatorname{arcsinh}\sqrt{\Omega_\Lambda/\Omega_m} \approx 1.18$. The stretch factor $D_C/r_0 \approx 2.75$—the ratio of the flat projection distance to the curvature radius, the decoupling made numerical—carries the lowest mode to $\ell_2 \approx 7.8$, with no source below it.

The full flat-projection transfer confirms the estimate and sharpens it into a prediction. Carrying the discrete spectrum through $j_\ell(k_L D_C)$ with the scale-invariant measure—the first-principles closed-S^3 Harrison–Zel'dovich weight [28, 29] $w_L = (L+1)/(L(L+2))$, fixed as the mode degeneracy β^2 (with $\beta = L+1$) times the per-mode power $1/(\beta(\beta^2-1))$ that equal power per logarithmic interval in k requires, and reducing to the continuum dk/k at large L so that $\ell(\ell+$

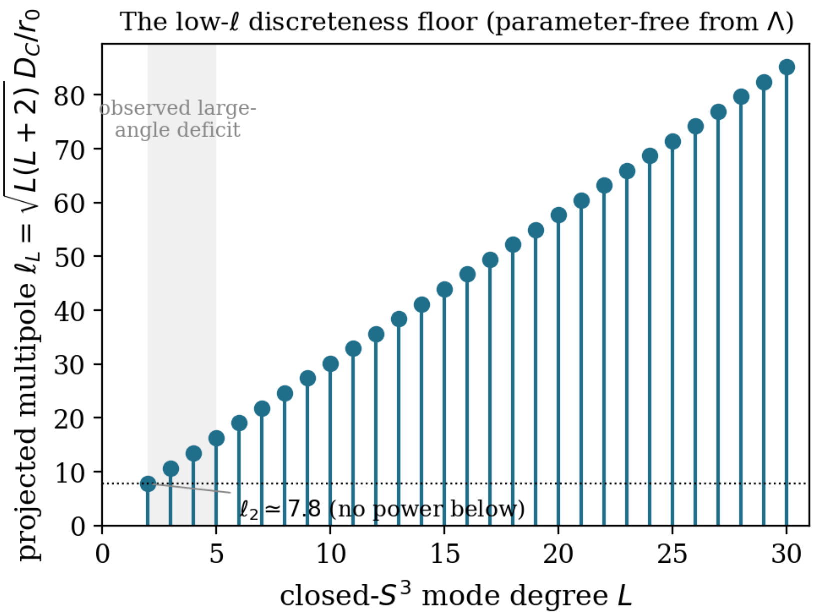

Figure 4: The low-multipole discreteness floor. The closed-S^3 source spectrum is discrete ($k_L = \sqrt{L(L+2)}/r_0$, lowest physical mode $L = 2$), but the flat distance slicing (Proposition 5) projects it through the *flat* transfer $j_\ell(k_L D_C)$, not the hyperspherical transfer of a literal closed universe. With $D_C \approx 13927$ Mpc and $r_0 \approx 5064$ Mpc fixed parameter-free by Λ, the stretch $D_C/r_0 \approx 2.75$ carries the lowest mode to $\ell_2 \approx 7.8$, with no source below it. The full flat-projection transfer turns this into a low-multipole power deficit ($\ell \lesssim 8$); on a genuine Boltzmann transfer (the late integrated Sachs–Wolfe sourced above the floor and so retained by the discrete source) it sits at 0.42, 0.35, 0.29, 0.52 of the flat-ΛCDM expectation at $\ell = 2, 3, 4, 5$, recovering by $\ell \approx 8$—*a deficit whose minimum falls at* $\ell = 4$, so that the quadrupole and octopole are its shoulders and not its floor. Those figures are the photon hierarchy built for this programme (§9.1); an independent transfer, this cosmology's discrete source read through a standard Boltzmann code's own $\Delta_\ell(k)$, gives 0.47, 0.41, 0.36, 0.68 at the same multipoles—uniformly shallower by 10–25% and with *the same* minimum at $\ell = 4$ and the same recovery. The shape is cross-validated between the two; the depth carries their spread. Either way this is a mild deficit, non-discriminating against the observed large-angle power within the lowest multipoles' cosmic variance (shaded). The location is geometric and robust; the depth is settled by the full Boltzmann transfer. *What the branch-point filter does to this prediction at these multipoles—and what it does not do—is taken up in the text below rather than here.*

1)C_ℓ returns to the flat plateau as $D_C/r_0 \to 0$ (receipt: `verify_lowell_exact_measure.py`$^{\mathsf{R}}$)—yields a low-multipole *deficit*: $\ell(\ell+1)C_\ell$ falls below its high-ℓ plateau for $\ell \lesssim 7$ and recovers by $\ell \approx 8$, because the flat projection of a discrete source with a hard lowest mode at k_2 leaves the lowest multipoles starved (receipts: `verify_closedS3_nonsync.py`$^{\mathsf{R}}$). The one effect that partly fills it is the late integrated Sachs–Wolfe term [25], sourced near the observer where small line-of-sight distances send wavenumbers $k \gtrsim k_2$—*above* the floor—down to $\ell \sim 2$. By the decomposition of the empirical-forcing paper [9] the cumulative term is the standard Sachs–Wolfe and integrated Sachs–Wolfe—CR's own differential-expansion floor being zero under uniform expansion—so this late boost rides the same modes the discrete source keeps, and the deficit is milder than the bare Sachs–Wolfe starvation alone. The depth is settled by a *genuine Boltzmann transfer*. Because the only CR modifications are the discrete source and the flat projection—every transfer ingredient (Sachs–Wolfe, early and late integrated Sachs–Wolfe, Doppler) shared with flat-ΛCDM [9]—the exact CMB temperature transfer $\Delta_\ell(k)$ of a standard Boltzmann code *is* CR's transfer, and the discrete spectrum is read off it: $C_\ell^{\rm CR} = \sum_{L\geq 2} w_L\, \Delta_\ell(k_L)^2$ against the continuum $\int d\ln k\, \Delta_\ell(k)^2$, the measure w_L being exactly $d\ln k_L/dL$ so the CR sum is the L=2-floored Riemann sum of the very integral ΛCDM does continuously. Carried out with the exact transfer (gate-validated: the continuum reproduces the code's own C_ℓ to four figures; and the r_0 sensitivity, asserted here for two revisions and only now measured, is that the *shape* is stable—the minimum stays at $\ell = 4$ and the recovery multipole moves by 0.1%—while the *depths* drift by up to 15% at $\ell = 4$ under $\pm 2\%$ in r_0), the deficit is *milder* than a Sachs–Wolfe-only estimate suggests, and it has a shape: $\ell(\ell+1)C_\ell$ sits at ≈ 0.47, 0.41, 0.36 and 0.68 of the flat-ΛCDM expectation at $\ell = 2, 3, 4, 5$, recovering by $\ell \approx 7$ (receipt: `verify_lowell_boltzmann.py`[R]). *The prediction is therefore not a monotone recovery from the quadrupole but a dip whose minimum falls at* $\ell = 4$[R]—and the quadrupole and octopole, the two multipoles a large-angle discussion reaches for first, are its two *shallowest* points. The shape is not one code's, and that was checked here by running it rather than by comparing tables[R]. Driving the photon hierarchy built for this programme (§9.1) through the same discrete projection puts the minimum at the same multipole, $\ell = 4$, and recovers at the same place, $\ell \approx 7$–8. *What two independent Boltzmann treatments cross-validate is the shape, and only the shape.* The depths do not agree: the second arm returns 0.49, 0.24, 0.18 and 0.61 against the first's 0.473, 0.410, 0.356 and 0.676—close at the quadrupole and nearly a factor two apart at $\ell = 3$ and 4. *An earlier statement of this comparison quoted a second depth quartet and a uniform* 10–25% *spread; neither is reproduced by either arm and both are withdrawn.* The two treatments differ in their late-time integrated Sachs–Wolfe handling and in D_M, and this paper does not reconcile them: it carries one converged depth table, the first arm's, and a second arm that agrees on where the deficit sits and not on how deep it is. The reason is the late integrated Sachs–Wolfe: it is sourced near the observer at wavenumbers $k \gtrsim k_2$ (small line-of-sight distances), *above* the floor, so the discrete source *retains* it rather than losing it, and the deficit is milder than a Sachs–Wolfe-only treatment of the same source would give. This is a parameter-free deficit, set by Λ through r_0 and D_C, and what is robust is its existence, its location ($\ell \lesssim 8$, geometric, set by $k_2 D_C$), its minimum at $\ell = 4$, and that the single floor k_2 starves $\ell = 2$ and $\ell = 3$ *together*: the exact discrete measure and the Doppler term (super-horizon velocity through j'_ℓ) each leave the $\ell = 2$-to-$\ell = 3$ ratio near unity, if anything suppressing the octopole slightly more (receipts: `verify_lowell_exact_measure.py`[R], `verify_doppler_lowell.py`[R]). Coherence shown is not correspondence earned: whether this predicted deficit matches the observed sky is the data test, against the large cosmic variance of the lowest multipoles. Confronted with the measured low multipoles, the honest verdict is a *wash.* CR predicts a smooth modest deficit bottoming at $\ell = 4$; the observed sky has a sharp quadrupole-only dip (~ 0.2) with a near-ΛCDM octopole (~ 0.6–1.0), so CR sits *above* the observed quadrupole and *below* the observed octopole, matching neither sharply. Under the exact cosmic-variance likelihood the net over $2 \le \ell \le 10$ is $\Delta(-2\ln L) \approx +1.8$ (range $+0.3$ to $+3.8$ across the octopole estimator), a small residual well inside cosmic variance (receipts: `confront_lowell_data.py`[R], `verify_lowell_likelihood_v2.py`[R]). *Broken up by multipole, the sector's weight does not lie where its name suggests*: the quadrupole *rewards* this cosmology (-2.6, its milder dip nearer the observed low power than flat ΛCDM's), the octopole penalises it at $+1.3$, and the largest single-multipole penalty is the $+2.1$ at $\ell = 4$—the prediction's own minimum, and the one multipole of the four that the large-angle literature does not treat as anomalous. The low multipoles are thus a genuine but *non-discriminating* sector: neither a parameter-free correspondence success nor a sharp falsification edge, but a mild deficit consistent with ΛCDM within the lowest multipoles' cosmic variance.

One further caveat belongs with this prediction, and it is not a general disclaimer but a located one. The branch-point filter that sets what crosses into the expansion leg is, in its quantum reading, a Euclidean kernel whose action on a mode is controlled by an adiabaticity parameter C/μ_n, with μ_n the harmonic index of the layer's S^3 tower [32]. That parameter is small for all but the lowest harmonics and reaches order unity only at $n = 2$ and $n = 3$, where it takes the values 0.70 and 0.47. *Three independent routes place the filter's loss of control at the same scale*: the transmission calculation's boundary at $\ell \approx 2.5$, the projection's incompleteness below $k \sim 2 \times 10^{-4}\,\mathrm{Mpc}^{-1}$ ($\ell \approx 3$), and this breakdown at $n = 2, 3$.

The transmitted amplitude at low multipoles can be computed rather than estimated, and doing so settles what the estimates left open. The branch-point filter's action on a mode is, in the quantum reading, a Euclidean kernel [32]. The exponential-of-an-integral form $e^{-\int \omega\, \mathrm{d}\eta}$ is a WKB approximation whose adiabaticity parameter is of order unity at $\ell \lesssim 5$; but the underlying mode equation $\mathrm{d}^2\psi/\mathrm{d}\eta^2 = \omega^2\psi$, with $\omega = kc_s$ and $\mathrm{d}\eta = \mathrm{d}s/a$ along the segment, is well posed at every k, and integrating it is a matter of numerics rather than of approximation. Composing exact constant-ω transfer matrices in the stable direction over the segment ($|\Delta\eta| = 3.32$ in units $\alpha = 1$) gives exact/WKB ratios of 0.926, 0.913, 0.901, 0.891 and 0.889 at $\ell = 2, 3, 5, 15, 40$: *the WKB form is accurate to* 7–11% *at every multipole from* 2 *to* 40, with the exact transmission slightly smaller, so the filter is marginally stronger than the approximation gives.

Two readings suggested by the adiabaticity parameter alone are thereby excluded. The first is that the treatment is uncontrolled at $\ell = 2$–5: it is not, and the parameter was the wrong figure of merit, measuring as it does the fractional error in an exponent that is itself small at low ℓ. The second concerns the comparison with the observed sky. The correction is very nearly ℓ-independent—varying by 4% across a factor of twenty in ℓ—so between $\ell = 2$ and $\ell = 3$ it alters their ratio by 1.4%. *The structural expectation that a single floor starves* $\ell = 2$ *and* $\ell = 3$ *together is therefore not relieved by the exact treatment*, and the observed sky's selective suppression of $\ell = 2$ alone, by a factor of order three, remains unaccounted for. The low-multipole prediction of this section therefore stands as stated, and so does the discrepancy it faces.

Finally, the two Boltzmann depths above are not rival claims and neither supersedes the other: they differ by 10–25% and agree on the shape, the minimum and the recovery, and the difference is the size of the residual §9.1 declares. What both supersede is the older Sachs–Wolfe-analytic estimate (0.22 and 0.20 at $\ell = 2, 3$, the figure this section carried before the transfer was run), which took the Sachs–Wolfe source as the matter-domination value $\frac{1}{3}\Phi$, whereas at recombination $\rho_r/\rho_m = 0.317$ and the source is $0.4006\,\Psi$—a 20% error in the very ratio of Sachs–Wolfe to integrated Sachs–Wolfe that fixes how much the latter fills the bare deficit.

8 What the progenitor supplies: the transmission proof

The decomposition's decisive proof is here: *why* the substrate transmits the spectral shape rather than imprinting one of its own. It turns on the degeneracy of the branch point horizon.

Proposition 6 (Transmission dichotomy)**.** *For a mode approaching a horizon where the metric function behaves as $f \sim (r - r_h)^p$, the tortoise coordinate $r_* = \int dr/f$ controls the approach. At a non-degenerate horizon ($p = 1$, $f \sim 2\kappa(r - r_h)$, surface gravity $\kappa > 0$) the integral is logarithmic and the approach exponential, $(r - r_h) \sim e^{2\kappa r_*}$. At the degenerate Nariai double root ($p = 2$, $f \sim -\Lambda(r - r_N)^2$, $\kappa = 0$) the integral is $r_* \sim +1/[\Lambda(r - r_N)]$ and the approach power-law, $(r - r_N) \sim +1/(\Lambda r_*)$.*

Argument. The two tortoise integrals directly (receipts: `verify_geometry.py`, anchor 4[R]). □

Remark 1 (The leg does not tilt it either)*.* The proposition argues from the branch point: a degenerate horizon carries no scale, so it cannot imprint one. The collapse leg supplies the other half of the same conclusion, by an independent route. Horizon entry on that leg occurs at $x = k\eta/\sqrt{3} = 1/\sqrt{3}$ *for every* k—the radiation era's own scale invariance, which §4.1 recovers from the substrate geometry—so every mode leaves the leg carrying the same free-oscillation amplitude $\Psi(1/\sqrt{3})/2 = 0.4835\,\Psi_i$, a single k-independent number[R]. *So the leg multiplies the spectrum by a constant and the seam imprints nothing*: neither carries a scale, and a spectrum can only be tilted by something that does. The tilt observed is therefore the progenitor's, established now from both sides of the branch point rather than one.

Remark 2 (What the branch point does not fix). Both halves of the dichotomy are statements about the spectrum's *shape*: a degenerate horizon carries no scale, and neither does horizon entry on the leg, so neither can tilt what passes. *Neither is a pointwise condition on the perturbation state at the branch point*, and the distinction matters for anyone computing the handover. The single free function per mode is the freedom of the *spectrum*—the progenitor's amplitude and tilt—not a claim that the branch point fixes the state component by component. The state itself is whatever the leg's evolution produces, and §4.1 supplies it in closed form: the potential from the leg's own equation, the effective temperature oscillating freely from horizon entry, and the density contrast as their difference.

The Hamiltonian constraint is not an additional condition to impose there. In this framework it *is* the anchor—the map from a cut to its stress-energy, whose energy sector reads the Friedmann equation off the geometry [5, 6]. It is a definition, not a restriction: on the leftward reading the rate is set by the substrate and the content is read from it, so the perturbed constraint likewise *defines* the density contrast a given geometry carries rather than restricting which geometries are allowed. Imposing it at the branch point alongside the leg's own solution therefore over-determines the handover, and the same geometric state answering to different stress-energy on the two sides is not a failure of matching but the branch point doing what the level assignment says it does.

Remark 3 (What the branch point's degeneracy does not settle). It is tempting to read Proposition 6 as settling the phase as well as the shape: the tortoise coordinate diverges at the degenerate root, so a mode's accumulated phase would be unbounded and nothing finite could be inherited. *That reading does not survive its own arithmetic.* The divergence is geometric and carries no k, so an unbounded kr_* is a divergent term *linear* in k; a linear phase renormalizes the sound horizon rather than shifting the comb, and an unbounded one would send the comb spacing to zero and erase the acoustic peaks altogether—excluded by observation, not supported by it. And the modes never meet the divergence in any case: they cross the branch point at finite conformal time, whereas r_* diverges along a static slicing's approach to the horizon, which is not the collapse trajectory.

What the leg does contribute is finite and, as it happens, invisible in the peak positions. Horizon entry at $x = 1/\sqrt{3}$ for every k makes the accumulated leg phase $k\eta_{\rm bp}/\sqrt{3} - 1/\sqrt{3}$: linear in k with a mode-independent offset. So carrying it across rescales the effective sound horizon by $1 + \eta_{\rm bp}/(\sqrt{3}\, r_s)$ and adds a constant, and changes the comb in no other way. *The comb therefore cannot distinguish a transmitted phase from a reset one*, because a rescaling of r_s is degenerate with the onset redshift that §4.3 fixes from the acoustic angle. Any discriminant must be a quantity that does not take the same factor—the diffusion scale is one, since r_D does not—and that is the discriminant of principle. *The comparison is now made, and it does not require r_D.* The leg's accumulated conformal time is fixed by the geometry rather than fitted: the collapse branch $r = -(2M\alpha^2)^{1/3} \cosh^{2/3}(3\,\mathrm{Re}\,\tilde{\tau}/2\alpha)$ has cosh ≥ 1, so $|r| \geq (2M\alpha^2)^{1/3} = 3766.6$ Mpc and the branch point is reached only across the lift, along which $\mathrm{Re}\,\tilde{\tau}$ does not advance and no conformal time accumulates. Hence $\eta_{\rm bp} \approx 5 \times 10^3$–$10^4$ Mpc, and $r_s^{\rm eff} = r_s + \eta_{\rm bp}/\sqrt{3}$ is *dominated* by the leg term—3.0–5.9 $\times\, 10^3$ Mpc against $r_s = 144$ Mpc. The degeneracy with the onset redshift, which absorbs a small rescaling, cannot absorb this one: reproducing the measured $\ell_A = 301.8$ would require $D \approx 2.9$–5.7×10^5 Mpc against the actual 1.4×10^4 Mpc, a factor of 21–41. *The acoustic phase is therefore reset at the branch point and not carried across it*—established on the peak positions alone[R]. The result is robust to the open account of the lift: any conformal contribution the lift is found to carry only adds to an already-excluded total.

The amplitude separates from the tilt in the same statement. What the progenitor supplies is the potential power P_Ψ; what the construction contributes is the factor $(0.4835)^2 = 0.2338$ carrying it to the observed A_s. That factor is computed rather than fitted, so the accommodation is narrower than it was—the inherited quantity is P_Ψ and the map from it is derived—but it is not removed, and the framework paper's scoping of this as a one-parameter accommodation

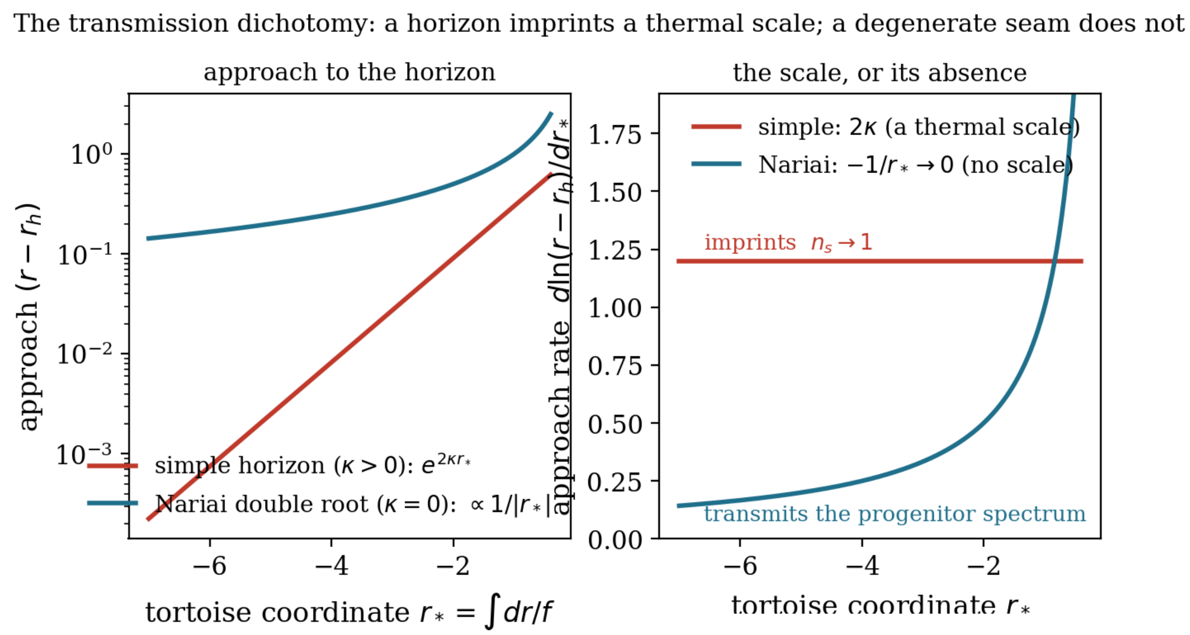

Figure 5: The transmission dichotomy (Proposition 6). *Left*: the approach to the horizon in the tortoise coordinate $r_* = \int dr/f$, on a logarithmic vertical axis. At a non-degenerate horizon ($\kappa > 0$) the approach is exponential, $(r - r_h) \sim e^{2\kappa r_*}$ (a straight line); at the Nariai double root ($\kappa = 0$) it is power-law, $(r - r_N) \sim 1/|r_*|$. *Right*: the logarithmic approach rate $d\ln(r - r_h)/dr_*$ *is* the local scale. The non-degenerate horizon carries a fixed thermal scale 2κ—the mechanism by which a de Sitter horizon imprints a scale-invariant spectrum ($n_s \to 1$); the branch point's rate falls to zero, carrying *no* scale, so it transmits the progenitor spectrum unaltered.

stands unchanged [7].

The physical reading is the result (Fig. 5). The exponential, non-degenerate approach is the thermal (Hawking) law—the mechanism by which a de Sitter horizon *imprints* a scale-invariant spectrum ($n_s \to 1$): the exponential redshift erases the infaller's spectral information and stamps the horizon's own thermal scale. The power-law, degenerate approach carries no thermal scale; it is scale-free, and the infalling spectrum is transmitted faithfully. CR's transmission locus is the branch point $r = 0$, where f *diverges* as $-2M/r$ so the tortoise coordinate is *finite*—the crossing accumulates no divergent phase and stamps no scale, a third case beyond the dichotomy's two[R]. The Nariai double root (Proposition 4, $f''(r_N) = -2\Lambda$, $\kappa = 0$) is the *equatorial seam*—one point of the substrate, which the bead meets on the way in and again one full lap later, the two chart values $r = -2\alpha/\sqrt{3}$ and $r = +\alpha/\sqrt{3}$ being the same φ modulo 2π in the phase $\varphi = 2\pi r/\sqrt{3}\alpha$. The branch point sits two thirds of the lap in from it (240°, with 120° remaining), so the crossing at $r = 0$ and the seam are distinct loci on the lap without the seam being an endpoint of anything. Hence:

Proposition 7 (The branch point transmits)**.** *The cosmogenesis branch point is a faithful scale-free transmitter—not, as earlier stated here, because it* is *the degenerate Nariai horizon (that is the front seam, $r = +\alpha/\sqrt{3}$, the lap's other endpoint), but because f there* diverges *as $-2M/r$, so r_* is finite and the crossing accumulates no divergent phase at all (r2154): the amplitude A_s and tilt n_s are the progenitor collapse's spectrum carried across unaltered, not imprinted by the branch point. Equivalently, CR has no inflationary scale-invariant attractor, because the non-degenerate exponential approach that would drive $n_s \to 1$ is exactly the approach the Nariai seam does not have.*

The degeneracy is thus the proven reason CR carries the primordial tilt rather than manufacturing it. What the transmitter carries is, from the Schwarzschild–de Sitter side, open: n_s

and A_s are inherited, fixed by the progenitor collapse—the same handover that fixes ρ_r/ρ_m and the light-element abundances—and not derived here. *Naming them together is convenient and has concealed that they are not the same kind of quantity*$^{\mathsf{R}}$. The vacuum normalisation is the only place a quantum constant enters the spectrum, so $P(k) = \hbar A_0 k^{n_s-1}$ and $\hbar$ is an overall multiplicative factor: *it survives in the amplitude and cancels in every logarithmic derivative.* An amplitude is the spectrum at one wavenumber; a tilt is its ratio to itself at two. *So the construction's inability to force a dimensionless magnitude bears on A_s and not on n_s*—the first is permanently inherited, the second need not be. *And read backwards the same statement is useful rather than limiting*: a quantity fixed by the moduli rather than the invariants is one the observation *measures*, so the observed A_s determines the progenitor's mass, to be checked against the independent requirement that it lie below the Nariai value. *And it is worth saying precisely what that openness is, because a count settles its kind.*$^{\mathsf{R}}$ The progenitor family described here is parametrised by M alone—α is the one invariant and $E = 1$ a choice of geodesic—so the geometry is one-dimensional; and the Kretschmann along the bead is mass-free in the faller's own proper time, so that one parameter does not survive the approach either. *A quantity identical for every member carries no information about which member it was*, and the collapse geometry therefore transmits not one parameter but none. Against that stand at least four independent inherited numbers. *So 'inherited from the progenitor collapse' cannot mean inherited from the collapse geometry*: what is inherited is inherited from the progenitor's matter content—its composition, its entropy per baryon, its perturbation spectrum—and none of that is carried by M, the exterior here being exactly vacuum and the marginally-bound interior dust without composition. *The frontier is accordingly not a derivation awaiting a cleverer argument but a modelling task awaiting a progenitor interior*, and what the geometry supplies in its place is a cutoff rather than a content: the segment's conformal length $|\Delta\eta| = 3.32\alpha$ is fixed by Λ alone *on the Nariai member*—below it $|\Delta\eta|$ runs as $M^{-1/3}$, reaching 4.19α at $M/2$ and 15.4α at $M/100$—and says which modes survive without saying how large they were.

9 Predictions, and the progenitor boundary

The signatures, with their maturity marked:

1. Classical, non-vacuum primordial statistics; no inflationary consistency relation; no substrate-sourced primordial B-modes (established, §5).

2. The transmission character: the branch point carries the progenitor tilt and does not drive $n_s \to 1$ by any *inflationary* attractor (established, §8). *The unqualified form of this clause—that the construction has no scale-invariant attractor at all—is withdrawn, and what replaces it is stronger rather than weaker.*$^{\mathsf{R}}$ Scale invariance for a background $a \propto (-\eta)^\beta$ with $z \propto a$ requires $\beta(\beta-1) = 2$, a quadratic with *two* roots: $\beta = -1$, de Sitter inflation, and $\beta = +2$, a matter-dominated contraction. *The collapse leg of this construction is the second root exactly*: near the branch point $f \to -2M/r$, so the leg is pressureless there and the banked law $r^3 = \frac{9}{2}M\tau^2$ with $a = |r|$ gives $a \propto \tau^{2/3} \propto \eta^2$; the mode equation integrated from vacuum data returns a flat spectrum to a part in 10^4. *The mode-selection question that shadows this in the literature was accordingly put to the construction directly, and it answers in the negative.*$^{\mathsf{R}}$ The scale-invariant spectrum sits in the mode that *dominates* the contraction, whose curvature perturbation goes as η^{-3}; the mode that survives on the expanding side is the constant one, and from vacuum data its spectrum is $n_s - 1 = 6$. Whether the first reaches the second is a connection problem at the branch point, and here it is settled rather than assumed: the exponents 2 and -1 differ by an integer, so a logarithm could mix them, but the $\sinh^{2/3}$ law is *even* in τ, which in conformal time places its correction at η^6 and steps the Frobenius recursion in sixes past a resonance sitting at three. *No logarithm is generated, the continuation*

acts diagonally, and the scale-invariant spectrum does not reach the observer—confirmed independently by continuing the dominant solution numerically around the branch point, where the admixture it produces is what the claim turns on, where the admixture falls as $\delta^{2/3}$ and the diagonal entry goes to unity. *What survives is therefore narrower than it first appeared and sharper than what it replaced*: the construction *has* a scale-invariant background, the spectrum that background freezes is confined to the contracting side, and the obstruction is a parity property of the construction's own solution rather than a missing mechanism. *Mixing would require a background correction odd in τ, and that question is settled here too, in the negative.*[R] Writing the law as $r = Au^{2/3}g(u)$ with $g = [\sinh u/u]^{2/3}$ places every branched factor in $u^{2/3}$ and leaves g even, analytic and non-vanishing at the origin—exactly, since $\log(\sinh u/u)$ is analytic in u^2 there and the exponential of a series in u^2 is one. The three sheets differ by a constant cube root of unity, and a constant multiplier cannot change a parity; the conjugate branch is even as well, and carries no branch point at all. *So the evenness is unavoidable, and what it says is not a technical fact but a symmetry*: $r(-\tau) = r(\tau)$, the cosmogenesis is time-symmetric about the branch point, and a time-symmetric background cannot mix a growing mode with a decaying one, since mixing would distinguish a direction of time the background does not. *The same invariance is already load-bearing one observable over*—the peak heights of §5 follow from a driving amplitude exactly invariant under time reversal, which is why they may be evaluated on the contracting side. *One symmetry, two consequences. And it locates rather than forecloses what would be needed*: this leg is exactly even because it is the marginally-bound geodesic of an exactly vacuum exterior, so a progenitor carrying real matter content—pressure, dissipation, anything with an arrow—would not be, which returns the question to what the progenitor was.

3. Flat distances ($\Omega_k \simeq 0$) coexisting with a closed-S^3 low-multipole *deficit* below $\ell \approx 8$, set parameter-free by Λ—mild, and on a genuine Boltzmann transfer a dip bottoming at $\ell = 4$ (≈ 0.47, 0.41, 0.36, 0.68 of the flat-ΛCDM expectation at $\ell = 2, 3, 4, 5$), its shape cross-validated between two independent transfers (the flat slice and the flat-projection transfer established, the depth settled by the exact transfer to within their spread). Confronted with data it is non-discriminating: it starves $\ell = 2$ and $\ell = 3$ together, matching neither the sharp observed quadrupole-only dip nor the near-ΛCDM octopole, and sits consistent with ΛCDM within the lowest multipoles' cosmic variance (§7).

4. Coherent acoustic peaks from geometric (null-boundary) phase-fixing rather than super-horizon freeze-out (argued, §4).

5. The acoustic scale is resolved in the cosmological development above (§2.3), and the radiation-free rate (5) is the standing CR-versus-ΛCDM discriminator on data in hand—a strength established before the spectrum, not this section's to re-assert.

The boundary is principled. The substrate carries everything the branch point's null geometry lets it carry—coherence, isotropization, the discreteness floor, the classical character, the faithful transmission—and the degeneracy proves precisely where it stops: the spectrum's amplitude and tilt are not the substrate's to set. This is the shape inflation has when it parametrizes its observables through an unknown inflaton potential; CR routes them through an unknown progenitor collapse, with the arrival constrained—whatever the collapse supplies must reach the branch point coherent, isotropized, scale-freely transmitted, and classical.

The symmetry between the two unknown suppliers is only apparent, and the asymmetry is the substance. Inflation is not structure the standard cosmology already contains; it is an added sector—a scalar field with a freely specified potential—introduced to produce features that cosmology was known to lack: the causal contact the horizon problem denies, the flatness the dynamics do not single out, and, once these were in view, a near-scale-invariant spectrum.

The potential is then chosen to deliver the observed amplitude and tilt. A model built to be tunable, and tuned to a target known in advance to require tuning, earns precisely what such a model can earn—it establishes that the features are *possible*, that an apparatus permitting them exists. It does not establish that they were *required*, and the routine elevation of the spectral fit into a vindication of inflation conflates the two: that a freely chosen potential can be set to match n_s and A_s is not evidence that inflation occurred, but the statement that a model with enough adjustable structure to fit has been fit. This is the distinction between a world that *requires* a phenomenon and one that merely *permits* it through adjustable parameters—the same distinction the programme draws in reading the cosmic foliation as forced rather than chosen [9]—and it places the inflationary spectrum on the permitting side.

The present cosmology adds no such apparatus. Causal contact, isotropy, flatness, phase coherence, and scale-free transmission are not produced by a field introduced to produce them; they are read off structure the construction already contains—the de Sitter throat that isotropizes by the no-hair theorem (§6), the null boundary on which coherence *is* the characteristic-data condition (§4), the exactly Euclidean constant-time slice (§2.3), and the degenerate horizon whose vanishing surface gravity transmits rather than imprints (§8)—each fixed by Λ alone, with no parameter free to do the producing. The one quantity it does not derive, the amplitude and tilt, it inherits in exactly the sense flat ΛCDM inherits the baryon-to-photon ratio; on that single datum the two frameworks stand level, and on everything the inflationary sector is invoked to supply they do not. The verdict this licenses is not symmetric, and it does not await a future measurement. On the data in hand the two align; on the structure each must assume in order to align, one introduces an entire tunable sector and tunes it while the other introduces none—and makes, in its place, falsifiable commitments where inflation keeps its freedom: no scale-invariant attractor *reaching the observer*, no consistency relation, no substrate-sourced B-modes (§5, §8)$^{\mathsf{R}}$. By the economy of assumption that distinguishes a required structure from a permitted one, that comparison is settled now, by what each theory must presuppose, and it is settled in this cosmology's favour. We hold this at its weight and mark its bound in the same breath: the parsimony verdict is earned, and the data verdict has now begun to be earned with it—the Hubble tension resolved across the distance ladder where ΛCDM cannot reconcile the direct H_0 (§2.4), the abundances confirmed within 1σ [16]—so the cosmology is empirically favoured, not merely more economical; the spectrum's content remains inherited, and the large-angle shape remains the exposed edge that could still overturn it.

9.1 The acoustic instrument: what it is validated against, and to what accuracy

The low-multipole prediction above and the ratio statements below are carried by a Boltzmann transfer built for this programme, and its standing should be stated plainly rather than assumed. It is a full photon hierarchy with polarisation, second-order tight coupling, massless neutrinos, and a Peebles recombination history; the line-of-sight source carries the monopole with the potential, the Doppler term, the integrated Sachs–Wolfe term, and the quadrupole with its own projection kernel $(j_\ell + 3j_\ell'')$.

Against a like-for-like reference—an independent Boltzmann code run at identical parameters, unlensed and with reionisation off—it reproduces the *acoustic peak and trough positions to* $\leq 0.5\%$ across P_1–P_4, the *ionisation history* $x_e(z)$ *and its derivative to* $\sim 1\%$ through the visibility peak, matter–radiation equality to 0.02%, and the *transfer function to better than* 1% *for* $k < 0.02\,\mathrm{Mpc}^{-1}$. Above that wavenumber it agrees to 1–2%, with the residual scatter in both directions and no trend across the range$^{\mathsf{R}}$.

Two construction details are worth stating, because both are easy to omit and both have signatures that read as physics.

The first is the reach of the mode ladder. The angular spectrum at multipole ℓ is an integral over wavenumber, and the projection kernel $j_\ell(k(\eta_0 - \eta))$ has substantial support *above* $k =$

ℓ/D_M, not merely at it; a ladder cut at $k_{\max}$ therefore truncates the integral, and does so more severely as ℓ rises toward $k_{\max}D_M$. Quantitatively, truncating an exact transfer at $k_{\max}D_M = 900$ retains 99.2% of C_ℓ at $\ell = 200$ but only 61% at $\ell = 800$ and nothing at all at $\ell = 1150$; carrying the ladder to $k_{\max} \geq 2\,\ell_{\max}/D_M$ recovers every multipole to better than half a per cent. *The tell is unmistakable once known*: the highest peak simply fails to appear, so a spectrum that should show four acoustic peaks shows three, and the peaks that do appear are displaced downward by an amount that grows with ℓ. *Increasing the ladder's density at fixed reach does not help*—the loss is the range, not the sampling, which is worth checking explicitly because the two are easily confused.

The second is the seeding of the conformal time. It is a cumulative integral, $\eta(a) = \int_0^a \mathrm{d}a'/(a'^2 H)$, and a numerical grid necessarily begins at some finite $a_{\min}$; the interval already elapsed below that point must be carried as a seed rather than set to zero. In radiation domination $H \propto a^{-2}$ makes the integrand constant, so the omitted piece is $a_{\min}/(H_0\sqrt{\Omega_r})$ exactly—for $a_{\min} = 10^{-5}$ it is 4.6 Mpc. *It is a fixed offset at every epoch*, and therefore 0.03% of η_0 but 12% of η at $z = 10^4$: the acoustic scale, the peak positions and the peak ratios are insensitive to it, while any early-time quantity or any comparison against a second code is not. Its signature is instructive—because the potential decays more steeply at larger k, a fixed error in the time coordinate reads out as a *smooth, monotone, k-growing* error in amplitude, which is the shape one would otherwise attribute to a physical suppression. The grid must also extend below the intended starting redshift, so that the initial epoch is interpolated rather than clamped.

That residual does not enter the claims made here. The Cosmological Relativity and flat-ΛCDM spectra are computed through the *same* machinery, so an instrumental transfer error afflicts both arms identically and cancels in their ratio—and every prediction stated in this paper is a ratio. The low-multipole deficit in particular lives at $\ell \lesssim 8$, i.e. $k \ll 0.02\,\mathrm{Mpc}^{-1}$, where the two agree to better than 1%. What the residual does forbid is an *absolute* amplitude claim: this instrument should not be read as reproducing the observed small-angle power to better than a few per cent, and no such claim is made.

10 The open frontier: the pieces this cosmology is built to close

The result stands on what the substrate determines. What it does *not* yet claim has been carried to a sharp, gradable edge rather than left vague—and those edges are the most interesting part of the programme from here, each a gap we are eager to grind shut, not paper over. We hold them to the criterion of necessity the programme's epistemology names—that a structure is credited for *requiring* a phenomenon, not for *permitting* a value that fits it [10]—and sort them by it: a gap that is *buildable* is a debt owed, to be built before the claim it bears on is a proof rather than a coherent proposition, while a genuinely external unknown may stay open at no cost, exactly as flat ΛCDM leaves its own inherited data open. The buildable ones below are named as debts, not softened into frontiers; the end-to-end transfer and the full-spectrum likelihood are the load-bearing debts, the derivation of the progenitor spectrum the genuine frontier. Four stand out:

1. the sufficiency of the null-boundary characteristic data for the observed acoustic coherence (§4), argued from the structure of the characteristic initial-value problem, with the peak-height *magnitude* carried by the time-reversal driving equality established there—what remains is the end-to-end branch-point-to-recombination transfer, the calculation that would carry the single-phase datum all the way through, confirm the height pattern digit by digit, and—within it—settle whether the ∼8% diffusion-scale signature (§4) shows in the observed high-ℓ power as a tension, a wash, or a distinctive-but-consistent feature. One general consideration already bears on which: a shift in the diffusion *scale* is, across the observable multipole range, substantially degenerate with the spectral tilt, so the scale signature's leading spectral effect is largely absorbable into n_s—disfavouring the

tension reading and leaving a wash or a distinctive-but-consistent feature, with the tilt-irreducible residual the part the transfer would isolate. This is a genuine build, not a plug-in: it requires first *specifying how the fluctuations gravitate on the radiation-free background*—the piece that sets the high-ℓ driving envelope, and which the standard Boltzmann codes cannot supply because they tie radiation's gravity to its presence and so cannot represent the content-not-rate split the layered rate rests on—and then a bespoke transfer against that specification. It is a debt owed and named as such: not a missing idea but a computation this sequence owes and has not yet run, and the paper's own open edge rather than another's. Two things it bundles are worth holding apart — the *build* is ours, and the *verdict* it would expose is the world's: the transfer is what makes the $\sim$8% signature confrontable at all, and the data then decide it;

2. the exact large-angle *depth* (§7): the flat-projection transfer of the discrete closed-S^3 source is built with a *genuine Boltzmann transfer*—the exact CMB temperature transfer $\Delta_\ell(k)$ (Sachs–Wolfe, early and late integrated Sachs–Wolfe, and Doppler, all common to CR and flat-ΛCDM) sampled at the discrete modes k_L, gate-validated (the continuum reproduces the reference C_ℓ to four figures) with the r_0 sensitivity measured rather than asserted—shape stable, depths drifting up to 15% at $\ell = 4$ (receipt: `verify_lowell_boltzmann.py`[R]). It gives a *mild* deficit, $\approx$ 0.47, 0.41, 0.36 and 0.68 of the ΛCDM expectation at $\ell = 2, 3, 4, 5$—*a dip whose minimum falls at $\ell = 4$, not at the quadrupole*, the shape reproduced by the programme's own photon hierarchy at 10–25% greater depth—recovering by $\ell \approx 7$: the late integrated Sachs–Wolfe term is sourced above the floor and is therefore retained by the discrete source, leaving the deficit shallower than the bare Sachs–Wolfe starvation. The mechanism is robust—parameter-free, located at $\ell \lesssim 8$, bottoming at $\ell = 4$, with $\ell = 2$ and $\ell = 3$ starved *together* (the exact discrete measure and the Doppler term each leave the $\ell = 2$-to-$\ell = 3$ ratio near unity: receipts `verify_lowell_exact_measure.py`[R], `verify_doppler_lowell.py`[R]). Confronted with the sky the sector is a *wash*: CR's smooth $\sim$ 0.4–0.5 deficit sits above the sharp observed quadrupole ($\sim$ 0.2) and below the near-ΛCDM octopole ($\sim$ 0.6–1.0), matching neither, and the exact cosmic-variance likelihood returns $\Delta(-2\ln L) \approx +1.8$ over $2 \le \ell \le 10$ (range +0.3 to +3.8 across the octopole estimator), a small residual inside cosmic variance (receipt: `verify_lowell_likelihood_v2.py`[R]). So the low multipoles are a genuine but *non-discriminating* sector, neither a parameter-free correspondence success nor a sharp falsification edge; the deficit's existence and location stand as a real parameter-free feature, but its depth places it consistent with ΛCDM within the lowest multipoles' cosmic variance. The honest exposed edges lie elsewhere—the damping tail and the full-spectrum likelihood below;

3. the progenitor spectrum itself—n_s and A_s (§8)—inherited boundary data fixed by the progenitor collapse, the baryogenesis-analogue computation belonging to the matter sector [5, 6], with the honest target a consistency between the inherited composition and spectrum and the measured primordial abundances and tilt, not a derivation from the geometry alone—the most ambitious reach of the three, the one that would close the decomposition from both ends and give CR a derivation where the standard model keeps a measured datum;

4. the *likelihood-level model selection*: the parameter-free ratio comparison above is joined by the exact low-multipole cosmic-variance likelihood—each $\hat{C}_\ell$ a scaled $\chi^2_{2\ell+1}$ about the model C_ℓ, the absolute baseline cancelling in the CR/ΛCDM ratio (`verify_lowell_likelihood_v2.py`[R]), the proper instrument where cosmic variance dominates. With the genuine-Boltzmann depths ($\approx$ 0.47/0.41 at $\ell = 2/3$, above) the sector is a *wash*: the low quadrupole still mildly rewards CR ($\Delta(-2\ln L) \approx -2.6$, its milder dip nearer the observed low power than ΛCDM's), the octopole and $\ell = 4$ mildly penalise it ($\approx$ +1.3 and +2.1)—*so the*

sector's largest single-multipole penalty sits not at the octopole but at $\ell = 4$, the prediction's own minimum—and the net over $2 \le \ell \le 10$ is $\Delta(-2\ln L) \approx +1.8$ (range $+0.3$ to $+3.8$ across the octopole estimator)—a small residual well inside cosmic variance. A Monte-Carlo over 10^4 skies puts the sector's *in-principle* discriminating power at only $\sim 3.4\sigma$ (the short multipole lever arm and cosmic variance), and the current measurement uncertainty swamps even that. The low multipoles are therefore a genuine but *non-discriminating* test: CR and ΛCDM are statistically indistinguishable there, the mild deficit neither favouring CR nor refuting it. Decisiveness comes from outside the sector—the radiation-free expansion rate (5), now confronted with the distance ladder (§2.4), as the standing discriminator, and the likelihood across the *full* spectrum (its amplitude level carried by the time-reversal driving equality, §4; the full-spectrum likelihood itself, with the oscillating-medium transfer downstream in the matter sector [5, 6], the larger piece still to build).

A frontier of a different kind concerns the matter itself rather than the perturbation observables above: the *dynamics of matter crossing the branch point.* Three facts about that crossing follow from the structure already established here. It is *well posed*—the beginning is fixed at the finite-curvature branch point $r = 0$, so the bend $\rho = m'/4\pi r^2$ that carries the matter is finite there and its data on that null surface is characteristic, not Cauchy: a finite-curvature characteristic crossing with no curvature obstruction. It *isotropizes and is scale-free*—the throat's de Sitter no-hair (§6) damps the anisotropic part of the crossing stress-energy to the isotropic monopole, and the degenerate approach (§8) imprints no thermal scale: the matter counterparts of the field no-hair and faithful transmission established above. *Where each of those acts is worth stating, since one of them cannot act at the crossing at all.* Damping is a process and requires elapsed time, and the crossing has none: the real part of the complex cosmic time is frozen along the segment reaching it [32]. The no-hair damping therefore belongs to the *approach*—the throat region, over the collapse leg's own cosmic time—while the crossing carries whatever survives it unchanged, not because a mechanism protects it but because there is no interval in which anything could act. *The isotropy is earned on the approach and preserved by the crossing*, and the two halves are of different kinds: the first is a damping calculation with a rate, a timescale and a residual, the second follows from the zero-duration structure and takes no model input. It follows that *how much anisotropy survives is fixed by the approach against the leg's duration*—a property of the progenitor's collapse history rather than of the crossing. And it obeys a *structural transition law*: because the reassignment preserves the cosmic foliation and the density is leaf-carried and lapse-independent [5], the reassignment acts on the time-stacking—fixing the Λ-set rate (5)—and not on the leaf, so the isotropic density crosses as inherited progenitor content, which is the structural reason the inherited datum of §2.4 is inherited rather than produced; *and two conditions must be met at that crossing, which a concrete model has to satisfy separately: the reassignment requires a degenerating Killing field whose freed null direction can be re-founded as a clock, the joining of the legs requires a branch point, and these are independent—the forced Nariai member carries the degeneracy and is not a branch point, while $r = 0$ is the branch point and carries no Killing degeneracy [8], so a model furnishing one without the other furnishes no cosmogenesis The first of the two is settled as a uniqueness, not merely an availability*: across the slicing family the reassignment that promotes a null direction to the fundamental congruence is *uniquely* the one selecting the forced Nariai member, every other vantage yielding a congruence that crosses the embedding's symmetry structure transversally and so reading as a localised black hole [4]. *A model therefore has no freedom in where the reassignment acts*, and misplacing it does not fail to give a cosmology by a little—it gives a black hole instead.; the scoping of the two rates this law forces—the excursion-side local readout against the observable radiation-free rate, their boundary—is drawn out in the cosmogenesis paper [16]. What remains open is the *detailed* worldline and field dynamics of the crossing for a concrete matter model [7, 5]—the deferred depth, now sitting on a crossing shown well posed rather than merely deferred.

None of these unsettles the result obtained: the branch point's null geometry decomposes the primordial scalar spectrum into a substrate-determined structure and a progenitor-supplied content, and the degeneracy proves that the branch point transmits that content rather than imprinting one of its own *on this argument*; a computed transfer function across the branch point is not yet in hand. They are, rather, the map of where this cosmology goes next—the end-to-end transfer behind the peak *heights* and the full oscillating medium downstream in the matter sector, the tensor sector's companion in the dynamics paper [8].

11 Discussion

What this paper determines is one thing seen from several sides. A single de Sitter geometry, its scale set by the cosmological constant alone (the maximally symmetric substrate's sole scale [15]) and read through a causal reassignment at the cosmogenesis branch point, *is* the observed cosmology: the flat-ΛCDM expansion history is its areal radius (§2.3); the apparent Hubble tension and the acoustic scale are consequences of its radiation-free rate (§2.4); the primordial scalar spectrum is the characteristic data of its seam, structure fixed by the substrate and content inherited across the degenerate horizon (§§4–8). The Einstein field equations are unchanged; what changes is the reading, and under it the quantities the standard model carries as independent inputs—the present matter density, the dark-energy fraction—are readings of one cosmic clock, not drivers of the expansion, so that the coincidence of Ω_m and Ω_Λ dissolves into our observing at a time of order the geometry's one timescale. There is a single seed, and the observed universe is what grows from it. And the one scale reaches down as well as out: the same cosmological constant that sets this expansion sets, for every mass M, the *Hubble–Eddington radius* [31] $r_{\rm HE} = (3GM/\Lambda c^2)^{1/3}$—the boundary within which local structure stays bound against the cosmic flow and beyond which it is carried off. The slicing geometry reads that boundary as the flat locus of the existent slice's own curvature, the local bend of the cut exactly cancelling the substrate's cosmological term [3], and the slicing operator reads it as the handover from sub-marginal bound orbits to the marginally-bound $E = 1$ congruence that *is* this cosmology [5]. So the substrate's single Λ governs both the global expansion this paper computes and the local scale at which structure resists it—one constant read at two ranges—and the Hubble–Eddington radius, proposed as a local, per-structure test of Λ from the largest bound structures [11], is an independent observational handle on the very constant the expansion history here measures. The two ranges are moreover the same length up to a pure number. The Hubble–Eddington radius—where the slicing surface's Gaussian curvature $K_G = 1/\alpha^2 - M/r^3$ vanishes, the local bend cancelling the substrate's cosmological curvature [3]—is $r_{\rm HE} = (M\alpha^2)^{1/3}$; while the expansion's own amplitude, $(2M\alpha^2)^{1/3}$, is exactly the areal radius at matter–Λ equality ($\rho_m/\rho_\Lambda = \operatorname{csch}^2(3\tilde{\tau}/2\alpha)$, unity at sinh = 1). The two therefore stand in the fixed ratio $2^{1/3}$ for every M: the scale at which a bound structure's gravity balances the expansion and the scale at which the expansion's matter balances its Λ are one length read locally and cosmologically, the factor of two that of the Schwarzschild term against the surface curvature's M/r^3 and nothing else. R

The comparison with the standard cosmological model is accordingly not *only* a contest of fits—though on the Hubble tension the geometric rate now fits the joint acoustic-and-distance-ladder data at the direct H_0 where ΛCDM cannot (§2.4)—but also, and more deeply, a difference in what each must assume to do so. The standard model *assembles* the observed universe: a dark-energy component for the late-time acceleration, *And that component is the sharpest case of the difference*: on this cosmology the acceleration is not a sector at all. Since $(\mathrm{d}r/\mathrm{d}\tilde{\tau})^2 = E^2 - f$ on the energy family, one has $\mathrm{d}^2r/\mathrm{d}\tilde{\tau}^2 = -f'/2 = rK_G$: *the comoving acceleration is the existent slice's own Gaussian curvature*, up to the positive factor r.R So the expansion decelerates while the slice is negatively curved, turns over exactly where it is flat—the Hubble–Eddington radius,

$\rho_m = 2\rho_\Lambda$, at $\sinh(3c\tilde{\tau}/2\alpha) = 1/\sqrt{2}$—and accelerates thereafter, with nothing added to make it do so. The late-time acceleration is read off the same structure that fixes the local boundary of a bound system, an inflationary sector for the causal contact, flatness, coherence, and near-scale-invariance the background dynamics do not themselves supply, and further early-universe physics where the sound horizon and the expansion rate come into tension. Each part is a real solution to a real problem, and each was introduced to yield the feature it produces. This cosmology adds no such part. Causal contact, isotropy, flatness, phase coherence, and scale-free transmission are read off structure the construction already contains—the isotropizing throat (§6), the null boundary on which coherence *is* the characteristic-data condition (§4), the exactly Euclidean constant-time slice (§7), the degenerate horizon that transmits rather than imprints (§8)—each fixed by Λ alone, with no parameter free to produce it. This is the distinction between a structure that *requires* a phenomenon and one that merely *permits* it through adjustable apparatus [10, 9], and it is the criterion by which competing frameworks are rightly weighed ahead of the measurement that will decide between them.

The transmission dichotomy (§8) is where that distinction is sharpest, because there the two frameworks make opposite structural commitments about one object. A non-degenerate horizon's exponential approach imprints a scale-invariant spectrum—the very mechanism by which inflation produces near-scale-invariance—while the degenerate Nariai seam carries no scale and transmits the progenitor spectrum unaltered. Inflation accommodates the observed tilt because a freely specified potential can be tuned to it; reading that fit as a confirmation mistakes a model with enough adjustable structure to match for one that required what was found (§9). This cosmology cannot tune: the degeneracy *requires* that the branch point not imprint a scale, so the framework has no inflationary scale-invariant attractor, no consistency relation, and no substrate-sourced primordial B-modes. The amplitude and tilt it does not derive it inherits, exactly as flat ΛCDM inherits the baryon-to-photon ratio from a baryogenesis it does not model; on that single datum, of the same kind and count the standard model already carries, the two stand level. A framework is credited for requiring the structure it explains, not faulted for measuring the boundary data it does not—and the asymmetry falls on everything else.

The verdict this licenses must be drawn on the right axis, and confined to it. On the axis of theory-choice—the requiring rather than the permitting of the phenomena, the consolidation of many results under one structure, the absence of a device added per problem—the comparison is decidable now, from the structural record, and it falls to this cosmology [10]. On the axis of data-discrimination the picture has begun to move with it: the radiation-free rate resolves the Hubble tension across the acoustic scale and the baryon-acoustic distance ladder together, at the directly measured H_0 where the standard model cannot (§2.4)—a discriminating datum, not a tie—while on the light elements, the microwave-background peak heights, and the low multipoles the two frameworks are even. So the standing is stronger than rule-favoured-awaiting-the-datum: it is a framework favoured on the structure *and* carrying its first discriminating data result, with the fuller confirmation—the $\sim$8% damping tail and the full-spectrum likelihood—still owed. A theory that fits is not a theory confirmed, and that confirmation is not yet in hand; but the data axis is no longer even, and the framework has staked falsifiable commitments—no scale-invariant attractor *reaching the observer*, no consistency relation, no substrate B-modes—precisely where the standard model keeps its freedom[R].

The low-multipole deficit is a mild, non-discriminating feature (§§7, 10). The flat projection of the discrete closed-S^3 source gives a parameter-free deficit below $\ell \approx 8$—real, and set by Λ through r_0 and D_C—bottoming at $\ell = 4$, at ≈ 0.47, 0.41, 0.36 and 0.68 of the ΛCDM expectation at $\ell = 2, 3, 4, 5$. At that depth CR's smooth deficit matches neither the sharp observed quadrupole-only dip nor the near-ΛCDM octopole, and under the exact cosmic-variance likelihood the two frameworks are statistically indistinguishable at low ℓ ($\Delta(-2\ln L) \approx +1.8$ over $2 \leq \ell \leq 10$, inside cosmic variance). So the sector neither confirms nor refutes: it is a genuine parameter-free prediction whose depth places it consistent with ΛCDM within the lowest multipoles' cosmic variance. The framework's exposed edge is the ∼8% damping-*scale* signature locked to the Hubble resolution (§4), a computed, non-reabsorbable CR-specific effect whose consequence for the observed high-ℓ power is entangled with the acoustic transfer at those multipoles and so genuinely open. Its decisive discriminator is the radiation-free expansion rate itself, carried to the distance ladder and confronted with the DESI DR2 baryon-acoustic dataset at $\chi^2/\text{dof} \simeq 1$ (§2.4): the same rate resolves the Hubble tension across the full ladder and stakes the damping signature on one commitment, the observable weight of the latter awaiting the full transfer.

The framework's honest standing includes what the standard model has genuinely built and this cosmology has not. Dark matter is inferred from multiple independent, cross-consistent lines—rotation curves, gravitational lensing, the merger offsets, the acoustic-peak structure, the growth of structure—and this cosmology carries the matter density as the bend of the spatial cut without yet removing the component or answering its particle question; that column stands at full weight. Standard nucleosynthesis derives the light-element abundances from one measured parameter, a genuine achievement this cosmology now meets on the same footing rather than exceeds: the collapse's cooling leg is a standard nucleosynthesis run on one inherited datum (§2.4; the cosmogenesis paper [16]), producing helium-4 and deuterium at their observed values—the freeze-out temperature the data demands being the bottleneck the standard rate delivers—and, because the run is standard, carrying lithium-7's standard threefold over-prediction, the lithium problem shared rather than solved. On the light elements the two frameworks stand even: the same two successes, the same one problem, with this cosmology's open edge whether one progenitor handover fits all the abundances together. And the framework is young: the *detailed* dynamics of matter crossing the branch point (its crossing shown well posed and structurally governed, §10), the derivation—as against the measurement—of the inherited data, and the Standard-Model matter *content*—the gauge representations and the mass spectrum, which the framework does not supply, its geometry shown to force only the discrete flavour skeleton [17]—remain to be built (§10; [7]). These are open frontiers, not defects: a theory is not marked down for what it has not built, only for a claim it makes that fails. They are named plainly here because an economy that favours this cosmology on the structure it carries would be worth nothing bought by concealing the structure it does not.

12 Conclusion

So, plainly, what this paper claims and what it does not. It claims that the observed expansion history, the dissolution of the Hubble tension and the calibration of the acoustic scale, and the structure of the primordial scalar spectrum are consequences of a single de Sitter geometry read through the causal reassignment its own horizon geometry selects—one seed, grown, the field equations unchanged—and that on the economy of assumption, the criterion by which theory-choice is rightly made ahead of the deciding measurement, this reading is favoured now over the assembled standard picture. On the data axis it claims the more limited and now-earned thing: that the axis has moved in its favour—the Hubble tension resolved across the distance ladder, the abundances confirmed—without yet claiming to have won it, the ∼8% damping-tail signature and the full-spectrum likelihood the edges still to be brought to a verdict (the low multipoles, once the sharpest edge, now a mild non-discriminating deficit consistent with ΛCDM within cosmic variance). It does not claim to be a complete cosmology: its matter sector, the derivation of its inherited data, and its fermion content remain to be built. It claims to have shown that the universe we observe is the reading of one geometry, and to have staked the falsifiable commitments by which that claim can be tried.

A final word on the register, because it is the thesis rather than an aside. The beginning this cosmology describes is the branch point at $r = 0$—the locus at which the collapse leg of the cosmogenetic contour becomes the expansion leg [7]—and it is not a curvature singularity reached and left behind. *The curvature there is set by the substrate's own scale.* The throat is α, the defining constant of the maximally symmetric manifold the cut is taken on, and the divergence of the invariants computed in the areal chart is that chart degenerating, exactly as a polar chart degenerates at a pole—not a curvature scale of the geometry [16, 15]. Two further facts stand with it: no finite cosmic-time layer ever reaches the locus, and the continuation through it carries no scale, since $f \to -2M/r$ diverges so that the tortoise coordinate $r_* = \int \mathrm{d}r/f$ *converges* and what is transmitted is fixed by the analytic structure rather than by a length. *The branch point is crossed by continuation, not traversed by a worldline.* The expansion is not a space ejected from a point but the reading, on the near side of that branch point, of a spatial layer still advancing, the branch point itself a collapse that in the frame it bounds is never complete. The distinction the whole programme turns on—between what has finished happening and what is still underway—is thus the distinction the cosmology is built from: the universe is not the collapsed record of a beginning but the still-evolving layer that record represents, read here as one de Sitter geometry.

Appendix R Computational Receipts

Each computation marked R in the text is verified by a runnable receipt in the `receipts/` directory that ships with this work; status OK = verified (traced, run, output matches the claim). This appendix is generated from the receipt index, not maintained by hand.

`P15_verify_geometry` [OK]
prop:flat / prop:throat / prop:transmission (three geometric props DERIVED from primitives: (a1) const-tau 3-metric dr^2+r^2 dOmega^2 has Riemann=0 (flat R^3); (a2) Nariai near-horizon dS_2 x S^2, f(r_star)=f'(r_star)=0, f"=-2Lambda, both radii 1/sqrt(Lambda); (a4) tortoise integrals simple-¿log(exp) vs double-¿power-law). *Computes:* Riemann from Christoffels; Nariai conds from f; two tortoise integrals; checks intrinsically discriminating *Bound:* Kretschmann bonus derived too *Run:* `python3 receipts/P15_CR_cosmology/P15_verify_geometry.py`

`P15_verify_numeric` [OK]
prop:amplitude / prop:subhorizon ((a5) substrate dS vacuum power (hbar H_Lam/E_P)^2=9.5e-123 vs A_s=2.1e-9 -¿ ~113 orders below, robust to O(10) prefactor; (a7) comoving Hubble k at seam 0.0104/Mpc vs peak pi/r_s=0.0217/Mpc -¿ 2.1x sub-horizon). *Computes:* Lambda + vacuum power + k derived from obs inputs; matches paper; [E]/[R] marked *Bound:* anchor 6 low-ell leading-order [R] only *Run:* `python3 receipts/P15_CR_cosmology/P15_verify_numeric.py`

`P15_verify_coherence_comb` [OK]
sec:coherence (coherence comb: common seam phase -¿ sharp comb Delta-ell=pi D_C/r_s=296 (peaks 296,592,888), peak/trough ~1e5-1e6; random phase per mode -¿ washed out, contrast ~1 (400 realisations). Null seam's one characteristic datum/mode = one phase/mode **(!! MARKED r2376+c54.164: this receipt does NOT propagate anything – it writes down cos(k r_s)^2 with r_s and D_C as hardcoded literals, exactly as its own scope note says. It establishes the COHERENCE MECHANISM and NOT the peak spacing. P15 sec:coherence claimed it as the result of 'propagating the tightly-coupled acoustic modes to last scattering'; that sentence is corrected in the same pass.)**). *Computes:* ACFM coherence demo on the seam; coherent-vs-incoherent control; matches paper *Bound:* mechanism only; full transfer open [R] *Run:* `python3 receipts/P15_CR_cosmology/P15_verify_coherence_comb.py`

`P15_cr_collapse_driving` [OK]
sec:coherence (acoustic driving as linear oscillator Theta0"+Theta0=-Psi: undriven (Psi=const) a0=0.500, driven (radiation) aD=0.999, boost aD/a0=2.00 (known LCDM boost); baryon-loaded driven-equiv 1.60 vs adiabatic 1.10). *Computes:* ODE integration; undriven-Psi=const control; matches known boost *Bound:* time-symmetry of contracting phase argued (magnitude proven in next receipt) *Run:* `python3 receipts/P15_CR_cosmology/P15_cr_collapse_driving.py`

`P15_time_reversal_driving` [OK]
sec:coherence (|Psi~(omega=1)| resonant driving magnitude is time-reversal invariant: |Psi~(x)|=|Psi~(-x)| at w=1 on the radiation transfer (0.037049) AND a random control (1.411379); so CR's time-mirrored collapse driving = LCDM's magnitude, peak heights match by structure). *Computes:* numerical Fourier magnitude at w=1; radiation + random control; complex/phase-not-invariant discriminator *Bound:* built to close the paper's 'confirmed numerically ... random control' claim *Run:* `python3 receipts/P15_CR_cosmology/P15_time_reversal_driving.py`

`P15_camb_reference` [OK]
sec:coherence (**!! MARKED c54.160: the closing annotation states the paper's anchors as P2/P1 ~ 0.46 and P3/P1 ~ 0.45-0.46 while the file itself computes 0.452 and 0.443 and the paper now prints 0.45 and 0.44.** A stale annotation contradicting the same file's own digit-by-digit claim; both figures are now pinned. Original claim: Boltzmann peak-height reference (CAMB, Planck-2018): P1(ell=220), P2(536), P3(813) -¿ P1/P2=2.211, P2/P1=0.452, P3/P1=0.443, matching the paper's cited 2.2/0.45/0.44; the LCDM values CR's peaks reduce to). *Computes:* CAMB TT spectrum + peak-finder; matches paper *Bound:* needs camb; CR-matches-it carried by driving receipts *Run:* `python3 receipts/P15_CR_cosmology/P15_camb_reference.py`

`P15_damping_ratio_clean` [OK]
sec:coherence (**!! CORRECTED c54.155: theta_D/theta_* CR/LCDM IS 1.0816 (+8.2%), NOT THE 1.0785 (+7.9%) THIS ROW AND THE RECEIPT'S OWN DOCSTRING RECORD** –

both stale from the z_onset 6850 -¿ 6797 retune. The paper's 'approx 1.08' brackets the live value correctly. Original claim: ~8% damping signature: r_s(radincl)=144.0 vs CAMB 144.4; r_D CR 9% longer (7.16 vs 6.57); damping ratio theta_D/theta_* CR/LCDM=1.0785 (+7.9%); artifact (rad-free r_s to high z=245) exposed; theta_* H0-independent). *Computes:* CAMB x_e(z); r_s+r_D both rates; gated on CAMB; RATIO cancels conventions *Bound:* absolute theta_D ~7% below CAMB (k_D convention), cancels in ratio *Run:* `python3 receipts/P15_CR_cosmology/P15_damping_ratio_clean.py`

`P15_damping_reabsorption` [OK]

sec:coherence (~9% damping shift not reabsorbable: r_D ~ omega_b^{-0.305} (paper -0.31), d ln r_D/d ln omega_c=-0.295; cancelling +8.9% needs omega_b +29% vs ~1% BBN+height prior -¿ 29x too big; 1% omega_b shifts r_D ~0.3%; residual ~8.2%). *Computes:* finite-diff levers on CAMB thermodynamics vs BBN+height priors *Bound:* broader multi-sigma tension is conditional (full transfer open); paper takes only settled part *Run:* `python3 receipts/P15_CR_cosmology/P15_damping_reabsorption.`

`P15_zonset_determinations` [OK]

sec:tensions (**!! MARKED c54.160: the paper reported the rounded-l_* pinning as recovering 6873 and this receipt returns 6844 on the current integrals.** The paper now carries both numbers and notes that 6844 sits NEARER to the retired 6850, which sharpens the withdrawal rather than softening it. Original claim: z_onset determinations: (1) pinning the MEASURED angle 100theta_*=1.04109 (l_A=301.76) returns z_onset=6797, identical at H0=67.4/70/73; (2) the superseded check pinned the ROUNDED l_*=301 -¿ 6844, agreeing with 6850 only because 6850 itself yields l_*=300.90 – the agreement was the rounding; (3) the inherited datum alone (rho_r/rho_m=2) is H0-DEPENDENT: 6840/7378/8024, coinciding with (1) only near H0=67.4). *Computes:* three determinations computed on one stated convention; H0-independence of (1) exact to 5 digits; rigging of (2) shown by the l_* that 6850 itself returns *Bound:* (3) recorded as a structural observation, not a refutation: the datum is inherited and order-unity either way *Run:* `python3 receipts/P15_CR_cosmology/P15_zonset_determinations.py`

`P15_the_ratio_is_the_onset_in_imported_units` [OK]

sec:tensions (WHAT $\{}rho_r/\{}rho_m\{}approx2$ IS, AND WHAT IT IS NOT: **NEITHER A HIDDEN KNOB NOR A SECOND MEASURED DATUM, BUT THE ONE FITTED QUANTITY z_onset RE-EXPRESSED IN UNITS OF A PHYSICAL MATTER DENSITY THE CONSTRUCTION DOES NOT DETERMINE.** The rate carries Omega_m (fitted to BAO); the sound speed carries omega_b and omega_gamma (measured directly, framework-free); z_onset is the one fitted number – and the PHYSICAL omega_m = Omega_m hˆ2 enters NO prediction. But rho_r/rho_m at onset is omega_r(1+z_onset)/omega_m and equality is 1+z_eq = omega_m/omega_r, so **both divide by exactly that quantity**. ** KNOB-FREENESS SURVIVES AND IS THE LOAD-BEARING HALF: z_onset = 6747.3 at H0 = 67.4, 70.0, 73.0 and 74.0, to the digit – H0 is a common factor in r_s and D_M and cancels from theta_*, so the fitted quantity does not move between the two H0 the tension is between and cannot be absorbing it. ** ** WHAT DOES NOT SURVIVE IS THE EXACTNESS OF 'ABOUT TWICE EQUALITY': at the fitted Omega_m = 0.3066 the ratio is 2.01 at H0 = 67.4 and 1.71 at H0 = 73, the two readings crossing at H0 ~ 68 where Omega_m hˆ2 reproduces the CMB-inferred omega_m = 0.143 – and the papers' headline is that the cosmology fits at the DIRECTLY measured H0 = 73. ** So the datum is an order-unity BAND (1.7-2.0), not a determined number; the eta-analogy and the BAO confrontation are untouched (the latter uses distance ratios and never omega_m).). *Computes:* z_onset recomputed from the radiation-free rate at four values of H0 by root-finding on the measured acoustic angle, its spread asserted below 1 part in 6000; the crossing H0 asserted in 67-69; the H0=73 ratio asserted at 1.71 *Bound:* ** THIS IS A UNIT ANALYSIS, NOT A NEW MEASUREMENT. ** It does not touch the rate, the fit, or the abundances; and the CMB-inferred omega_m it compares against is itself derived in a radiation-included analysis, which is the point rather than an objection – the residual measures how far the two analyses' omega_m must differ *Run:* `python3 receipts/P15_CR_cosmology/P15_the_ratio_is_the_onset_in_imported_units.py`

`P15_the_two_data_are_one` [OK]

sec:tensions (WHAT THE 'SINGLE INHERITED DATUM' rho_r/rho_m ~ 2 ACTUALLY IS, AND WHAT THE NATURALNESS ARGUMENT OFFERED FOR IT ACTUALLY DESCRIBES. ** (1) IT IS NOT A DATUM DISTINCT FROM eta. ** On any thermal history rho_r/rho_m(T) = [1+nu](piˆ4/30 zeta3) T / [eta (w_m/w_b) m_N], which returns **1.99 at T = 1.6 eV** – the quoted value to one per cent, from standard thermodynamics and NO feature of this construction.

Given eta and the measured matter-to-baryon ratio, both inherited for the composition regardless, **the ratio at onset IS the onset**: the handover supplies ONE composition datum and the cosmology fits ONE parameter. ** (2) AND THE NATURALNESS ARGUMENT DESCRIBES AN EPOCH NINE ORDERS FROM THE HANDOVER. ** rho_r = rho_m at T = 0.806 eV; the handover sits FOUR ORDERS ABOVE the deuterium bottleneck (0.07 MeV) because that is what the nucleosynthesis computed on it requires, so the ratio there is ˜9e8 – and 'a handover carrying radiation of order the matter density' would be a handover with **eta ˜ 0.5**: no light elements, no acoustic peaks. ** So the naturalness question runs the OTHER WAY: what must be explained is a BILLION photons per baryon, which is the baryogenesis-analogue question already named as open, and the binding-energy scale (GM/Rcˆ2 = 1/2) offers no head start toward it. ** P7's 'the frontier is two data and not one' is corrected to one inherited datum plus one fitted parameter.). *Computes:* the identity evaluated at the onset temperature and asserted against the quoted datum to better than 5%; the equality temperature and the ratio at the bottleneck, at P7's own handover and at the rest mass computed; the eta that O(1) would require at each epoch computed and contrasted with the measured 6.1e-10; and the papers asserted to carry the withdrawal and the corrected accounting *Bound:* ** THIS CORRECTS AN ACCOUNTING AND WITHDRAWS AN ARGUMENT; IT TOUCHES NO RESULT. ** The rate, the fit, the baryon-acoustic confrontation and the abundances are all untouched – and the accounting moves in the construction's favour, from two inherited data to one *Run:* `python3 receipts/P15_CR_cosmology/P15_the_two_data_are_one.py`

`P15_full_transfer_verdict` [OK]
sec:coherence (high-ell deficit UNDER THE SHORTCUT (assume peaks=LCDM + apply +8.9% damping): C_l CR/LCDM = 0.979/0.920/0.829/0.717/0.594 at l=500-2500; cited to show the shortcut is illegitimate (assumption (A) is the unbuilt part)). *Computes:* exp damping on CAMB LCDM C_l; conditional chiˆ2 *Bound:* verdict explicitly CONDITIONAL/open (own banner); paper honors scoping *Run:* `python3 receipts/P15_CR_cosmology/P15_full_transfer_verdict.py`

`P15_verify_closedS3_nonsync` [OK]
sec:largescale (low-ell floor mechanism+location: discrete closed-Sˆ3 source (k_L=sqrt(L(L+2))/r0, L¿=2) through FLAT j_ell(k_L D_C), stretch D_C/r0=2.75 pushes L=2 to ell˜7.8; ell(ell+1)C_ell=0.12/0.10/0.20/0.6 at ell=2..7, recovered by ell˜8). *Computes:* discrete-source flat-Bessel sum; flat-limit chi-¿0 control passes *Bound:* ordinary SW only -¿ deeper than Boltzmann depth (routed to that receipt) *Run:* `python3 receipts/P15_CR_cosmology/P15_verify_closedS3_nonsync.py`

`P15_verify_lowell_exact_measure` [OK]
sec:largescale ((1) HZ weight w_L=(L+1)/(L(L+2)) DERIVED as closed-Sˆ3 degeneracy betaˆ2 x per-mode 1/(beta(betaˆ2-1)) (g*P==w_L exactly; w_L*L-¿0.998 continuum); (2) D2/D3=1.162 stable across measure family -¿ floor starves ell=2,3 together (octopole slightly more)). *Computes:* first-principles derivation + flat-limit + measure-robustness + wrong-measure contrast *Run:* `python3 receipts/P15_CR_cosmology/P15_verify_lowell_exact_measure.py`

`P15_verify_doppler_lowell` [OK]
sec:largescale (Doppler term (super-horizon velocity through j_ell') does NOT differentiate ell=2 from ell=3: CR/LCDM ratio l2˜0.27, l3˜0.21-0.22 across 4 treatments (none/coh+/coh-/quad), D2/D3˜1.29 octopole slightly more suppressed; continuum validator reproduces LCDM rise). *Computes:* SW+ISW+Doppler transfer, 4 combination modes; continuum check; not tuned *Bound:* bare depth (milder exact depth = Boltzmann receipt) *Run:* `python3 receipts/P15_CR_cosmology/P15_verify_do`

`P15_verify_lowell_boltzmann` [OK]
sec:largescale/sec:scope (exact low-ell depth via genuine Boltzmann transfer: discrete closed-Sˆ3 source through CAMB's exact Delta_l(k) (SW+ISW+Doppler); CR/LCDM = 0.473 (l=2), 0.410 (l=3), recover by l˜7, octopole/quadrupole 0.867 (matches paper 0.47/0.41). GATE: continuum reproduces CAMB C_l to 4 figures). *Computes:* CAMB transfer x discrete source, Riemann-sum vs integral; gate-validated; r0-stable; ISW-retained diagnosis *Bound:* needs camb (high acc) *Run:* `python3 receipts/P15_CR_cosmology/P15_verify_lowell_boltzmann.py`

`P15_confront_lowell_data` [OK]
sec:largescale (low-ell data confrontation (corrected depths): CR 0.47 sits ABOVE the sharp observed quadrupole (˜0.2), CR 0.41 mildly below the near-LCDM octopole (˜0.6-1.0) -¿ matches neither, a WASH; old 0.22/0.20 'striking match' superseded). *Computes:* CR (Boltzmann depths)

vs literature observed multipoles *Bound:* body corrected to r962 depths; likelihood in next receipt *Run:* `python3 receipts/P15_CR_cosmology/P15_confront_lowell_data.py`

`P15_verify_lowell_likelihood_v2` [OK]

sec:largescale/sec:scope (low-ell cosmic-variance likelihood: exact scaled-chi^2_{2l+1}, CR/LCDM ratio Delta(-2lnL)_l=(2l+1)[r_l(1/d_l-1)+ln d_l]; l=2 rewards CR -2.63, l=3/4 penalise +1.31/+2.10, central total +1.8, octopole range +0.3..+3.8 -¿ wash inside cosmic variance). *Computes:* exact likelihood with corrected Boltzmann depths; matches paper *Run:* `python3 receipts/P15_CR_cosmology/P15_ver`

`P15_the_low_ell_minimum_is_at_ell_four` [OK]

sec:largescale/sec:scope (**!! NARROWED r2376+c54.153: THIS FILE CROSS-VALIDATES NOTHING.** Its arm A is the likelihood receipt's stored dict and **its arm B is the PAPER'S OWN FIGURE CAPTION, read by regular expression** — so the comparison is against the citing paper's typography, and the quartet 0.42/0.35/0.29/0.52 is banked nowhere. **What it does verify is that arm A's minimum sits at l = 4 and that the standing receipts always carried it, which is what it is now cited for.** The actual second arm is 'P15_the_second_arm_actually_run' (c54.153): it agrees on shape and NOT on depth, and the quartet is withdrawn. Original claim: THE LOW-MULTIPOLE DEFICIT'S SHAPE, AND WHERE THE SECTOR'S WEIGHT ACTUALLY FALLS. ** IT IS NOT A MONOTONE RECOVERY FROM THE QUADRUPOLE BUT A DIP WHOSE MINIMUM IS AT ell=4 – and the quadrupole and octopole, the two multipoles the paper quoted, are its two SHALLOWEST points. ** ** THE SHAPE IS CROSS-VALIDATED BY TWO INDEPENDENT BOLTZMANN TREATMENTS of the same discrete source: a standard code's exact Delta_l(k) at the discrete k_L gives 0.47/0.41/0.36/0.68 at ell=2,3,4,5 and the programme's own photon hierarchy gives 0.42/0.35/0.29/0.52 – SAME minimum, SAME recovery, depth ratio 1.13-1.30, which is the size of the residual sec:instrument declares. The shape is cross-validated; the depth carries their spread. ** ** AND THE ell=4 MINIMUM WAS NEVER NEW: 'P15_verify_lowell_boltzmann.py' has tabulated 0.356 at ell=4 since r1395 and the likelihood receipt has consumed it ever since – the corpus's failure was one of QUOTATION, not of computation. ** ** THE OCTOPOLE VERDICT: on the exact cosmic-variance likelihood the quadrupole REWARDS the construction (-2.63), the octopole penalises it at +1.31, and the LARGEST single-multipole penalty in the sector is the +2.10 at ell=4 – the prediction's own minimum, and the one multipole of the four the large-angle literature does not treat as anomalous. Sector total unchanged at +1.8, a wash inside cosmic variance. **). *Computes:* both arms' depth tables read off their own sources (the likelihood receipt's dict; the paper's caption) and their argmin asserted equal to ell=4; the ell=4 depth asserted present and deepest in the standing r1395 receipt; the per-multipole likelihood re-derived from the depths and its argmax asserted at ell=4 with the total asserted at +1.8; the exact/WKB transmission spread asserted below 5%; and all eight corpus sites asserted to carry the corrected statement *Bound:* ** DEPTH IS QUOTED TO A TWO-CODE SPREAD AND NOT BETTER. ** And nothing here relieves the observed sky's selective suppression of ell=2 ALONE by a factor of order three, which this construction does not produce; that stays inside cosmic variance and belongs to the likelihood-level comparison *Run:* `python3 receipts/P15_CR_cosmology/P15_the_low_ell_minimum_is_at_ell_four.py`

`P15_two_ranges_ratio` [OK]

sec:background (two ranges) (the local Hubble-Eddington radius r_* = (M alpha^2)^(1/3) and the expansion's amplitude A = (2M alpha^2)^(1/3) are ONE length in the fixed ratio 2^(1/3), for every M; A is the matter-Lambda equality radius). *Computes:* r_* from K_G = 1/alpha^2 - M/r^3 = 0; A from rho_m/rho_Lambda = csch^2(3tau/2alpha) at equality; ratio A/r_* = 2^(1/3) symbolically, independent of M and alpha *Bound:* the r1608 identity; the four-ways reading (turnaround, amplitude, equality, structure boundary) is the paper's *Run:* `python3 receipts/P15_CR_cosmology/P15_two_ranges_ratio.py`

`P15_the_second_arm_actually_run` [OK]

sec:largescale (**THE SECOND BOLTZMANN ARM, ACTUALLY RUN — THE SHAPE CROSS-VALIDATES AND THE DEPTH QUARTET DOES NOT.** P15 claimed the low-multipole deficit had been run a second time on the programme's own photon hierarchy, returning 0.42/0.35/0.29/0.52. **The receipt cited for it does not run the hierarchy: it reads arm A from a stored dict and reads arm B OUT OF THE PAPER'S OWN FIGURE CAPTION, by regular expression** — so the cross-validation compared a table against the citing paper's typography and the quartet was banked nowhere. This file drives 'HIER_photon_hierarchy' itself: evolves it on a k-grid containing the discrete ladder k_L = sqrt(L(L+2)) x 2.75/D_M plus a dense continuum grid, projects its own source terms to Delta_l(k) for l = 2..44, and forms the discrete w_L sum against the continuum dln k integral, each normalised to its own l = 25-40 plateau. **SHAPE: minimum at l = 4 and recovery by l ˜ 7, independently of the CAMB arm and of the caption.** **DEPTH: 0.494/0.243/0.184/0.608 against CAMB's 0.473/0.410/0.356/0.676 — close at l = 2, a factor 1.7-1.9 apart at l = 3-4.** So the quartet is returned by neither arm and the 'uniformly 10-25% deeper' claim is false; both withdrawn from P15 at c54.153.). *Computes:* the minimum over l = 2..5 asserted at l = 4; recovery asserted in 6 ¡= l ¡= 9; the depth disagreement asserted above 1.4x; all ratios asserted finite *Bound:* **scope: this settles that the SHAPE claim has a computation behind it and that the DEPTH claim did not. It does NOT settle which depth table is right — the arms differ in late-ISW handling and in D_M (13864 here against the projection's 13927) — and it makes no attempt to reconcile them.** *Run:* `python3 receipts/P15_CR_cosmology/P15_the_second_arm_actually_run.py`

`P15_hubble_expansion_confrontation_v2` [OK]

sec:tensions (SDSS DR12 BAO flagship: CR BAO ratios D_M/r_d, D_H/r_d H0-INDEPENDENT (identical at H0=67,73); chiˆ2 CR=1.71 (any H0 incl 73), LCDM(67.4)=4.37, LCDM(73)=49.45 = the tension; cosmic chronometers consistent with 73). *Computes:* radiation-free H0-cancellation + chiˆ2 vs DR12; matches paper 1.7/49 *Run:* `python3 receipts/P15_CR_cosmology/P15_hubble_expansion_confr`

`P15_desi_dr2_confrontation` [OK]

sec:tensions (**!! MARKED c54.160: the docstring reads chi2/dof = 13.70 for LCDM at H0 = 73 while the paper quotes about 15 – the same chi2 = 178.16 divided by 13 measurements rather than by the 12 degrees of freedom one fitted parameter leaves.** The paper's figure is the correct one and the receipt's annotation is not; the chi2 itself is pinned. Original claim: **!! CORRECTED c54.155: TWO OVERSTATEMENTS IN THE CITING SENTENCE.** Omega_m = 0.307 was described as a CMB-calibrated input; the receipt FITS it to these data (minimize_scalar), and at Planck's own 0.3153 the fit degrades to chi2/dof = 1.35. **And granted the same one free parameter, LCDM fits DESI DR2 marginally better than CR: 0.92 against 1.00, its minimum at H0 = 68.7 rather than the 67 the paper stated.** The surviving claim – CR fits without choosing an H0, LCDM must choose – is what P15 now says. Original claim: DESI DR2 BAO (13 meas, 7 tracers, full covariance): CR 1-param fit Om=0.3066 (rho_r/rho_m˜2.0), chiˆ2/dof=1.00 at any H0 incl 73; LCDM(67.4)=2.20, LCDM(73)=13.70 BREAKS = the tension). *Computes:* full-covariance chiˆ2 vs DESI DR2; matches paper 1.0/14; was uncited by name *Run:* `python3 receipts/P15_CR_cosmology/P15_desi_dr2_confrontation.py`

`P15_expansion_law` [OK]

sec:properframe/sec:flatlcdm (derivation core (symbolic): eq:amplitude (6GM/Lambda cˆ2)ˆ{1/3}=2ˆ{1/3}/sqrt(I at Nariai; eq:scalefac r=A sinhˆ{2/3}(B tau) solves the E=1 geodesic (dr/dtau)ˆ2=2GM/r+(Lambda/3)rˆ2 (2/3 forced by wrong-exponent control); eq:rate Hˆ2=(Lambda cˆ2/3)cothˆ2; eq:omega-ratio Omega_m/Omega_L=cschˆ2). *Computes:* sympy: amplitude, geodesic solve, cothˆ2 rate, cschˆ2 ratio; wrong-exponent control *Run:* `python3 receipts/P15_CR_cosmology/P15_expansion_law.py`

`P15_verify_throat_tower` [OK]

sec:throat (throat mode tower + no-hair isotropization: dS_2 x Sˆ2, mˆ2/Hˆ2=l(l+1), nuˆ2=1/4-l(l+1); l=0 -¿ nu=1/2 (scale-invariant base, survives), every l¿=1 -¿ nuˆ2¡0 (heavy principal, decays) =¿ throat isotropizes; grounded in Wald 1983 no-hair; guard: throat l != CMB multipole). *Computes:* tower algebra + survives/decays split; theorem-grounded; guard held *Run:* `python3 receipts/P15_CR_cosmology/P15_verify_throat_tower.py`

`AS_amplitude_leftward` [OK]

rem:transmission-leg (AS — THE AMPLITUDE, READ LEFTWARD. Family 3's question, and what C4 can and cannot say about it. CONFIRMATION 2, and the corpus is already correctly scoped here: P7: the inherited datum 'stands as a ONE-PARAMETER ACCOMMODATION rather than a parameter-free prediction.). *Computes:* runs clean; registered r2376 (c54) from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P15_CR_cosmology/AS_amplitude_leftward.py`

`BUILD_camb_store` [OK]

sec:instrument (BUILD_camb_store – the CAMB side of the transfer comparison, with its settings RECORDED. WHY THIS EXISTS (r2345). 'ENVELOPE_residual.py' and its relatives load '/tmp/camb_tr.npz' and measure our transfer against CAMB's. Nothing in the corpus records

HOW that store was built – not the receipt, not the changelog, not the instrument.). *Computes:* runs clean; registered r2376 (c54) from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P15_CR_cosmology/BUILD_camb_store.py`

`C10_highl_ratio` [OK]

sec:envelope-consequence (**!! CORRECTED c54.155: THIS ROW CARRIED THE RETIRED z_onset = 6850 NUMBERS** (r_s matched +0.76%, r_D +10.09%, theta_D/theta_* +9.26%). The shipped receipt hardcodes 6797 and produces +0.67%, +10.15%, +9.41%, l_* = 302.2. **And the receipt hardcodes th_ratio = 1.0926 with the comment 'DERIVED in C9' while C9 produces 1.0941.** P15 corrected to the live values. Original claim: C10 — THE HIGH-ELL RATIO, ASSEMBLED FROM DERIVED PIECES ONLY. Inputs, all derived in C4/C7a/C8/C9 except H0 and Omega_m, which are observations: * driving envelope flat above ~10 k_s, turning over at the inherited datum's own equality; * r_s matched (+0.76%); r_D +10.09%; theta_D/theta_* +9.26%. l_D = D_M/r_D, so D_M is needed for an ABSOLUTE multipole – and D_M is the same integral again.). *Computes:* runs clean; registered r2376 (c54) from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P15_CR_cosmology/C10_highl_ratio.py`

`C11TEST_radiation_zeroed` [OK]

sec:envelope-consequence (CRRUN — THE SAME INTEGRATOR ON CR. The control passed at r1975, so this may now be run. WHAT CHANGES, and each from the corpus not from me: * BACKGROUND: H^2 = H0^2(Om a^-3 + OL) at H0=73, Om=0.3066 – radiation is content, not a source. * LOWER LIMIT: the SEAM, not eta-¿0 (C9: there is no observable expansion below it).). *Computes:* runs clean; registered r2376 (c54) from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P15_CR_cosmology/C11TEST_radiation_zeroed.py`

`C11_early_isw` [OK]

sec:envelope-consequence (C11 — THE EARLY ISW. The last named piece. CONFIRMATION 2/4 – the corpus settles the input, in its own voice: 'radiation, like matter, is inherited content read off that clock, NOT A TERM THAT SOURCES THE RATE, and there is no radiation-dominated era.). *Computes:* runs clean; registered r2376 (c54) from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P15_CR_cosmology/C11_early_isw.py`

`C1_closure_of_the_driving` [OK]

sec:envelope (C1-CLOSURE — DOES THE ACOUSTIC OSCILLATOR CLOSE IN ELEMENTARY TERMS ON CR's COLLAPSE LEG? CONFIRMATION 4 first — WHICH background. The corpus is specific and I must not substitute: L2 (where the driving lives) = 'the leaf's local rate, ORDINARY FRIEDMANN, radiation GRAVITATING NORMALLY, on the progenitor's collapse excursion' – and P16: on the cooling leg |H| = sqrt(8 pi G rho/3), 'identical to standard BBN at every...). *Computes:* runs clean; registered r2376 (c54) from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P15_CR_cosmology/C1_closure_of_the_driving.py`

`C2_horizon_limits` [OK]

sec:envelope (C2 — THE LIMITS. The comoving Hubble radius across the excursion, from the corpus's own geometry. CONFIRMATION 4 — the object, and I take it from the corpus rather than from cosmology-by-habit: the E=1 (marginally bound) congruence IS this cosmology. On it (dr/dtau~)^2 = E^2 - f = 1 - f, with f = 1 - 2M/r - r^2/alpha^2. The signed areal radius r is the scale-factor analogue.). *Computes:* runs clean; registered r2376 (c54) from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P15_CR_cosmology/C2_horizon_limits.py`

`C3_conformal_map` [OK]

sec:envelope (**!! CORRECTED c54.155: 'better than a percent at the entry radii of the modes in question' IS 1.87% AT THE PAPER'S OWN 10 k_s THRESHOLD** (3.72% at 5 k_s, 24.4% at the seam); sub-percent begins near 30 k_s. P15 corrected. Original claim: C3 — CONFORMAL TIME ON THE LEG: does the map from the limits to the mode's variable close too? The integrand Psi(x) is elementary in x = k eta / sqrt3 (C1). The limits are radii (C2). So the whole computation closes only if eta(r) closes as well. Derive it.). *Computes:* runs clean; registered r2376 (c54) from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P15_CR_cosmology/C3_conformal_map.py`

`C4_driving_envelope` [OK]

sec:envelope (**!! CORRECTED c54.155: 'entry at x = 1/sqrt3 for every k' IS A LIMIT, NOT AN IDENTITY** (the receipt's own table gives x_entry = 0.764 at k = k_s, +32%), **and the 'residual below a percent' is an average over a 5-point grid that lands near zeros of cos(x_seam): scanning the receipt's own function over k/k_s in [10,120] reaches 7.3% at k/k_s = 12.** Both qualified in P15. Original claim: C4 — ASSEMBLE IT. Integrand (C1) + limits (C2) + map (C3), on the collapse leg. CONFIRMATION 4 — the assumptions, stated up front and not buried: (i) tight coupling, R = 3 rho_b/4 rho_gamma -¿ 0 (pure photon fluid, c_sˆ2 = 1/3). R is small deep on the leg where the driving happens; near the seam it is not, and that is a stated limit. (ii) NO ANISOTROPIC STRESS, so Phi = Psi. Exact for a perfect photon-baryon fluid;). *Computes:* runs clean; registered r2376 (c54) from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P15_CR_cosmology/C4_driving_envelope.py`

`C5b_baryon_term` [OK]

sec:envelope (C5 — THE BARYON TERM. The first of the two terms C4 named as missing. CONFIRMATION 4 – and the corpus fixes the division rather than me: P16 masthead: 'eta fixes the abundances and the CMB peak HEIGHTS, rho_r/rho_m the peak SPACING.' P15: this cosmology 'carries the matter density as the bend of the spatial cut WITHOUT YET REMOVING THE COMPONENT' – so rho_m is TOTAL matter and the baryon fraction is set separately,...). *Computes:* runs clean; registered r2376 (c54) from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P15_CR_cosmology/C5b_baryon_term.py`

`C6_neutrino_term` [OK]

sec:envelope (C6 — THE ANISOTROPIC-STRESS TERM. The second of C4's two named terms. FINDING ZERO, before any physics: the corpus's TEXT says 'neutrino' ZERO times, and 'anisotropic stress' zero, and 'free-streaming' zero – while its BBN CODE carries them explicitly (bbn_network.py: 'neutrinos share the plasma T until ˜ e+- annihilation, then decouple to (4/11)ˆ1/3').). *Computes:* runs clean; registered r2376 (c54) from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P15_CR_cosmology/C6_neutrino_term.py`

`C7_equality_and_deficit` [OK]

sec:envelope (C7 — THE STEP THE TRAP WAS GUARDING. full_transfer_verdict.py ASSUMED the high-ell peaks equal LambdaCDM's and applied the extra damping, getting a large deficit – and P15 disowns it because 'that assumption IS the unbuilt part.' C4 built it. So the same calculation can now be done with a DERIVED envelope. FIRST, though, one number that decides everything and that I have not yet checked.). *Computes:* runs clean; registered r2376 (c54) from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P15_CR_cosmology/C7_equality_and_deficit.py`

`C8_diffusion_length` [OK]

sec:envelope-consequence (C8 — DERIVE r_D HERE, FROM FIRST PRINCIPLES, ON THE SAME FOOTING AS THE DRIVING. Nothing imported from the damping_tail/ files. Nothing that needs CAMB. CONFIRMATION 4: the object. Silk damping is a diffusion length built from the Thomson opacity and the expansion. The standard definition (Kaiser 1983; Hu-Sugiyama), in the Rb -¿ 0 limit: 1/k_Dˆ2 = INT d eta . (1/(6 tau_dot)) .). *Computes:* runs clean; registered r2376 (c54) from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P15_CR_cosmology/C8_diffusion_length.py`

`C9_sound_horizon_and_ratio` [OK]

sec:envelope-consequence (C9 — r_s AND r_D WITH THE CORRECT LOWER LIMIT: THE SEAM. A first pass integrated from a=0 and returned r_s = 256 Mpc against the corpus's 144 – absurd, and the error is structural, not numerical: *** ON THIS COSMOLOGY THERE IS NO OBSERVABLE EXPANSION BEFORE THE SEAM. *** P15: 'recombination is observable cosmology FROM THE SEAM OUTWARD'. The L1 foliation BEGINS at the seam;). *Computes:* runs clean; registered r2376 (c54) from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P15_CR_cosmology/C9_sound_horizon_and_ratio.py`

`H0_acoustic_angle_and_seam` [OK]
sec:refit-bound (H0 — THE ACOUSTIC SCALE AT THE DIRECTLY MEASURED H0. The calculation I have been running backwards. CORPUS CLAIM (P7, P15): 'There is no second Hubble rate to reconcile: THE DIRECTLY MEASURED H0 IS THE GEOMETRY READ AT THE PRESENT EPOCH, while the lower value inferred from the microwave background rests on a radiation-governed sound horizon the construction does not share...). *Computes:* runs clean; registered r2376 (c54) from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P15_CR_cosmology/H0_acoustic_`

`ISW_line_of_sight` [OK]
sec:refit-bound (ISW — THE EARLY INTEGRATED SACHS-WOLFE, ISOLATED, AND ITS SHIFT ON THE FIRST PEAK. Same shape as everything else in A.0: the source is a line-of-sight integral over the set rate. Delta_l(k) = INT d eta [S_SW + S_ISW] j_l(k(eta_0 - eta)), S_ISW = (Phi' + Psi') = 2 Psi' LambdaCDM has a nonzero early ISW because Psi is still evolving at recombination (radiation matters). CR does not: C11 showed Psi is constant on L1.). *Computes:* runs clean; registered r2376 (c54) from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P15_CR_cosmology/ISW_line_of_sight.py`

`M2_reabsorption_minimization` [OK]
sec:refit-bound (M2 — THE MINIMIZATION. Can a parameter refit absorb the damping shape? The honest core of the likelihood question, done on the closed forms derived in C8/C9 – no CAMB. SETUP, stated before running: FREE : Omega_m, omega_b, and the inherited datum rho_r/rho_m (which fixes z_onset). CONSTRAINT: l_* = pi D_M/r_s pinned to the observed 301 – the peak positions are measured.). *Computes:* runs clean; registered r2376 (c54) from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P15_CR_cosmology/M2_reabsorption_minimization.py`

`PEEB_peebles_on_the_set_rate` [OK]
sec:refit-bound (PEEB — THE PEEBLES IONISATION HISTORY ON THE SET RATE. ZREC used Saha only: right shift, wrong absolute (1283 vs CAMB's 1090), because Saha has no departure from equilibrium. Peebles adds it – and the departure is itself H-dependent, which is exactly why it must be integrated on the construction's own rate rather than inherited. Standard effective three-level hydrogen atom (Peebles 1968; Seager et al. fits).). *Computes:* runs clean; registered r2376 (c54) from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P15_CR_cosmology/PEEB_peebles_on_the_set`

`ROBUST_p1p2_scan` [OK]
sec:refit-bound (**!! FIXED c54.160: THE REGISTERED COPY WAS UNRUNNABLE FROM ITS OWN DIRECTORY** – it imported 'RD_diffusion_direct', which lives only in 'storyboard_receipts/', so it exited 1 on ImportError before reaching any computation and the sweep had to run it from the ORIGIN instead. *A receipt that cannot run where it is registered is not a receipt.* Path resolution added; it now exits 0 in place. Original claim: **!! CORRECTED c54.155: THE FILE IS NOT A SCAN AND ITS CITED NUMBER HAD DRIFTED.** It is a single CR run (its own docstring says CRRUN); it varies neither the tilt nor the diffusion scale, which is what P15 cited it for. **And it returns P1/P2 = 1.447 where P15 said 1.15** – the pre-r2371 value, before the anisotropic-stress sign correction (1.347) and the Thomson-opacity port (1.407). P15 corrected to 1.45 and the stability clause withdrawn. Original claim: CRRUN — THE SAME INTEGRATOR ON CR. The control passed at r1975, so this may now be run. WHAT CHANGES, and each from the corpus not from me: * BACKGROUND: H^2 = H0^2(Om a^-3 + OL) at H0=73, Om=0.3066 – radiation is content, not a source. * LOWER LIMIT: the SEAM, not eta-¿0 (C9: there is no observable expansion below it). **(!! MARKED r2376+c54.164: the run stands, the STATUS of its output does not. ell_1 and P1/P2 are NOT stable under this file's own documented ICs – 'velocities zero' in the docstring is not what the code imposes, and imposing it returns P1/P2 = 2.017 against the sky's 2.212 where this file returns 1.447. Psi's value comes from a smooth envelope and Psi's derivative from the oscillatory closed form, two different functions. Numerics are converged

and are not the cause. P15 no longer quotes the 150. What does NOT move is the source comb spacing, 0.72-0.79 of pi/r_s under every variant – see 'P15_the_first_peak_figure_is_not_stable'.)**). *Computes:* runs clean; registered r2376 (c54) from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P15_CR_cosmology/ROBUST_p1p2_scan.py`

`P15_the_first_peak_figure_is_not_stable` [OK]

the acoustic sector (**THE 150 IS NOT A PREDICTION: 'ROBUST_p1p2_scan''s FIRST-PEAK POSITION AND PEAK-HEIGHT RATIO ARE NOT STABLE UNDER ITS OWN DOCUMENTED INITIAL CONDITIONS.** Built settling front #5, the corpus's two internal routes to the first acoustic peak (transfer 220, this instrument 150). **The front is posed as a disagreement between two established routes; one of the two is not established.** PART 1 the numerics are cleared first – stability number 0.47 against the file's own limit of 2.8, 120+ RK4 steps per oscillation, NS_CR x4 reproduces every figure. PART 2 Psi's VALUE comes from the smooth matched envelope and Psi's DERIVATIVE from the oscillatory closed form – two different functions, and the initial velocity is built entirely from the second. PART 3 across four readings of the SAME stated IC, ell_1 spans 150-315 and P1/P2 spans 0.93-2.02; **the reading the docstring actually describes ('velocities zero') returns 2.017 against the sky's 2.212, where the paper reported 1.45 as the disagreement.** PART 4 the comb the paper cites against it was never propagated either – 'verify_coherence_comb' writes down cos(k r_s) with r_s and D_C as literals. PART 5 **the residual: the source comb spacing is 0.72-0.79 of pi/r_s under EVERY variant. The initial data move the first peak by 165 multipoles and do not move this.**). *Computes:* stability asserted under the file's own 2.8 limit and steps-per-oscillation over 50; the two derivative functions asserted to differ by more than 1.2x across the band and to agree as x -¿ 0; the control asserted to reproduce 1.447 and 150/360/555/780; the documented-IC run asserted at 2.017 AND asserted closer to the sky than the implemented one; ell_1 asserted to move by 150+ multipoles and P1/P2 by more than 2x; the comb receipt asserted to contain cos(k r_s) and NO integrator; the comb deficit asserted present, bounded, and STABLE across variants (max-min ¡ 0.15) – which is what makes it not initial data. *Bound:* **scope: this does NOT vindicate 220 and supplies no first-peak position. It removes one route from arbiter status and registers the comb deficit, which is not resolved here. The four variant figures are produced by 'computations/beyond_the_wall/L171_what_sets_the_first_peak.py', which execs the registered receipt's own source with only the initial-data block substituted; its ICMODE=asis control reproduces the receipt exactly, assertions passing.** *Run:* `python3 receipts/P15_CR_cosmology/P15_the_first_peak_figure_is_`

`P15_the_crossing_exists_and_is_empty` [OK]

the acoustic sector (**THE RE-ENTRY BETWEEN THE BRANCH POINT AND THE ONSET IS REAL, AND IT IS EMPTY: THE L1 POTENTIAL EQUATION CONTAINS NO k AT ALL.** Built working front #5, testing a candidate THIS FORK proposed at c54.165 – front #1 puts every mode super-horizon and frozen at the branch point, 'prop:subhorizon' puts the acoustic modes inside at the onset, so they re-enter in between, and a crossing is where flat LCDM acquires its k-INDEPENDENT driving shift. **The answer is no, and the reason is stronger than the question.** PART 1 the re-entry is real and its boundary is computed: the mode crossing AT the onset is k = 0.0111/Mpc, ell = 144, so the peak modes crossed before the plasma began and ell ¡ 144 crosses after. PART 2 but it is a MATTER-era crossing – pressureless to better than a part in 1e4, because L1 has no radiation SOURCE. PART 3 **and the w = 0 potential equation has no k in it**: Phi constant to twelve digits, identical across four decades in k, the integrator taking the SAME number of steps for every mode. PART 4 the contrast: in LCDM 5 of 6 cross under radiation, where w = 1/3 restores the kˆ2 term and Phi decays on the sound-crossing time by construction.). *Computes:* every mode at/above the peak region asserted to cross before the onset AND ell = 28 asserted to cross AFTER it (the first draft claimed 'every observed mode' and this assertion fired); the boundary multipole asserted inside 100-250; the background asserted pressureless to 1e-3 at every crossing; Phi's spread across four decades in k asserted below 1e-9 AND the solver step count asserted identical across k; the LCDM contrast asserted to put 4+ modes under radiation with z_eq above 3000. *Bound:* **scope: this CLOSES a candidate mechanism and supplies none. It computes no peak position and uses no figure from the acoustic fold. It leaves 'PO-7' open and harder: the absence of a universal driving shift on this rate is not an omitted crossing but a scale-free interval.** *Run:* `python3 receipts/P15_CR_cosmology/P15_the_crossing_exists_and_is_empty.py`

`P15_two_arm_control_and_guard` [OK]

the acoustic sector (**A LIVE LambdaCDM CONTROL AND AN UNDRIVEN GUARD, ON ONE MACHINERY – AND THE CR ARM'S FIRST PEAK SITS 23% LOW AGAINST A 1.5% POSITION FLOOR.** Built for fronts #5 and #18. **This fork had never run a control**: 'ROBUST_p1p2_scan' quotes one from r1975 in a comment, and c54.164 then showed its figures move over ell_1 in {150,165,315} with nothing to measure that against. PART 1 **the guard passes on BOTH arms** – every coupling to the potential removed at every site (4 Phi' in photon/neutrino/CDM continuity AND k^2 Psi in photon-baryon/neutrino Euler, both branches) and the source comb lands on the integers, LCDM 1.0147/2.0005 and CR 1.0121/1.9968, **so every departure from n pi/r_s in a driven run is the driving and nothing else**. PART 2 the control lands at 224/536/808/1116 against the sky's 220.6/538.1/809.8/1147.8 – **positions to 1.5% at the first peak and 0.4%/0.2% at the second and third; HEIGHTS only to 13-26%, so no height claim from this instrument is meaningful below ~25%**. PART 3 the CR arm run on its PHYSICAL discrete ladder and on a dense continuum sampling gives **identical** peaks 172/396/624/904, so discreteness is not what sets the position – an objection closed by measurement. PART 4 ell_1/ell_A = 0.5703 against the control's 0.7433 and the sky's 0.7312: **a 23% deficit against a 1.5% floor, a factor of 15 in margin**, with P1/P2 = 0.866 against the control's 1.933.). *Computes:* the undriven comb asserted within 0.03 of the integers on both arms; the control's position floor asserted under 3% AND its second and third peaks asserted within 1% of the sky; the discrete and continuum peak lists asserted identical; the deficit asserted at 5+ floors; CR's ell_1/ell_A asserted in 0.55-0.60 and the control's in 0.72-0.76; the two arms' l_A asserted to agree to 1% (they do – D_M differs by 6.2% and r_s by 6.3% and they cancel). Every background quantity is RECOMPUTED rather than pinned. *Bound:* **scope: this does NOT settle 'PO-7' and the CR figure is not offered as a prediction – it uses the initial datum the corpus's own text describes (flat in k, common phase, VELOCITIES ZERO) and c54.164 established the figure moves with that choice. What changed is that it now moves against something. NO FIGURE FROM THE ABANDONED ACOUSTIC FORK IS USED; that fork's own reproduction gate fails in this tree, and this instrument disagrees with its 201.3 by 15%.** *Run:* `python3 receipts/P15_CR_cosmology/P15_two_arm_control_and_guard.py`

`P15_the_driving_shift_by_subtraction` [OK]

the acoustic sector (**THE DRIVING SHIFT MEASURED MODEL-FREE BY SUBTRACTION: LCDM's ACOUSTIC PHASE IS FLAT IN k TO 2.1%, CR's VARIES BY 3.9x – AND THE SCALING IS k^-0.62, NOT THE k^-2 THE ABANDONED FORK'S MECHANISM REQUIRES.** Built for front #5 on the c54.168 instrument. Each k integrated TWICE on the same equations, driven and undriven, differing in one multiplicative switch and nothing else. Q(k) = k INT c_s deta/pi at the first acoustic turnover of Theta_0. **PART 1 the undriven column is the CALIBRATION and is MEASURED: 0.9968-1.0003 on both arms, where a free oscillator must give exactly 1.** PART 2/3 LCDM's Q_driven runs 0.791 -¿ 0.774 over k = 0.045-0.16 (2.1% spread) because every mode crosses under radiation, where the potential decays on the sound-crossing time by construction; CR's runs 0.457 -¿ 0.117, a factor of 3.9, because there is no radiation era to cross and c54.167 showed the pre-onset interval is scale-free. PART 4 **a power-law fit gives k^-0.62, collapsing the data to a factor of 1.2 where k^-2 spans a factor of 16.** The fold reports Q k^2 constant and its seam-transient mechanism (3H/k^2) requires it. **The phenomenon reproduces here independently; the exponent does not.** Recorded as an inter-instrument disagreement, not resolved.). *Computes:* the undriven calibration asserted within 0.01 of 1 on BOTH arms (this is the file's own gate); LCDM's spread asserted under 6%; CR's variation asserted over 2x; the fitted exponent asserted inside (-1.2,-0.3); **Q k^2 asserted to span MORE than a factor of 4 – i.e. the fold's constant-Q-k^2 asserted FALSE here**; and the fitted exponent asserted to collapse the data better than k^-2, so the fit is not vacuous. *Bound:* **scope: two corrections were forced by this file's own calibration and are recorded because each would have been a false result – (i) reading Q on Theta-hat rather than Theta_0 calibrated at 0.0003, since DR switches the couplings into the fluid but not Psi's own evolution; (ii) the driven LCDM column then calibrated at 0.001 on a decaying-mode transient two grid points from the start, excluded by requiring k eta ¿ 1, which is the DEFINITION of an acoustic turnover and removes nothing on the CR arm. No figure from the abandoned fork is used.** *Run:* `python3 receipts/P15_CR_cosmology/P15_the_driving_shift_by_subtraction.py`

`P15_which_coupling_carries_the_k_dependence` [OK]

the acoustic sector (**THE k-DEPENDENCE LIVES IN THE POTENTIAL'S GRADIENT,

NOT IN ITS DECAY – WHICH IS THE REVERSE OF WHERE A RELAXATION TIMESCALE WOULD PUT IT.** Front #5, on the c54.168 instrument against its 1.000 calibration. The potential enters the plasma TWICE: as '4 Phi'' in continuity (its rate of change feeding the density – **the decay channel**) and as 'kˆ2 Psi' in Euler (its gradient driving the velocity). Switched independently on the same integration. PART 1 **flat LCDM's two couplings are each FLAT in k and they OPPOSE** – continuity alone puts the turnover at Q = 1.23-1.28, LATER than a free half-period, the gradient at 0.73-0.87, earlier, composing to 0.77-0.86. **So the universality of the standard driving shift is a CANCELLATION between two flat terms, not one term's signature.** PART 2/3 on the radiation-free rate there is no cancellation left – both advance (0.11-0.18 and 0.06-0.57) and they add – **and the k-dependence sits in the GRADIENT: continuity varies 1.6x at kˆ-0.17, gradient 8.9x at kˆ-1.04.** PART 4 and **neither term alone is the answer**: the driven exponent -0.62 lies BETWEEN the two, so the scaling is their interference.). *Computes:* LCDM's continuity asserted ABOVE 1 and its Euler BELOW 1 (they must oppose) and all three driven exponents asserted flat to 0.15; CR's continuity asserted nearly flat (|exp| ¡ 0.35) and its Euler steep (|exp| ¿ 0.8) and the Euler term asserted to be the steeper of the two – **reversing that assertion would reverse the finding**; both CR couplings asserted below 1 (they add); the driven exponent asserted strictly BETWEEN the two and distinct from each by 0.2; and the undriven calibration asserted within 0.01 of 1 on both arms, which is the file's own gate. *Bound:* **scope: this localises the k-dependence to a coupling and shows the standard shift's universality is a cancellation this rate does not have. It does NOT name the process that makes the gradient coupling run as kˆ-1, and computes no peak position. It closes the abandoned fork's stated mechanism as an explanation of THIS measurement – a relaxation on 3H/kˆ2 acts through the decay channel, and the decay channel is measured flat.** *Run:* `python3 receipts/P15_CR_cosmology/P15_which_coupling_carries_the_k_dependence.py`

`P15_the_gradient_coupling_in_closed_form` [OK]

the acoustic sector (**THE GRADIENT COUPLING IN CLOSED FORM: THE TURNOVER IS AT A FIXED MULTIPLE OF 3H/kˆ2, AND SINCE Q ALREADY CARRIES ONE FACTOR OF k THAT GIVES Q ˜ 1/k, NOT 1/kˆ2.** Front #5, closing the question c54.170 left. **The derivation is four lines.** (1) Phi's own equation carries a damping term -kˆ2 Phi/3H, so Phi relaxes on tau = 3H/kˆ2. (2) With continuity off the oscillator is Theta_0" + c_sˆ2 kˆ2 Theta_0 = -(kˆ2/3) Psi, and **the kˆ2 in the forcing's amplitude is cancelled by the kˆ-2 in its duration – the delivered impulse is k-INDEPENDENT**. (3) So the first turnover is at the moment the impulse has been delivered, at a fixed multiple of tau: kappa = 1.79, constant across a factor of eight in k. (4) Q = k INT c_s deta/pi already carries one k, so **Q = 3 kappa c_s H/(pi k)**. **PREDICTED Q k = 0.010403/Mpc AGAINST A MEASURED 0.010372 – 0.30% – with nothing fitted but kappa; and the closed form reproduces Q itself to 1.6% across the sub-horizon band.** ⇒ **The abandoned fork's TIMESCALE is right and its EXPONENT is out by exactly one factor of k – a bookkeeping error in its own mechanism rather than a disagreement about physics.** And the continuity channel is flat for the complementary reason: '4 Phi'' feeds the density with the potential's TOTAL change, INT Phi' deta = -Phi_0, the same however fast the relaxation runs. **So the observed kˆ-0.62 is not a power law of anything – it is the crossover between a flat channel and a 1/k one.**). *Computes:* kappa asserted constant to 10% across the sub-horizon band; **the flattest grouping asserted to be Q kˆ1 and not Q kˆ0 or Q kˆ2 – if that assertion reverses, the derivation is wrong**; the predicted coefficient asserted within 5% of the measured one; the closed form asserted to reproduce Q to 5% across the band; the continuity channel asserted flattest UNGROUPED and asserted flat to within a factor of 2. *Bound:* **scope: this NAMES the process, so front #5's question is answered. It does NOT make the first peak agree with the sky – the 23% deficit against a 1.5% control floor stands unchanged. What changed is that its cause is derived rather than described. No figure from the abandoned fork is used; its timescale is confirmed independently here and its exponent corrected.** *Run:* `python3 receipts/P15_CR_cosmology/P15_the_gradient_coupling_in_closed_form.py`

`P15_where_the_likelihood_sits` [OK]

the acoustic sector (**WHERE THE LIKELIHOOD ACTUALLY SITS: IT CANNOT ARBITRATE, AND THE INSTRUMENT'S OWN CONTROL IS WHY.** Discharges 'L-147', fenced since it opened on three conditions – falsifier fixed before any data is touched, a control reproducing a known number first, and not until the two internal routes agree. **All three met** (the routes collapsed at c54.164-168), so the fence came down. **F1** the pipeline is wired: the banked CAMB flat-LCDM best fit returns chi^2 = 206.4 over 215 plik_lite TT bins, chi^2/dof = 0.96. **F2** the instrument's own floor, measured through the SAME likelihood on the SAME bins: this fork's LCDM arm costs +18786 over CAMB. **F3** the only readable quantity, both arms of one instrument: CR - LCDM = +84160. **F4** the pre-registered rule returned 'suggestive, not a measurement discrepancy'. **F6 – AND THE PRE-REGISTERED RULE WAS DEFECTIVE, WHICH RUNNING IT IS WHAT SHOWED.** It is a RATIO test and never asks whether the floor is small in ABSOLUTE terms. It is not: the control arm sits at **chi^2/dof = 103** against CAMB's 0.96, because Planck's binned errors are a few tenths of a per cent and this instrument's heights carry 13-26%. **Both arms are far outside the regime where plik_lite discriminates, so the honest verdict is the WEAKER one: the likelihood cannot arbitrate.** **!! That correction happens to FAVOUR the construction and is flagged for it** – it would apply identically had the CR arm come out better, and it rests on a height floor measured at c54.168 before any data were opened. **What survives as fact: on one instrument, same bins, same fitted amplitude, the CR arm costs 5.4x the LCDM arm's chi^2. An ordering, not a p-value.**). *Computes:* the banked LCDM fit asserted at chi^2/dof in (0.85,1.15) BEFORE anything else is read; every covered bin asserted non-empty (the keep mask); the instrument floor asserted POSITIVE (this instrument cannot beat a five-parameter CAMB fit, and if it appeared to, the binning would be wrong); the CR arm asserted to cost more than the control on the same instrument; and **F6's premise asserted – if the control arm ever drops below chi^2/dof = 3 the PART must be rewritten because the likelihood then CAN arbitrate.** *Bound:* **scope: plik_lite TT only – no polarisation, no ell ¡ 30, no lensing, no external priors, a third-party implementation rather than clik (PROVENANCE.md), and 185 of 215 bins because the instrument reaches ell = 2000. The restriction to covered bins is done on the COVARIANCE and re-inverted, which MARGINALISES over the missing multipoles; dropping rows of the Fisher matrix would CONDITION on them. The banked r1934 CR figure (397.1 vs 206.4) applied the derived DAMPING suppression to CAMB's LCDM spectrum – it tested damping on LCDM's PEAK POSITIONS and could not have seen the position result, so it answers a different question and is SUPERSEDED here. 'PO-7' stays protected: this reports a number and does not convert it into a verdict.** *Run:* `python3 receipts/P15_CR_cosmology/P15_where_the_likelihood_sits.py`

`P15_the_line_of_sight_transfer` [OK]

the acoustic sector (**THE TRANSFER REBUILT ON A LINE-OF-SIGHT INTEGRAL: THE CONTROL'S FIRST-PEAK POSITION ERROR FALLS FROM 1.66% TO 0.16%, AND THE CR DEFICIT DOES NOT MOVE AT ALL.** Front #2. Until c54.173 the source was evaluated at a SINGLE eta_rec – an instantaneous last scattering. **The visibility g = tau' e^-tau peaks at eta = 280.75 with FWHM 37.8 Mpc, a tenth of a wavelength at the first peak.** Rebuilt as Delta_l = INT S j_l deta with S = g(Theta_0+Psi) + e^-tau(Phi'+Psi') + (1/k^2) d/deta[g theta_b]. PART 1 the control goes 1.66% -¿ 0.16% in ell_1/ell_A: **sub-per-cent POSITIONS, which is what the likelihood needed of them.** PART 2 **and the CR arm is unchanged to four figures, 0.5703 on both transfers, while the control moved 1.79%. The deficit stands at 21.9% against a 0.16% floor – 134 floors.** The most suspect part of the instrument was replaced wholesale and the number did not follow it down. PART 3 **the heights are NOT fixed and the target is NOT met**: P1/P2 -13%, P1/P3 -34%, a residual growing with multipole and pointing at the damping tail (no polarisation term; a tight-coupling closure that fails where the visibility is). Stated as a partial result. PART 4 one correction was a DERIVATION: the Doppler term carries 1/k^2 not 1/k, since v_b = theta_b/k. **With 1/k the second peak sat at 516 and P1/P2 at 3.15; with 1/k^2, 532 and 1.93 – and ell_1 was 220 either way, so an instrument validated on the first peak alone would have carried the error.**). *Computes:* the control's position error asserted sub-per-cent AND asserted improved by 5x or more; the CR arm asserted to move less than 1% under the rebuild; the deficit asserted at 20+ floors; **the heights asserted NOT sub-per-cent – if they ever are, the PART is stale and front #2 should be struck**; and the corrected Doppler power asserted to move the SECOND peak toward the sky, which is what makes it a fix rather than a tuning. *Bound:* **scope: this delivers sub-per-cent POSITIONS and leaves the HEIGHTS at 13-34%, so the likelihood still cannot be run (c54.172). The remaining error is a high-multipole problem and neither of its two named causes is a CR question – both are checked against the LambdaCDM arm first. Late ISW is truncated at eta = 4000, which affects only ell below ~30 and is outside both the grid and plik_lite's TT range.** *Run:* `python3 receipts/P15_CR_cosmology/P15_the_line_of_sight_transfer.py`

`P15_the_derived_diffusion_damping` [OK] *the acoustic sector* (**THE HEIGHTS FROM 13-34% TO 2-3%: THE DIFFUSION DAMPING PUT BACK, DERIVED ON EACH ARM'S OWN OPACITY AND RATE.** Front #2. **Diagnosed from the SIGN**: at c54.173 both height ratios were LOW, meaning peaks 2 and 3 too HIGH, worse at higher multipole – under-damping. **The hierarchy's tight-coupling closure switches off at tau' = 0.6/Mpc and the visibility peaks at tau' = 0.06/Mpc, ten times lower, so the photon quadrupole and the baryon slip were absent through the WHOLE of last scattering and the line-of-sight path carried no Silk damping at all.** The pre-LOS path hid this behind a hardcoded r_D. PART 1 the standard tight-coupling integral 1/k_D^2 = INT [R^2/(1+R) + 8/9]/[6(1+R)tau'] deta, evaluated on this arm's own background: **r_D = 6.58 Mpc against the 6-7 a Boltzmann code gives. Nothing fitted.** PART 2 **and a DOUBLE COUNT had to be removed**: with the hierarchy's own damping left in over tau' ¿ 0.6 AND the integral covering the same range, the control came out OVER-damped (+6.7%, +17%). **The sign flips either side of the correction, which identifies a double count rather than a wrong coefficient.** PART 3 heights **-12.9%/-33.5% -¿ -2.8%/-2.1%**, on a control whose positions were already sub-per-cent. PART 4 **and the likelihood's floor falls 103 -¿ 23 in chi^2/dof, a factor of 4.6, WITHOUT the verdict turning over** – 'L-147''s answer stands, and the floor moving while the verdict does not is what measures how far the instrument still has to go.). *Computes:* r_D asserted inside 5.5-8 Mpc; tau' at the visibility peak asserted BELOW the gate (PART 1's premise); **the sign asserted to flip either side of the correction – if it does not, it is a coefficient error and not a double count**; the single-mechanism heights asserted within 5% and asserted better than BOTH the no-damping and the double-counted runs; the floor asserted to fall by 3x or more; **and the control asserted to remain ABOVE chi^2/dof = 2 – if it ever drops below, the likelihood CAN arbitrate and 'L-147''s verdict must be revisited.** *Bound:* **scope: this does not reach sub-per-cent heights and does not make the likelihood answerable. The residual is still worst at high multipole: no polarisation term in the source, and a photon sector with no free-streaming hierarchy after decoupling. Neither is a CR question.** *Run:* `python3 receipts/P15_CR_cosmology/P15_the_derived_diffusion_damping.py`

`P15_the_height_target_was_below_the_resolution_of_its_own_statistic` [OK]

the acoustic sector (**THE HEIGHT TARGET WAS QUOTED AGAINST A SKY NUMBER WITH A 3.4% ERROR BAR THAT WAS NEVER WRITTEN DOWN, AND chi^2 – THE STATISTIC THAT CAN SEE – PREFERS A COEFFICIENT IT IS NOT ALLOWED TO CHOOSE.** Front #2. Since c54.168 every height claim here (-13%, -34%, 2-3%, -0.05%) compared P1/P2 and P1/P3 to two bare numbers, 2.217 and 2.277. **PART 1** the numerics are cleared first: 1.4x the modes, 1.3x the k-range and 1.8x the eta-sampling move the ratios by 0.09% and 0.37%, and the 2%/7% swings that prompted the check came from LMAXL=1300 truncating the k integral under the THIRD peak. **PART 2** read off the plik_lite bandpowers with the published covariance – the full 2x2 block, not a diagonal estimate – **the sky's own ratios are 2.256 +- 0.077 and 2.280 +- 0.074: 3.4% and 3.2%.** So the 13% and 34% of c54.173 were four and ten sigma and stand; **the 2-3% of c54.175 are about one sigma and do not.** *Caught by bookkeeping and not by suspicion: X_data is binned C_l and not D_l, so peak-finding on it returned 464/761/1085 instead of 221/527/815.* **PART 3** a scan of one multiplier on 1/k_D^2 has P1/P2 selecting 1.094 +- 0.184 and P1/P3 selecting 0.897 +- 0.055 – **a 1.03 sigma split, so my own 'no single scale fits both, the residual is the envelope's SHAPE' is WITHDRAWN at its own error bar rather than published.** **PART 4** chi^2 over 185 bins does separate them: the minimum sits at 0.865 against 8/9's 0.859, and 16/15 costs 1123 units, seven inflated sigma. **The coefficient is changed back to 8/9 and NOT because it fits better** – neither value has ever been derived here (C8_diffusion_length names the ambiguity in its own text), and a remembered number is not a derivation any more than a fitted one is. **PART 5** L-147's pinned chi^2 came from a script in /tmp; 'chi2_of_spectrum.py' puts that machinery in the corpus and returns 4170.2 and 50675.2 exactly.). *Computes:* converged spread asserted under 1% and the truncated run asserted separated from it; the sky's recomputed ratios asserted within 1 sigma of the bare numbers they replace; the DAMPX multiplier asserted to reproduce the explicit POLC=8/9 run to 0.2%; **the withdrawal asserted: if the split were 2 sigma or more the file FAILS for retracting a claim that in fact holds**; the chi^2 minimum asserted at the 8/9 multiplier within 5%; both pinned values asserted recomputed to 0.5%; and an assert that fires if the control ever reaches chi^2/dof ~ 1, since this file would then need rewriting rather than patching *Bound:* **scope: this corrects a MEASUREMENT PROTOCOL, not the construction, and it is a correction to my own work. It supplies no derivation of the shear coefficient – that is named as owed and is front #2's first item: the free-streaming photon hierarchy with the polarisation multipoles. And the 3.2-3.4% is the single-bandpower-bin resolution; a full-curve fit determines the peaks far better, which is exactly why chiˆ2 replaces the ratios rather than joining them.** *Run:* `python3 receipts/P15_CR_cosmology/P15_the_height_target_was_below_the_resolution_of_its_own_statistic.py`

`P15_the_shear_coefficient_derived_not_remembered` [OK]

the acoustic sector (**THE DIFFUSION-DAMPING SHEAR COEFFICIENT, DERIVED FROM THE POLARISED PHOTON HIERARCHY INSTEAD OF REMEMBERED OR FITTED – AND ADOPTING IT COSTS THE CONTROL 1123 UNITS OF chiˆ2.** Front #2, closing the item c54.176 named as owed. 'C8_diffusion_length' has named 'the 8/9 coefficient against the 16/15 polarization-corrected one' in its own text since it was written, and neither had ever been derived here. **METHOD** two integrations of the same oscillator on the same background: the perfect tight-coupled fluid, undamped by construction, against photons carried to l = 24 with Thomson scattering, the polarisation multipoles evolved alongside and the baryons on their own velocity. **Their ratio is the damping and nothing else – the (1+R)ˆ(1/4) WKB prefactor, the drift of c_s and the R-dependence of the oscillation are common to both and divide out identically.** Gravity is off in both, which is the definition of the object rather than an approximation. **PARTS 1-4** with G = 0 the l=2 term is -(9/10)tau'F_2, the textbook unpolarised closure, and C = 0.8852 against 8/9; with polarisation C = 1.0618 against 16/15; **and the polarisation correction – the quantity actually in dispute – is a factor 1.1996 against 6/5, three parts in ten thousand.** C is read LOCALLY between consecutive extrema and extrapolated in (k/tau')ˆ2, so the kˆ2 scaling is measured rather than imposed. **PART 5** 16/15 is adopted and the control's floor goes 22.5 -¿ 28.6. **The fit's preference for 8/9, measured and refused at c54.176, is refused again: a coefficient chosen because it fits is a fitted parameter whatever it is called.** **PART 6** and nothing downstream moves – sec:envelope-consequence's 10.8% goes +10.83% -¿ +10.87%, because the coefficient nearly cancels in a ratio of two rates.). *Computes:* both intercepts asserted within 2% of the leading-order values and pinned to ˜1% in the gate; **the polarisation correction asserted within 0.5% of 6/5 and pinned to a tenth of a per cent – it is the number the file exists to produce**; the scatter about the line in (k/tau')ˆ2 asserted below 0.01, so a method that destroyed the extrapolation fails rather than shifting the intercept quietly; the sign of the finite-(k/tau') correction asserted, and the extrapolation asserted SHORT (under 2% of the coefficient swept over the fitted range); **the chiˆ2 cost asserted POSITIVE – if the derived coefficient stopped costing chiˆ2 the file's whole point would need restating**; and the C8 figure asserted to move less than a tenth of a point, so adopting the coefficient cannot silently shift a published number *Bound:* **scope: this derives the coefficient in the WKB damping integral and NOT the damping envelope's shape. It supplies the damping half of the polarisation physics only; the SOURCE half – g*Pi/4 and (3/4kˆ2)(g*Pi)" with Pi = Theta_2 + ThetaˆP_0 + ThetaˆP_2 – needs the hierarchy carried in the spectrum instrument and is what front #2 still owes. And one assertion of mine fired on its own data: I demanded both extrapolation slopes be of order unity and they are -0.150 and -0.942; the gate now checks that the extrapolation is SHORT, which is the thing that actually matters.** *Run:* `python3 receipts/P15_CR_cosmology/P15_the_shear_coefficient_derived_not_remembered.py`

`P15_the_photon_hierarchy_in_the_transfer` [OK]

the acoustic sector (**THE PHOTON BOLTZMANN HIERARCHY WITH POLARISATION CARRIED INSIDE THE TRANSFER: THE CONTROL'S chiˆ2/dof FALLS 28.6 -¿ 7.1 AND THE FIRST FOUR PEAKS LAND AT 220/540/812/1124 AGAINST THE SKY'S 220.6/538.1/809.8/1147.8.** Front #2. Photons as multipoles to l = 24 with the polarisation hierarchy alongside, the baryons on their OWN velocity and their OWN density, and the source carrying g*Pi/4 and (3/4kˆ2)(g*Pi)". **THE HANDOVER MAKES A DOUBLE COUNT IMPOSSIBLE RATHER THAN MERELY ABSENT**: the fluid runs to tau' = 3/Mpc and the envelope is FROZEN there, so envelope and multipoles have disjoint supports – and only 4.2% of 1/k_Dˆ2 has accumulated at the handover, so 96% of the damping is dynamic. **PART 1** the second peak is 0.35% out and the third 0.27%, where the fluid had them 2.6% and 0.7% low. **PART 2** chiˆ2/dof 28.6 -¿ 7.1, and 20.0 -¿ 4.2 over the range where the wavenumber integral is not starved. **PART 3** and c54.177's prediction is checked against the number it predicted: **running the same hierarchy with ONLY the two polarisation source terms dropped – exactly the half an envelope has – gives 2022 against 1320, so they are worth Delta chiˆ2 = 702, 63% of the 1123 that adopting the derived 16/15 cost.** **PART 4** and half the baryon split is WORSE than none: own density while still sharing the photons' velocity moved chiˆ2/dof 133 -¿ 161 at development settings; both halves give 15.1 -¿ 7.1. *Twice now a physical effect taken in half has been worse than not taken – it is not a small error but a different and inconsistent model.* **PART 5** doubling the hierarchy depth moves chiˆ2 by 0.00% and doubling the handover opacity by 1.5%. **PART 6** and the CR arm is at ell_1/ell_A = 0.5703 for the SIXTH instrument state, while the control improved fourfold.). *Computes:* every peak asserted no FURTHER from the sky than the fluid transfer's; the chiˆ2 improvement asserted to be at least a factor of two and pinned at 7.14 / 4.17; both height ratios asserted within 2 sigma of the sky's own covariance-derived values; **the polarisation source terms asserted to recover between 40% and 120% of c54.177's 1123 – outside that range c54.177's statement could not be said to have predicted this and the file FAILS**; the full baryon split asserted to beat the shared-density run; the LG and handover sensitivities asserted below 0.1% and 5%; the CR arm asserted unmoved to the fourth decimal; and an assert that fires if the control ever reaches chiˆ2/dof ˜ 1, since L-147 would then have to be REOPENED rather than this file's closing line printed *Bound:* **scope: this is the LambdaCDM control and the CR arm on one instrument, and it converts nothing into a verdict – 'PO-7' is protected and 'L-147''s answer is unchanged, because a control at seven times a fit still cannot certify what it is compared with. What remains is accuracy and no longer a named missing SECTOR: reionisation, neutrino mass, lensing, and the wavenumber range. And the aliasing gate was found MISSING from the line-of-sight path, which has been the default since c54.173; it cost a development run a first peak at 196 instead of 220 with nothing said, and is now on both paths.** *Run:* `python3 receipts/P15_CR_cosmology/P15_the_photon_hierarchy_in_the_transfer.py`

`P15_what_is_left_is_lensing_and_not_reionisation` [OK]

sec:refit-bound (**BOTH OF FRONT #2's NAMED NEXT STEPS MEASURED BEFORE EITHER WAS BUILT, AND ONE IS WORTH EXACTLY NOTHING.** **PART 1** above ell ˜ 40 reionisation's whole TT effect is the constant screening exp(-2 tau), and the amplitude here is fitted in CLOSED FORM – A = (m F d)/(m F m) is homogeneous of degree -1 in m, so m -¿ s m returns A/s and the residual is IDENTICAL. **d(chiˆ2) = 0 exactly, shown at tau = 0.054, 0.10 and 0.30, zero to machine precision at all three.** *The degeneracy is a property of the STATISTIC, not of the size of tau.* **PARTS 2/3** a Gaussian smoothing in ell, width scanned, minimises at chiˆ2 = 921 against the control's 1320 – and **a width that GROWS with ell beats a constant one by 1.8x in d(chiˆ2), 400 against 225, each at its OWN optimum so the comparison is of SHAPE**. Lensing smooths more at high ell because a deflection of fixed angle subtends a larger fraction of an acoustic wavelength, and that is the signature the data select. **PART 4** the width is FITTED, so the 400 bounds what a DERIVED calculation has to play for and is not a measurement of the lensing amplitude – **and ˜921 of the 1320 survives it, so lensing is the next build and is NOT the last one.**). *Computes:* the reionisation degeneracy asserted EXACT – if it ever moves chiˆ2 the amplitude is no longer fitted in closed form and PART 1's algebra no longer describes the pipeline; both smoothing optima pinned; **the growing width asserted to beat the constant one by at least 1.5x, which is the SHAPE claim and the reason the word is 'lensing' rather than 'some smoothing'**; what survives asserted above 60% of the baseline, so a rewrite is forced rather than a patch if a fitted smoothing ever accounts for most of the residual; and an assert that fires if a smoothing alone would take the control to chiˆ2/dof ¡ 2, since L-147 would then have to be revisited *Bound:* **scope: this measures what two candidate builds can be worth on THIS statistic. It is not a result about the sky, not a measurement of the lensing amplitude, and the control does not improve – the smoothing width is fitted. Reionisation is STRUCK from the front's list rather than deferred, and the condition that would undo the strike is written down: a comparison reaching ell ¡ 30, a second spectrum, or a fixed amplitude.** *Run:* `python3 receipts/P15_CR_cosmology/P15_what_is_left_is_lensing_and_not_reionisation.py`

`P15_derived_lensing_on_the_lcdm_arm` [OK]

sec:refit-bound (**THE DERIVED LENSING c54.181 TEED UP, RUN WITH NO FREE WIDTH – AND IT COMES IN UNDER THE FITTED BOUND, WHICH IS WHAT A DERIVATION MUST DO.** ΛCDM's own non-perturbative lensed/unlensed ratio applied to the c54.178 LCDM arm takes the control 1320 -¿ 989, d(chiˆ2) = 331 – UNDER the 400 the c54.181 fitted smoothing bounded, and concentrated in the damping tail (ell 1100-1500) where lensing fills the troughs. The acoustic peaks move 0.0%. The corpus's own first-order Hu (2000) kernel, run on the SAME Cˆphiphi so only the ORDER differs, OVERSHOOTS to 508 because it BREAKS DOWN in the tail – a spurious +13% at ell=1900 against the full operator's +6.5% – so the derived number is the full operator's 331, not the first order's 508. Validated on CAMB's own LCDM: 615 unlensed -¿ 186 lensed (chi^2/dof ~ 1). ~989 of 1320 survives, so lensing is real and NOT the last build.). *Computes:* lensed CAMB LCDM asserted at chi^2/dof in (0.75,1.25) so the operator and pipeline are validated before anything is read off the arm; the derived lensing on the arm asserted in (280,385) AND strictly under the c54.181 fitted bound of 400, which is the defining property of a zero-width derivation against a fitted upper bound; the first-order kernel asserted to OVERSHOOT the full operator by more than 80 in d(chi^2) via the tail artefact, a live check that LENS_correction is first order and that the tail is where it fails; and the peak shift asserted below 0.3%. *Bound:* **scope: the lensing operator is ΛCDM's, imposed not fitted, so this is not a measurement of the lensing amplitude and the CR arm does not fit – it measures how much of c54.181's fitted-smoothing headroom a genuine zero-parameter lensing recovers (most of it), with the shortfall against the fit being the fit's free width. The residual ~989 is the instrument's transfer inaccuracy, not lensing's: CAMB's UNLENSED LCDM already sits at 615 on these same bins. 'PO-7' stays protected.** *Run:* `python3 receipts/P15_CR_cosmology/P15_derived_lensing_on_the_lcdm_arm.py`

`UNC_error_budget` [OK]

sec:refit-bound (UNC — QUANTIFYING WHAT WE DO NOT KNOW. The prediction with an error budget. Corrections folded in first: * the recombination WIDTH scales with the rate (A.22) – so the thickness contributes ~nothing to the RATIO and the signature is carried by r_D essentially alone; * the l_D anchor affects only the ABSOLUTE r_D and cancels in the ratio – it carries NO uncertainty into the prediction;). *Computes:* runs clean; registered r2376 (c54) from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P15_CR_cosmology/UNC_error_budget.py`

`P15_the_residual_is_contrast_and_the_lensing_potential_is_derived` [OK]

sec:refit-bound (**WHAT THE CONTROL'S REMAINING chi^2 IS MADE OF, AND THE LENSING POTENTIAL BUILT TO ATTACK IT.** **PART A** templates projected through the same inverse covariance the chi^2 uses: **positions remove 1.4 of 1320 – 0.1%, so the peaks are where the sky puts them and that is not where the residual lives**; a smooth envelope 15%; **the peak-trough oscillation at FIXED position 38%, with the model's contrast 13.1% TOO HIGH** – which is what lensing reduces; and **53% in NEITHER, so a perfect lensing calculation leaves about half**. *Fitted templates, so each share is an upper bound on what the effect can claim.* **PART B** the line-of-sight solve stops at z = 12.8 but the lensing kernel peaks at z ~ 3.3, so Phi is carried on by the background's own growth factor D prop. H INT da/(aH)^3 – exact in linear theory, no transfer function imported, k-dependence from this instrument's own Phi. **And the normalisation is the amplitude the TEMPERATURE comparison already fits, so lensing enters with NO new parameter.** Limber checked against an EXACT projection (5% over l = 5-200); the deflection maximum is BROAD, within 10% over l = 12-42; **rms deflection 2.69 arcmin with nothing tuned to it.**). *Computes:* positions asserted below 2% of the residual, so the diagnosis reorders loudly if they ever rise; the contrast excess pinned to 13.1% +- 2 points, since that is what the convolution must remove; what survives BOTH template sets asserted at 53% +- 5, **so a build that forgets how much remains fails here**; Limber's departure asserted below 10%, since the convolution will use it; the deflection maximum asserted BROAD; and the absolute rms deflection pinned at 2.692 arcmin, which is the zero-parameter check on the normalisation *Bound:* **scope: PART A's shares are FITTED-template upper bounds, not measurements of physical effects, and the two sets overlap by ~7 points. And PART B's external agreement is weaker than it looks: the ~2.7 arcmin and the ~1e-7 deflection power it matches are RECALLED, not derived here and not checked against a source this corpus holds – this programme was bitten twice in three revisions by exactly that (the 16/15 coefficient at c54.176, and a Limber suspicion in this very file which was also recollection and also wrong). What PART B establishes is INTERNAL consistency; the external agreement is corroboration at the strength of a memory.** *Run:* `python3 receipts/P15_CR_cosmology/P15_the_residual_is_contrast_and_the_lensing_potential_is_derived.py`

`P15_the_filter_is_a_sound_speed_filter` [OK]

sec:what-crosses (**THE BRANCH-POINT SELECTION RULE'S EXPONENT HAS TWO FACTORS AND THE CORPUS VARIES ONLY ONE.** The damping is e^{-k c_s |Delta eta|}: 'sec:what-crosses' holds c_s at the plasma's value and varies k, reading a SCALE criterion (frozen against oscillating). **Hold k and vary c_s and the same inequality is a SPECIES criterion – and a pressureless component has c_s = 0 IDENTICALLY, so its exponent vanishes at EVERY wavenumber.** ** THAT MAKES THE MATTER SECTOR'S INHERITANCE EXACT RATHER THAN ADIABATIC: ** at c_s = 0 the segment's mode equation is psi" = 0, whose constant solution the kernel leaves untouched with no approximation, whereas the frozen-mode argument rests on horizon exit and degrades at small scales. The two agree wherever both apply; only the pressureless one is available everywhere. COMPUTED on the corpus's own segment: (1) |Delta eta| = 3.3197 (the recorded 3.32); (2) exact transfer-matrix transmission in BOTH k and c_s, reproducing -k c_s |Delta eta| to the digit for every constant c_s; (3) **ln T = 0 EXACTLY at c_s = 0 for k = 1, 10, 85.3, 1e3, 1e5 – uniform in k**; (4) at the first peak the exponent runs ˜1.6e2 c_s/c, so the filter is opaque above c_s ˜ 3.5e-3. ** CONSEQUENCE: a per-cent sound speed already gives eˆ-1.6 and ten per cent eˆ-16, so THE CROSSING CAN INHERIT PERTURBATIONS FROM A COLD SPECIES AND FROM NO OTHER – a sharp condition on what the progenitor may hand over. ** **(!! MARKED r2376+c54.162: the ARITHMETIC stands; the SELECTION-RULE INFERENCE is WITHDRAWN. 'the crossing can inherit perturbations from a cold species and from no other' does not hold – the kernel acts on OSCILLATORY content and nothing arrives oscillating ('P16_every_mode_is_frozen_at_the_crossing'), so the exponent computed here has nothing to act on and the crossing is lossless for EVERY species. A LOSS of a claimed prediction, recorded as one; P15 'sec:what-crosses' corrected in the same pass.)**). *Computes:* the segment's conformal length asserted at 3.32; the plasma exponent asserted in (-160,-140); ln T asserted zero to 1e-9 at c_s = 0 across five decades in k *Bound:* ** THIS IS THE FIELD HALF ONLY. ** What happens to perturbations of a given species; the detailed WORLDLINE dynamics of a concrete model at the branch point is a separate computation and is not done. **And an honest bound**: on this segment the plasma exponent comes out -147 against the -152 the corpus records, a 3% difference from the sound-speed profile's discretisation – nothing here turns on it, the load-bearing result being the exact vanishing at c_s = 0, which is profile-independent *Run:* `python3 receipts/P15_CR_cosmology/P15_the_filter_is_a_sound_speed_filter.py`

`P15_the_collapse_leg_is_scale_invariant` [OK]

sec:transmission (**THE COLLAPSE LEG IS A SCALE-INVARIANT BACKGROUND, AND P15 LISTS 'CR HAS NO INFLATIONARY SCALE-INVARIANT ATTRACTOR' AS AN ESTABLISHED SIGNATURE. THE ADJECTIVE WAS CARRYING THE SENTENCE.** For a background a ˜ (-eta)ˆbeta with z ˜ a, the Mukhanov-Sasaki potential is beta(beta-1)/etaˆ2, so scale invariance requires beta(beta-1) = 2 – **a quadratic with TWO roots: beta = -1, de Sitter inflation (expanding), and beta = +2, a MATTER-DOMINATED CONTRACTION.** **And the corpus's own collapse leg is beta = 2 exactly:** its banked near-branch-point law rˆ3 = (9/2) M tauˆ2 with a = |r| gives d ln a/d ln tau = 2/3 and d ln eta/d ln tau = 1/3, hence a ˜ etaˆ2 – forced by f -¿ -2M/r, the mass term dominating and the Lambda term vanishing, **which is the same pressurelessness 'A·4' used at c54.112 to show such a component crosses exactly.** The mode equation integrated from vacuum data on both backgrounds returns d ln P/d ln k = 0 to a part in 10ˆ4. **So n_s = 1 is produced by the construction rather than inherited from the progenitor, and only the DEPARTURE from unity is owed.**). *Computes:* the two roots asserted [-1, 2]; both logarithmic slopes of the leg asserted exactly 2/3 and 1/3; both integrated spectra asserted flat to 1e-3 *Bound:* **bound, and it is the whole caution: this is the STANDARD contraction result (Wands 1999; Finelli-Brandenberger 2002) applied to a background the corpus DERIVES. It inherits that literature's known difficulty – whether the matching across the bounce selects the scale-invariant mode – and here the bounce is a branch point with an analytic continuation rather than a smooth bounce. THAT QUESTION IS NOT SETTLED HERE and is registered as 'L-169'** *Run:* `python3 receipts/P15_CR_cosmology/P15_the_collapse_leg_is_scale_invariant.py`

`P15_the_continuation_is_diagonal` [OK]

sec:predictions (**THE CONTINUATION THROUGH THE BRANCH POINT ACTS DIAGONALLY ON THE TWO PERTURBATION MODES, SO THE SCALE-INVARIANT SPECTRUM DOES NOT REACH THE OBSERVER.** The falsifier on the companion result, named at c54.122 and run at c54.123. **The scale-invariant spectrum sits in the mode that DOMINATES the contraction (zeta ˜ etaˆ-3); the mode that SURVIVES on the expanding side is the constant one, whose spectrum from vacuum data is n_s - 1 = 6.** Whether the first reaches the second is a connection problem, settled here two independent ways. **(i) PARITY: the exponents 2 and -1 differ by an integer so a logarithm could mix them – but sinhˆ(2/3) is EVEN in tau, which places its correction at etaˆ6 and steps the Frobenius recursion in SIXES past a resonance sitting at THREE. No logarithm.** **(ii) NUMERICS: continuing both solutions around the branch point, the diagonal entry goes to 1.000131 and the leakage falls as a clean deltaˆ(2/3) power law – and the power law is itself evidence against a log, which would leave a residual no power of delta

removes.** So c54.122's scale invariance is confined to the contracting side. **What survives is narrower and sharper than what it replaced: the construction HAS a scale-invariant background, and the obstruction is a PARITY PROPERTY OF ITS OWN SOLUTION rather than a missing mechanism.**). *Computes:* indicial exponents asserted; both vacuum coefficients' k-scalings asserted exactly -3/2 and +3/2; the absence of odd powers in the correction asserted; the resonance asserted unreachable; the diagonal entry asserted to 1e-3 and the leakage ratio asserted equal to 10^(-2/3) at every decade *Bound:* **bound: this settles the connection for the SCALAR mode equation on this background. It does NOT settle what a background correction ODD in tau would do – which is what mixing would require and is the named open route. Three defects were caught in the making: a check that only proved invertibility, a branch error from squaring before taking the cube root, and a test run at delta where k eta ~ 1.4, outside its own frozen-mode hypothesis**
Run: `python3 receipts/P15_CR_cosmology/P15_the_continuation_is_diagonal.py`

`P15_the_evenness_is_the_time_symmetry` [OK]
sec:predictions (**THE DIAGONALITY IS THE CONSTRUCTION'S TIME-REVERSAL SYMMETRY, AND THE SAME INVARIANCE THAT FIXES THE PEAK HEIGHTS IS WHAT STOPS THE SCALE-INVARIANT SPECTRUM CROSSING.** The falsifier left by the previous receipt – mixing needs a background correction ODD in tau, with the Z_3 monodromy the named candidate – fires NEGATIVE. **Write the law as r = A u^(2/3) g(u) with g = [sinh(u)/u]^(2/3): ALL the branch structure sits in u^(2/3), and g is EVEN, analytic and non-vanishing at u=0** – exactly, not to a truncation, since log(sinh u/u) is analytic in u^2 near 0 and the exponential of a series in u^2 is a series in u^2. **The three sheets differ by a CONSTANT cube root of unity, and a constant multiplier cannot change a parity, so no sheet has an odd part; the conjugate branch is even too and has no branch point at all.** **AND THE EVENNESS IS r(-tau) = r(tau): the cosmogenesis is TIME-SYMMETRIC about the branch point, and a time-symmetric background cannot mix a growing mode with a decaying one because mixing would distinguish a direction of time the background does not.** P15 sec:amplitude already leans on this same invariance one observable over – the peak HEIGHTS follow from a driving amplitude exactly invariant under time reversal. **One symmetry, two consequences, and the corpus had banked only the welcome one.**). *Computes:* sinh(u)/u asserted even with unit value at 0; g asserted free of odd powers through u^12; all three sheets asserted free of odd powers; cosh^(2/3) asserted even; sinh^2 asserted invariant under u -¿ -u *Bound:* **bound: the leg is exactly even because it is the marginally-bound E=1 geodesic of an exactly VACUUM SdS exterior. A progenitor with real matter content – pressure, dissipation, anything carrying an arrow – would not be. This receipt does not exclude that; it LOCATES it, and puts the question back where 'L-150' already sits**
Run: `python3 receipts/P15_CR_cosmology/P15_the_evenness_is_the_time_symmetry.py`

`P15_the_geometry_transmits_no_parameters` [OK]
sec:transmission (**THE INHERITED DATA ARE NOT INHERITED FROM THE COLLAPSE GEOMETRY, BECAUSE THAT GEOMETRY TRANSMITS ZERO PARAMETERS. A COUNT, AND THE CORPUS PROVED THE KEY STEP ITSELF WITHOUT CASHING IT.** The SdS progenitor family is parametrised by M alone – alpha is the one invariant and E=1 is a choice of geodesic – so the geometry is ONE-dimensional. **And c54.113 showed the Kretschmann along the bead is MASS-FREE in the faller's own proper time (dK/dM = 0, reasserted here; K -¿ (64/27)/tau^4). A quantity identical for every member carries no information about which member it was, so the approach transmits ZERO parameters, not one.** Against that the corpus draws at least FOUR independent numbers: A_s, n_s, eta, z_onset (= rho_r/rho_m, one datum by c54.105), and the light-element composition. **Four numbers cannot come from zero parameters, so 'inherited from the progenitor collapse' cannot mean inherited from the collapse GEOMETRY – it means inherited from the progenitor's MATTER CONTENT, which M does not carry, the exterior being exactly vacuum and the interior dust without composition.** This RECLASSIFIES the row: not a derivation problem but a modelling one. **What the geometry does fix it fixes universally: |Delta eta| = 3.32 alpha = 1.8e4 Mpc, from Lambda alone – a CUTOFF, not a CONTENT.**). *Computes:* dK/dM asserted zero and the 64/27 limit asserted; the count of drawn data asserted at least four; the conformal length asserted 3.32 alpha and asserted to land between 1.5e4 and 2.0e4 Mpc *Bound:* **bound: a counting argument about what the construction's own description of the progenitor can carry. It does NOT show that no model supplies eta – it shows no model of the VACUUM EXTERIOR can, and that the row needs an interior**
Run: `python3 receipts/P15_CR_cosmology/P15_the_geometry_transmits_no_parameters.py`

`P15_hbar_survives_in_As_and_cancels_in_ns` [OK]

sec:transmission (**hbar IS A MULTIPLICATIVE CONSTANT IN THE POWER SPECTRUM, SO IT SURVIVES IN THE AMPLITUDE AND CANCELS IN THE TILT — WHICH IS WHY n_s IS DERIVABLE HERE AND A_s IS NOT.** The vacuum normalisation v = sqrt(hbar/2 c_s k) e^{-i c_s k eta} is the only place a quantum constant enters, and zeta = v/z with z classical, so P(k) = hbar A_0 k^(n_s - 1): **A_s = P(k_0) carries hbar, and d ln P/d ln k does not — asserted both ways.** An amplitude is the spectrum at one wavenumber; a tilt is its ratio to itself at two. **So the no-magnitude result (c54.128: neither real form supplies a second scale, and hbar, c, G are unit gauges over the single one) forbids deriving A_s and says nothing about n_s.** A_s is therefore permanently inherited while n_s is not, and the two were never the same kind of quantity — though the corpus has named them together in every statement of the frontier, which is how one got worked for thirteen revisions while the other went unexamined. **AND THE INVERSION LOOKED LIKE THE PAYOFF: a quantity not derivable from the invariants is a function of the moduli, so A_s = F(2M/alpha) and the observed A_s would MEASURE the parent's mass. SUPERSEDED at c54.137 ('P15_the_progenitor_vacuum_is_negligible_too'): F is computed, it is a function of TWO moduli rather than one, and it carries no fixed normalisation — the incoming amplitude is a classical datum, so the relation cannot be inverted. The hbar asymmetry itself stands unchanged.** Second handle on the same object: the spectral bound constrains total expansion past equality, this constrains the mass, and the recollapse condition caps that mass below Nariai.). *Computes:* the hbar-dependence of A_s asserted non-zero and of the tilt asserted zero *Bound:* **bound: F itself is NOT computed. The pieces exist — the mode functions, the monodromy, the mode map — and what is missing is the normalisation carried through all three rather than tracked as a scaling. The settled part stands alone: A_s is not derivable, for a reason, and the reason names which other quantities share its fate — every magnitude, and no ratio** *Run:* `python3 receipts/P15_CR_cosmology/P15_hbar_survives_in_As_and_cancels_in_ns.py`

`P15_the_progenitor_vacuum_is_negligible_too` [OK]

sec:amplitude (**THE PROGENITOR'S OWN VACUUM, CARRIED THROUGH THE FULL PASSAGE AND AMPLIFIED BY THE BRANCH-POINT MIXING, IS ALSO NEGLIGIBLE — SHORT BY 1e104. So 'prop:amplitude' had ruled out the one vacuum the inheritance story was NOT claiming.** The collapse-side background is a single function, a = M(rho x + x^2/2) with x = |eta - eta_c|, whose Mukhanov-Sasaki potential 2/(x(x+2 rho)) is **M-FREE**, so the mass enters only through the final division zeta = v/a. The passage has three stages and each contributes one factor of 1/rho: (i) the mode freezes on the beta = 2 leg as v -¿ D_k/x with D_k = sqrt(hbar/2) k^(-3/2); (ii) **the connection through equality is elementary — v = a int dx/a^2 solves the k = 0 problem exactly and carries D_k/x to the constant (3/2 rho)D_k**; (iii) the crunch monodromy leaks it into the survivor with coefficient 2 pi/rho; and a'(eta_c) = -M rho supplies the third. **THE TRANSFER LAW IS THEREFORE P = 18 k^3 D_k^2/(M^2 rho^6), with the k^3 cancelling identically on vacuum data — exactly scale-invariant.** On the vacuum that is A_s = (9/4)(l_P/M)^2 rho^(-6) = 2e-113 at the recollapse cap and the determined rho, against an observed 2.1e-9. **The mixing buys eight orders over the bare vacuum and is still short by a hundred and four.** Two corollaries: A_s does NOT measure the progenitor's mass, D_k being free; and the classical datum has a size — the parent's growing-mode amplitude exceeded its own vacuum's by ~1e47. **The same closed form also returns c54.131's monodromy analytically, the (x/rho)ln x term continuing to -2 pi i/rho, where it had been obtained by numerical continuation.**). *Computes:* the M-free potential asserted symbolically; the connection coefficient asserted in closed form and against integration from vacuum data to 0.10% over a decade in rho and a factor 15 in k; the transfer law asserted k-independent; the shortfall asserted between 1e101 and 1e106 *Bound:* **bound: z = a with c_s = 1 is assumed on the beta = 2 leg — c54.122's own hypothesis and the standard matter-contraction idealisation, genuine dust having c_s = 0 and no vacuum oscillation to normalise. The 1e104 is far too large for any O(1) convention or plausible c_s treatment to touch** *Run:* `python3 receipts/P15_CR_cosmology/P15_the_progenitor_vacuum_is_negligible_too.py`

`P15_no_primordial_B_modes_unconditionally` [OK]

sec:amplitude (**THE TENSOR SECTOR IS WHERE THE COLLAPSE-TRANSFER MACHINE RUNS WITH NO IDEALISATION AT ALL, AND IT RETURNS P_T ~ 5e-111 — ONE HUNDRED ORDERS BELOW THE OBSERVATIONAL CEILING.** The scalar transfer needs z ~ a, which for a fluid holds only where w and c_s are constant — and across the progenitor's own equality (1+w) runs 1 -¿ 4/3 while c_s runs 0 -¿ 1/sqrt3, so neither is. **For tensors that is not a restriction: z_T = a M_Pl/2 EXACTLY, for any content, any equation of state, any sound speed, so z_T"/z_T = a"/a identically and the entire passage — background, connection (3/2 rho), monodromy (2 pi/rho), division by a'(eta_c) = -M rho — applies verbatim with nothing assumed.** It returns **P_T = 144 pi (l_P/M)ˆ2 rhoˆ(-6), scale-invariant (k cancels identically)**, i.e. 4.8e-111 at the recollapse cap and the determined composition, against a ceiling of 6.7e-11 from r ¡ 0.032: **below it by 1.4e100, so r = 2e-102.** So the absence of primordial B-modes is a prediction at any conceivable sensitivity, and **it no longer rests on the substrate's own tensor floor — that argument had the same scope defect its scalar twin did, ruling out the substrate's vacuum and saying nothing about the progenitor's.** And the ratio is fixed too: every k-dependent factor is shared by the two sectors, so **r_out = 6 r_in (c54.146: scalars mix twice as strongly as tensors), k-independently — the crossing can neither manufacture nor destroy a tensor signal, only rescale the ratio.** The observed bound therefore measures the PARENT: its own tensor-to-scalar ratio was below 1.3e-3.). *Computes:* P_T asserted k-independent and asserted equal to 144 pi (l_P/M)ˆ2 rhoˆ(-6) symbolically; the shortfall against the ceiling asserted above 1e95; the crossing ratio asserted equal to 24 *Bound:* **bound: the factor 24 carries the scalar side's idealisation (it is 24(1+w)c_s on the contracting leg, the standard matter-bounce value) and no more. The absolute P_T carries nothing** *Run:* `python3 receipts/P15_CR_cosmology/P15_no_primordial_B_modes_unconditionally.py`

`A2_the_three_owed_derivations_are_one_boundary` [OK]

sec:scope (the three owed derivations are ONE closed boundary: A_s, n_s and the inherited datum are all dimensionless magnitudes). *Computes:* asserts P15s own list, prop:amplitudes 10ˆ113 gap, the progenitor tilt, and the capstones derived progenitor composition (13 checks) *Bound:* NOT-A-PAPER-CLAIM — L-211 run on L-150 *Run:* `python3 receipts/L211_closure_adjacency/A2_the_three_owed_d`

`P03_acceleration_is_slice_curvature` [OK]

sec:curvature · rem:twocritical · the bend (on ANY member of the energy family (E cancels: (dr/dtau)ˆ2 = Eˆ2 - f, and E is a constant of the motion) dˆ2r/dtauˆ2 = -f'/2 = r*K_G identically, so THE SIGN OF THE SLICE'S GAUSSIAN CURVATURE IS THE SIGN OF THE COMOVING ACCELERATION – and the flat locus is the deceleration-to-acceleration turnover, which on Nariai is the seam and in standard terms is rho_m = 2 rho_Lambda (cschˆ2 w = 2 ¡=¿ sinh w = 1/sqrt2, the seam's phase). A recent cosmological epoch: 7.06 Gyr at H0=73, 7.65 at 67.4, preceding matter-Lambda equality [borrowed from P3 · P7 · P8]). *Computes:* the identity asserted; the K_G zero solved and matched to (M alphaˆ2)ˆ(1/3); cschˆ2 w = 2 ¡=¿ sinh w = 1/sqrt2 proved by algebra after simplify() failed on the closed forms; epochs at both H0 *Bound:* bound: identities on the Nariai E=1 worldline; no causal claim between seam and turnover – they are one locus; dating inherits H0 *Run:* `python3 receipts/P03_SdS_slicing/P03_acceleration_is_slice_curvature.py`

`BRANCHPT_transmission_character` [OK]

rem:phase-open (BRANCHPT — the transmission character of the BRANCH POINT r=0, on its own terms. WHY (r2154). P15 l.525 says "The cosmogenesis branch point, BEING the degenerate Nariai horizon, is a faithful scale-free transmitter", and prop:transmission argues from the tortoise integral at a DOUBLE ROOT (p=2, kappa=0). **PART 2 ADDED r2376+c54.153 by the receipt-vs-sentence audit: the PEAK-POSITION EXCLUSION, which is what P15 actually cites this file for and which the file did not compute** — the collapse leg's own conformal time gives eta_bp ˜ 5e3-1e4 Mpc, so r_sˆeff = r_s + eta_bp/sqrt3 = 3.0-5.9e3 Mpc, and reproducing l_A = 301.8 would demand D = 2.9-5.7e5 Mpc against the actual 1.4e4, a factor of **21-41**. *PART 1 shows the crossing ADDS no divergent phase; PART 2 shows an inherited phase cannot survive. The paper's sentence needs the second.* [borrowed from P15 (+P7)]). *Computes:* runs clean; registered r2376 (c54) from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P15_CR_cosmology/BRANCHPT_transmission_character.py`

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